CAMPAIGN OPTIMIZATION IN TELECOMMUNICATION WITH COMMUNICATION LIMITATIONS AND NETWORK EFFECTS

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To my family

ABSTRACT

Marketers aim to effectively introduce products or services to both existing and potential customers. While previous studies have focused on identifying the target audience for marketing campaigns, this study addresses the next critical step: devising optimal communication strategies to engage the selected audience. Balancing timely, relevant communication with the risk of customer fatigue/irritation is essential, and such efforts must also account for channel capacities and governmental marketing regulations. We propose a novel approach that enhances Call Detail Record (CDR)-based social networks with tariff-change data, constructing a comprehensive customer network. This enriched social network is crucial for identifying key customers and maximizing campaign reach. A mathematical model integrates communication constraints and channel capacities to determine the optimal allocation of campaign messages, specifying who should receive information about which campaign, through which channel, and when. To solve this complex problem, we introduce a heuristic that combines a small-scale linear programming model with a customer-ranking mechanism based on network influence. Computational results show that the heuristic quickly provides high-quality solutions, even for large-scale realistic instances.

ÖZETÇE

Pazarlamacılar, ürün veya hizmetleri mevcut ve potansiyel müşterilere etkili bir şekilde tanıtmayı amaçlamaktadır. Önceki çalışmalar, pazarlama kampanyaları için hedef kitleyi belirlemeye odaklanırken, bu çalışma bir sonraki kritik adımı ele almaktadır: Seçilen kitleyi etkilemek için en uygun iletişim stratejilerini geliştirmek. Zamanında ve alakalı iletişim kurma ihtiyacı ile müşteri yorgunluğu ve rahatsızlığı riskini dengelemek önemlidir ve bu çabalar aynı zamanda kanal kapasiteleri ve hükümetin pazarlama düzenlemelerini de dikkate almak zorundadır. Bu çalışmada, bir sosyal ağ oluşturmak için tarife değişiklik verileriyle zenginleştirilmiş bir Çağrı Detay Kaydı (Call Detail Record-CDR) tabanlı yaklaşım öneriyoruz. Bu zenginleştirilmiş sosyal ağ, kilit müşterileri belirlemek ve kampanyanın ulaşımını maksimize etmek için kritik öneme sahiptir. İletişim kısıtlamalarını ve kanal kapasitelerini entegre eden bir matematiksel model, kampanya mesajlarının en uygun dağılımını belirler; kimin hangi kampanya hakkında, hangi kanaldan ve ne zaman bilgilendirilmesi gerektiğini gösterir. Bu karmaşık problemi çözmek için küçük ölçekli bir doğrusal programlama modelini, ağ etkisine dayalı müşteri sıralama mekanizmasıyla birleştiren bir sezgisel algoritma sunuyoruz. Hesaplama sonuçları, bu sezgisel algoritmanın, büyük ölçekli ve gerçekçi örnekler için bile hızlı bir şekilde yüksek kaliteli çözümler sağladığını göstermektedir.

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CHAPTER I

INTRODUCTION

Outbound marketing in telecommunications is a constrained decision problem: firms must decide whom to contact, through which channel, and on which day, while respecting legal requirements, channel capacities, and customer-contact rules. Unlike inbound marketing, outbound communication is firm-initiated, so poor targeting or excessive frequency can directly increase annoyance and churn [1, 2]. As digital channels expand, campaign planning requires optimization models that jointly handle campaign priorities, eligibility, capacity, and frequency constraints.

This thesis studies campaign optimization under communication constraints using operational data from a telecommunications setting. The baseline model captures direct campaign delivery decisions across customers, channels, and days. The extended model adds network-aware effects using customer adjacency constructed from Call Detail Records (CDRs) and tariff-change behavior. This extension supports decisions that account for both direct reach and one-hop peer awareness.

Terminology and scope. Throughout the thesis, we use *communication constraints* as the canonical operational term (instead of interchangeable alternatives). We use *network effects* in managerial and computational discussion, and *network externalities* when referring to the underlying mechanism. The modeled mechanism is strictly *one-hop*: a customer may be affected by directly connected neighbors only. The thesis does not model multi-hop cascade diffusion or viral spread.

1.1 Contributions

The thesis makes four main contributions:

- **COPL formulation:** A binary optimization model for campaign planning under communication constraints, including eligibility, per-customer limits, category limits, channel capacities, and rolling-horizon controls.

- **Scalable solution methods:** A sequence of heuristic/matheuristic approaches (Basic Greedy, Relax+Greedy, Aggregate+Greedy) designed to produce feasible high-quality solutions for realistic instance sizes.

- **Network construction and prioritization:** A practical customer-network construction pipeline from CDR and tariff-change data, together with CustSortA/B/C prioritization strategies.

- **COP-LN and HANSA:** An extension that incorporates one-hop network effects and a hybrid adaptive metaheuristic (HANSA) for improved solution quality in large instances.

The remainder of this thesis is organized as follows: Chapter 2 reviews the relevant literature on campaign optimization and network-aware targeting in telecommunications. Chapter 3 formalizes the campaign optimization problem with communication constraints and presents the mathematical model along with solution methods. Chapter 4 develops network construction methodologies using CDR and tariff-change data, and introduces three customer prioritization strategies that leverage network structure to identify influential customers. Chapter 5 extends the optimization model to incorporate network effects, utilizing the customer prioritization strategies from Chapter 4 within a simulated annealing-based metaheuristic framework. Finally, Chapter 6 concludes with a concise recap and directions for future research.

CHAPTER II

LITERATURE REVIEW

The optimization of outbound marketing campaigns has become a central concern in data-driven marketing research, particularly in the context of identifying high-response customer segments and effectively allocating communication resources. Traditional approaches have predominantly leveraged machine learning algorithms, statistical inference models, and heuristic business rules to forecast customer responsiveness to specific campaign offers [3, 4]. In parallel, the database marketing literature has focused on estimating the expected economic value of offer-channel-customer combinations to maximize return on marketing investment [5]. Despite precise targeting, even optimally selected customers may disregard marketing messages if the communication channel or timing is misaligned with their preferences or routines.

The proliferation of digital communication platforms has necessitated a shift toward omni-channel campaign strategies. Contemporary consumers expect cohesive brand interactions across web, mobile, social media, and in-person touchpoints [6]. Consequently, firms must design campaigns that transcend siloed communication streams and consider the cumulative impact of channel interactions [7, 8]. Notably, customers often engage with information in one channel but convert via another, underscoring the importance of modeling cross-channel influence in campaign optimization efforts [8].

While much of the existing literature concentrates on optimizing targeting strategies—maximizing click-through or action rates—many such studies are framed from the publisher's or platform's perspective, rather than that of the advertiser [9, 10]. Our approach diverges by foregrounding the advertiser's objective: to achieve efficient customer engagement within organizational and regulatory constraints, including communication frequency caps and resource limits.

Formally, campaign-to-customer allocation problems have been modeled as variants of the multidimensional knapsack problem (MDKP) or generalized assignment problems [3, 4, 11]. Solving such formulations at industrial scale poses substantial computational challenges, particularly in the presence of mixed-integer variables and complex constraint sets. Classical optimization methods such as branch-and-bound (B&B) [12, 13] are often computationally prohibitive for large datasets, motivating research into parallelized extensions [14, 15]. Alternatively, heuristic techniques—including greedy algorithms and Lagrangian relaxations—have shown efficacy in approximate solution generation under resource constraints [16].

Notwithstanding these methodological advancements, the rapidly evolving marketing landscape—characterized by abundant behavioral data, channel heterogeneity, and temporal dynamics—demands novel analytical approaches. A growing body of research has therefore begun to integrate machine learning with optimization models to better forecast response and optimize campaign strategies [17, 18].

A critical extension of the traditional framework is the integration of social network analytics (SNA). Particularly in the telecommunications sector, Call Detail Records (CDRs) have been exploited to construct temporal and spatially rich interaction graphs, revealing latent patterns of customer influence and behavior diffusion [10, 19]. These networks enable the identification of structurally influential individuals—often central nodes—whose inclusion in outbound campaigns can catalyze viral propagation effects [20]. However, as recent studies demonstrate, social graph data alone is insufficient. Effective models must also incorporate demographic attributes, purchasing behavior, and historical campaign interactions to accurately model engagement propensity [21, 22].

Moreover, campaign fatigue—manifested in declining responsiveness due to oversaturation—has emerged as a critical consideration in the design of outbound strategies. The literature increasingly recognizes the need for fatigue-aware scheduling and frequency capping mechanisms to maintain customer receptivity over time [23, 24]. Campaign optimization thus evolves from a static allocation problem into a dynamic, multi-objective control challenge across rolling horizons.

While earlier models tend to assume fixed campaign durations, single-product offerings, or homogeneous customer behavior, more recent contributions advocate for adaptive frameworks capable of managing multi-product campaigns under time-varying conditions [25, 26, 27]. These models must reconcile short-term performance metrics with long-term engagement objectives, incorporating behavioral decay functions and influence dynamics.

Taken together, the reviewed studies reveal a clear gap: integrated models that jointly address communication constraints, multi-channel campaign planning, rolling-horizon controls, and network-aware customer influence remain limited in telecommunications applications. This gap motivates the formulations and methods developed in the following chapters.

CHAPTER III

CAMPAIGN OPTIMIZATION PROBLEM WITH COMMUNICATION LIMITATIONS

3.1 Problem Definition

In this section, we formalize the campaign optimization problem encountered in outbound marketing within the telecommunications sector, under strict communication and operational constraints. We present a mathematical programming formulation that aims to maximize the dissemination of high-priority campaigns to eligible customers while ensuring full adherence to organizational communication policies and infrastructural limitations.

The telecommunications industry operates within a highly dynamic and competitive environment, offering a broad portfolio of services such as data bundles, value-added services, and promotional tariffs. In this context, marketing plays a pivotal role in maintaining customer engagement and driving revenue through targeted campaigns. These campaigns, typically executed on a daily or weekly basis, constitute structured outreach initiatives designed to offer incentives—such as bonus data, discounted rates, or exclusive services—to selected customer segments. The pre-campaign phase involves sophisticated data analytics to define the optimal target audience. Once identified, customers are contacted via various communication modalities, including but not limited to SMS, mobile push notifications, and call centers. Although such marketing efforts are often beneficial from a value-offering perspective, they risk being perceived as intrusive or annoying if the messaging frequency, channel, or timing does not align with customer preferences. As a result, telecommunication firms have instituted comprehensive communication governance policies. These include daily and weekly contact frequency caps, per-campaign and per-category limits, and channel-specific capacity thresholds. Importantly, our formulation does not impose monetary budget constraints directly; instead, we model campaign budgets by restricting the number of targetable customers. Furthermore, campaign messages are categorized into various groups based on their strategic purpose (e.g., upselling, retention, or usage stimulation) and assigned relative priority levels that reflect their organizational significance. We denote this formalized decision problem as the *Campaign Optimization Problem with Communication Limitations* (COPL), following the terminology used in prior work and chapter titles. In operational discussion, however, we use the equivalent term *communication constraints*. The core objective is to allocate communication opportunities optimally to maximize exposure for high-priority campaigns while complying with operational, regulatory, and experiential constraints. To accommodate longer planning cycles and mitigate message fatigue, we adopt a rolling horizon strategy that considers historical contact records in addition to future planning decisions.

The primary objective of this study is to optimize the marketing communication process by maximizing the number of communications for the highest-priority campaigns while upholding the aforementioned rules and guidelines. This optimization seeks to strike a balance between effective marketing outreach and customer satisfaction, ensuring that the most significant campaigns receive ample exposure while mitigating customer annoyance associated with excessive or untimely communications. To address this marketing communication challenge, we employ a mathematical model that operates on a weekly decision-making basis. To manage the frequency of customer contact over longer periods of time, we adopt a rolling horizon approach and regulate the number of messages/calls per customer over consecutive weeks.

Next, we present the notation and our mathematical model. The term "communication limit" refers to the maximum number of times a customer can be contacted, and "message" refers to any marketing contact with a customer.

Sets

C: Set of campaigns

U: Set of customers

H: Set of communication channels

D: Set of days (7 days of a week)

I: Set of campaign categories

P: Set of priority categories

In this chapter, campaign categories (I) represent the business type of a campaign (e.g., upsell, retention), while priority categories (P) represent its strategic importance level used in the objective function.

Decision Variables

$X_{cuhd} = 1$, if a message on campaign $c \in C$ will be sent to customer $u \in U$ through channel $h \in H$ on day $d \in D$, and 0 otherwise.

Parameters

$e_{cu} = 1$, if customer $u \in U$ is eligible (i.e., in the target group) for campaign $c \in C$, and 0 otherwise.

$s_{cuhd} = 1$, if customer $u \in U$ is contacted about campaign $c \in C$, through channel $h \in H$, on day $d \in D$ the previous week and, 0 otherwise.

$q_{ic} = 1$, if campaign $c \in C$ is in campaign category $i \in I$, and 0 otherwise.

r_{cp} priority value of campaign $c \in C$ in priority category $p \in P$.

b communication limit per customer for the planning horizon (one week).

k communication limit per customer for each day.

l_c communication limit per customer for campaign $c \in C$.

m_i communication limit for campaign category $i \in I$ for the planning horizon.

n_i communication limit for campaign category $i \in I$ for each day.

t_{hd} capacity for channel $h \in H$ on day $d \in D$.

$$\textbf{BIP}: \qquad \text{Max} \sum_{p \in P} \sum_{c \in C} \sum_{u \in U} \sum_{h \in H} \sum_{d \in D} X_{cuhd} r_{cp} \tag{1}$$

subject to

$$X_{cuhd} \leq e_{cu} \qquad \forall c \in C, \forall u \in U, \forall h \in H, \forall d \in D, \tag{2}$$

$$\sum_{h \in H} X_{cuhd} \leq 1 \qquad \forall c \in C, \forall u \in U, \forall d \in D, \tag{3}$$

$$\sum_{h \in H} \sum_{c \in C} X_{cuhd} \leq k \qquad \forall u \in U, \forall d \in D, \tag{4}$$

$$\sum_{h \in H} \sum_{c \in C} \sum_{d \in D} X_{cuhd} \leq b \qquad \forall u \in U, \tag{5}$$

$$\sum_{d \in D} \sum_{h \in H} X_{cuhd} \leq l_c \qquad \forall c \in C, \forall u \in U, \tag{6}$$

$$\sum_{d \in D} \sum_{h \in H} \sum_{c \in C} X_{cuhd} q_{ic} \leq m_i \qquad \forall u \in U, \forall i \in I, \tag{7}$$

$$\sum_{h \in H} \sum_{c \in C} X_{cuhd} q_{ic} \leq n_i \qquad \forall u \in U, \forall d \in D, \forall i \in I, \tag{8}$$

$$\sum_{h \in H} \sum_{c \in C} \left(\sum_{d=1}^{w} X_{cuhd} + \sum_{d=w+1}^{7} s_{cuhd} \right) \leq b \qquad \forall u \in U, \forall w \in [1,6], \tag{9}$$

$$\sum_{h \in H} \left(\sum_{d=1}^{w} X_{cuhd} + \sum_{d=w+1}^{7} s_{cuhd} \right) \leq l_c \qquad \forall c \in C, \forall u \in U, \forall w \in [1,6], \tag{10}$$

$$\sum_{h \in H} \sum_{c \in C} q_{ic} \left(\sum_{d=1}^{w} X_{cuhd} + \sum_{d=w+1}^{7} s_{cuhd} \right) \leq m_i \qquad \forall u \in U, \forall i \in I, \forall w \in [1,6], \tag{11}$$

$$\sum_{u \in U} \sum_{c \in C} X_{cuhd} \leq t_{hd} \qquad \forall d \in D, \forall h \in H, \tag{12}$$

$$X_{cuhd} \in \{0,1\} \qquad \forall c \in C, \forall u \in U, \forall h \in H, \forall d \in D. \tag{13}$$

The objective function (1) maximizes the prioritized campaign communication. Constraints (2) ensure that only eligible customers are contacted for each campaign. Constraints (3) say that a customer should not be targeted for the same campaign from different channels on the same day. Constraints (4) define an upper bound for the total number of contacts made with each customer over all campaigns in a single day. Constraints (5) and (6) limit the number of messages sent to a customer within a planning horizon (a week) without considering the campaign categories, or the rolling horizon. More specifically, constraints (5) limit the total number of times a customer can be contacted in a week, and constraints (6) define an upper-bound for the total number of communications to each customer per campaign for the whole week. Campaigns are grouped by their marketing purpose, and for each of these groups we have combined limitations on campaign categories. Constraints (7) and (8) limit the messages sent to a customer considering the campaign categories; such that constraints (7) draw a limitation on the number of communications about campaigns that fell in specific groups ignoring the rolling horizon. q_{ic} equals to 1 if campaign c is in category i. Constraints (8) ensure that the number of communications for campaign categories for each day is not exceeded. Constraints (9), (10), and (11) are similar to the constraints (5), (6), and (7), and consider the previous week when limiting the number of contacts with the customer. Constraints (9) consider the total number of times a customer can be contacted, constraints (10) define an upper-bound for the total number of communications to each customer per campaign and, constraints (11) limit the number of communications for campaign categories. Constraints (12) ensure that the capacity of each communication channel per day is not exceeded. Finally, constraints (13) determine the domain of the X_{cuhd} variables.

Given the inherent complexity of the problem, large instances are computationally expensive to solve using exact methods within a desired time frame. To overcome this limitation, we develop heuristic approaches that can achieve high-quality solutions in a reasonable amount of time. We strive to achieve a balance between computational feasibility and solution quality. This approach allows us to efficiently manage larger instances of the problem while generating solutions that are of practical relevance and high quality within acceptable time constraints.

3.2 Solution Methods

In this section, we propose three heuristic approaches to solve the campaign optimization problem. In §3.2.1, we introduce the Basic Greedy heuristic that focuses on the campaign priorities. Next, in §3.2.2, we propose the Relax+Greedy matheuristic that first utilizes the linear programming (LP) relaxation of the original BIP model by redefining the X_{cuhd} variables as real numbers between 0 and 1, and then uses the greedy approach. Finally, in §3.2.3, we define the Aggregate+Greedy matheuristic which first considers the aggregate number of eligible individuals for each campaign and then uses the greedy approach concerning the campaign priorities. Thus, we improve the basic greedy algorithm with two different initial processes.

Although decomposition-based exact methods, such as Benders decomposition, are widely used for large-scale mixed-integer programs, they are not a natural fit for the COPL. Classical Benders decomposition requires partitioning the variables into a (typically integer) master problem and a (typically continuous) subproblem whose dual can be used to generate strong feasibility and optimality cuts. In COPL, the core decisions are the binary assignment variables X_{cuhd}, and fixing any plausible subset of these decisions does not yield a tractable continuous subproblem; the remaining problem still contains a large number of binary variables and retains essentially the same combinatorial structure due to the per-customer limits and the channel-capacity coupling constraints (12). As a result, a Benders subproblem would remain integer and would not provide informative dual-based cuts, limiting the computational benefits of Benders in our setting.

3.2.1 Basic Greedy heuristic

As a first approach, we consider the basic greedy algorithm, Algorithm 1, that prioritizes campaign priority values. It starts with the campaign with the highest r_{cp} value and sets the corresponding X_{cuhd} variable to 1 for $c = 1, h = 1$ and $d = 1$. Subsequently, the algorithm examines the feasibility of the BIP model with $X_{c111} = 1$ with respect to constraints (2) to (13). If a violation is detected, the variable value is modified to 0. The campaign priority value has precedence in this algorithm, thus the outer loop begins with the randomly sorted campaign list. Other dimensions, such as customers, channels, or days are incremented in the algorithm. The algorithm stops when all X_{cuhd} values are considered.

Algorithm 1 Basic Greedy Heuristic

Input: r_{cp}, $c \in C$, $p \in P$, Constraints (2) to (13) of BIP
Output: X_{cuhd}, $c \in C$, $u \in U$, $h \in H$, $d \in D$ feasible for Constraints (2) to (13)

Sort campaigns ($c \in C$) such that ($r_{c_i p} \geq r_{c_{i+1} p}$)
for $c \leftarrow 1$ **to** C **do**
    for $d \leftarrow 1$ **to** D **do**
        for $h \leftarrow 1$ **to** H **do**
            for $u \leftarrow 1$ **to** U **do**
                $X_{cuhd} = 1$
                if any one of the constraints (2) to (13) are violated **then**
                    $X_{cuhd} = 0$
                end if
            end for
        end for
    end for
end for

3.2.2 Relax+Greedy Matheuristic

In this section, we propose the Relax+Greedy matheuristic, Algorithm 2, which uses LP-relaxation of the BIP model described in §3.1 as a first step, and then utilizes the greedy algorithm to ensure the feasibility of the solution.

In the first part, the original BIP model is relaxed by changing Constraints (13) from $X_{cuhd} \in \{0,1\}$ to $X_{cuhd} \in [0,1], \forall c \in C, \forall u \in U, \forall h \in H, \forall d \in D$ and the LP-relaxation solution is obtained. Then, variables that took an integer value in the optimal solution are fixed to their current values. We set the other variables, which did not have an integer value, equal to 0 and kept them in a separate set $NonIntegralSet$. In the second part of the algorithm, we apply the Basic Greedy heuristic. However, this time we only use the X variables from the $NonIntegralSet$. We sort the campaigns from the $NonIntegralSet$

with respect to their priorities and attempt to improve the objective value. Starting from the most important campaign, we set the corresponding X variable in $NonIntegralSet$ to 1 and assess whether any of the constraints from (2) to (13) are violated. If a violation occurs, the variable is reset to 0. This step effectively moves the LP-relaxation solution from an infeasible region to a feasible region while accommodating the possibility of some non-integer variables becoming 1. This approach strikes a balance between feasibility and optimization.

Algorithm 2 Relax+Greedy Matheuristic

Input: r_{cp}, $c \in C$, $p \in P$, Constraints (2) to (13) of BIP

Output: X_{cuhd}, $c \in C$, $u \in U$, $h \in H$, $d \in D$ feasible for constraints (2) to (13)

```
Solve the LP-relaxation to obtain X_{cuhd}, c ∈ C, u ∈ U, h ∈ H, d ∈ D
NonIntegralSet ← {}
for c ← 1 to C do
    for d ← 1 to D do
        for h ← 1 to H do
            for u ← 1 to U do
                if X_{cuhd} > 0 and X_{cuhd} < 1 then
                    X_{cuhd} = 0
                    NonIntegralSet ← NonIntegralSet ∪ {X_{cuhd}}
                end if
            end for
        end for
    end for
end for
Sort campaigns (c ∈ C) such that (r_{c_i p} ≥ r_{c_{i+1} p})
for c ← 1 to C do
    for d ← 1 to D do
        for h ← 1 to H do
            for u ← 1 to U do
                if X_{cuhd} ∈ NonIntegralSet then X_{cuhd} = 1
                    if any one of the constraints (2) to (13) are violated then
                        X_{cuhd} = 0
                    end if
                end if
            end for
        end for
    end for
end for
```

3.2.3 Aggregate+Greedy Matheuristic

In this section, we enhance the basic greedy heuristic by incorporating a new linear programming model as the first step. We acknowledge that campaigns with a higher priority may have a smaller eligible audience or campaigns with lower priority may have a larger audience. So, we may reach a higher objective value by focusing on campaigns with lower priorities since we may have a higher number of customers to send them to. We revise our initial BIP model and consider the number of eligible individuals that can be reached through each campaign, disregarding other constraints specific to each individual, i.e., how many messages an individual receives, or how often they do so. Thus, we aggregate the customers and by prioritizing campaigns based on their potential audience reach, we can tailor the communication plan to maximize customer engagement and achieve the desired marketing objective using a linear programming model.

Next, we introduce the notation for the new decision variables, additional parameters, and then present the mathematical model for the new model.

Y_{cd} total number of messages sent on campaign $c \in C$ at day $d \in D$, $\forall c \in C$, $\forall d \in D$

e_c number of eligible customers $(\sum\limits_{u \in U} e_{cu})$ for campaign $c \in C$

$$\textbf{LP}: \quad \text{Maximize} \sum_{c \in C} \sum_{d \in D} Y_{cd} \tag{14}$$

subject to

$$\sum_{d \in D} Y_{cd} \leq l_c e_c \qquad \forall c \in C, \tag{15}$$

$$\sum_{c \in C} \sum_{d \in D} q_{ic} Y_{cd} \leq \sum_{c \in C} q_{ic} e_c m_i \qquad \forall i \in I, \tag{16}$$

$$\sum_{c \in C} q_{ic} Y_{cd} \leq \sum_{c \in C} q_{ic} e_c n_i \qquad \forall d \in D, \forall i \in I, \tag{17}$$

$$\sum_{c \in C} \sum_{d \in D} Y_{cd} \leq \sum_{c \in C} e_c b, \tag{18}$$

$$\sum_{c \in C} Y_{cd} \leq \sum_{c \in C} e_c k \qquad \forall d \in D, \tag{19}$$

$$Y_{cd} \geq 0 \qquad \forall c \in C, \forall d \in D, \tag{20}$$

The objective function (14) maximizes the total number of people contacted over the planning horizon. Constraints (15) ensure that the number of communications for each campaign can not exceed the number of eligible customers for the campaign regarding the campaign limit (similar to the constraints (6) in the BIP model). Constraints (16) limit the number of communications for a campaign category (similar to the constraints (7)). Constraints (17) repeats the previous constraint for each day (similar to constraints (8)). Constraints (18) and (19) ensure that the total number of communication does not exceed weekly and daily limits (similar to the constraints (5), and (4), respectively). Finally, constraints (20) ensure that the variable Y_{cd} is a positive number. This mathematical model is used for finding an estimated number of the possible number of communications for each campaign. Then, in Algorithm 3 we use Y_{cd} as an input (in addition to campaign priority r_{cp}) to sort the campaigns and apply the basic greedy algorithm, i.e., prioritize both the campaign priority and the number of customers reached.

Algorithm 3 Aggregate+Greedy Matheuristic

Input: X, $c \in C$, $u \in U$, $h \in H$, $d \in D$, Constraints (2) to (13) of IP

Output: X_{cuhd}, $c \in C$, $u \in U$, $h \in H$, $d \in D$ feasible for constraints (2) to (13)

Solve the LP Model to obtain Y_{cd} $c \in C$ $d \in D$ (14) to (20)
for $d \leftarrow 1$ **to** D **do**
    Sort campaigns ($c \in C$) such that ($Y_{c_i d} r_{c_i p} \geq Y_{c_{i+1} d} r_{c_{i+1} p}$)
    for $c \leftarrow 1$ **to** C **do**
        for $h \leftarrow 1$ **to** H **do**
            for $u \leftarrow 1$ **to** U **do**
                $X_{cuhd} = 1$
                if any one of the constraints (2) to (13) are violated **then**
                    $X_{cuhd} = 0$
                end if
            end for
        end for
    end for
end for

3.3 Computational Study

In this section, we present the outcomes of our computational study to evaluate the performance of the proposed heuristics. All computational processes were executed on a computer running a 64-bit Windows 10 operating system, equipped with an Intel(R) Core(TM) i7-3630QM CPU and 16 GB of RAM. We implemented both the BIP model and the heuristics using the Python programming language. For optimization purposes, we utilized CPLEX 20.1 integrated with Python.

3.3.1 Test Instances

In real-world scenarios, campaigns are usually planned every week. On each day, the telecommunication company launches an average of around 10 campaigns, targeting more than 500,000 customers through 3 communication channels. There are typically three campaign categories: data, voice, and value-added services campaigns. The priorities assigned to campaigns can vary between 1 and the total number of campaigns in the system.

In accordance with the provided specifications, we have systematically generated a set of test instances tailored for this study. A succinct summary of these instances is presented in Table 1. The columns represent the number of campaigns (c) and the count of customers (u), respectively. We set the number of communication channels (h) as 3, and the planning horizon (d) as 7 days.

Our dataset encompasses a total of 45 instances: small-sized (Instances 1-15), medium-sized (Instances 16-30), and large-sized (Instances 31-45) instances. Notably, instances falling within the customer size range of Instances 39 to 45 closely mirror the challenges encountered by telecommunication companies in real-world scenarios, with high problem complexity.

For each instance, we generated the additional parameters according to the following guidelines:

- The eligibility matrix e_{cu} of size $|C| \times |U|$ was populated with random binary values $\in \{0, 1\}$ to indicate the eligibility of customers for each campaign.
- The campaign category matrix q_{ic} of size $|I| \times |C|$ was filled with random binary values $\in \{0, 1\}$ to specify the campaign categories for each campaign.
- The campaign priority array r_p of size $|P|$ was generated with random integer values ranging from 0 to 100, indicating the priority level of each category.
- The matrix r_{pc} of size $|P| \times |C|$ was created, with each campaign assigned to one of the $|P|$ priority values using random binary values $\in \{0, 1\}$. Each campaign's individual priority r_c was calculated by multiplying r_p and r_{pc}.
- We set the weekly limitation value b to 7 (representing 7 days in a week).
- The daily limitation value k was set to 3 (representing the maximum number of communications per customer per day).
- The campaign limitation array l_c of size $|C|$ was populated with randomly selected integer values from the set $\{2, 3, 4\}$, indicating the maximum number of communications for each campaign.

- The campaign category limit per week array m_i of size $|I|$ was filled with randomly selected integer values from $\{3, 4, 5\}$, representing the maximum number of communications for each campaign category per week.
- The campaign category limit per day array n_i of size $|I|$ was generated with randomly selected integer values from $\{1, 2, 3\}$, indicating the maximum number of communications for each campaign category per day.
- The channel capacity matrix of size $|H| \times |D|$ was populated with random values from the set $\{0.01|U|, 0.02|U|, 0.03|U|\}$.
- The solution from previous weeks of size $|C| \times |U| \times |H| \times |D|$ was generated for s_{cuhd} with random binary values $\in \{0, 1\}$.

The generated values for these parameters are given in Appendix A.1. These test instances allow us to evaluate the performance of the algorithms under different conditions, providing insights into the behavior of the proposed heuristics and their ability to handle real-world scenarios.

Table 1: Test instances.

Small			Medium			Large		
Instance	c	u	Instance	c	u	Instance	c	u
1	1	10	16	10	25000	31	10	100000
2	2	100	17	10	30000	32	10	150000
3	5	100	18	10	35000	33	10	200000
4	5	200	19	10	40000	34	10	250000
5	5	500	20	10	45000	35	10	300000
6	5	700	21	10	50000	36	10	350000
7	5	1000	22	10	55000	37	10	400000
8	10	1000	23	10	60000	38	10	450000
9	10	2000	24	10	65000	39	10	500000
10	10	3000	25	10	70000	40	10	550000
11	10	4000	26	10	75000	41	10	600000
12	10	5000	27	10	80000	42	10	650000
13	10	10000	28	10	85000	43	10	700000
14	10	15000	29	10	90000	44	10	750000
15	10	20000	30	10	95000	45	10	800000

3.3.2 Heuristic Performances

We can find the exact solutions to the original binary integer programming model using the CPLEX solver only for small (Instances 1 to 15) and two medium-sized (Instances 16 and 17) instances. As expected, the computational time to reach the optimal solution for the mathematical model increases as the customer size gets larger. To evaluate the performance of the proposed heuristics, we compare the results obtained with each heuristic with the optimal solution for the small-sized instances and with an upper bound for the medium- and large-sized instances. For the instances that could be solved to optimality, Table 2 shows the objective value gap (%) (calculated by $[(Z_e - Z_h)/Z_e] \times 100$, where Z_e gives the exact objective value found by CPLEX solver and Z_h gives the result for the corresponding heuristic) and the time required. CPLEX takes an average of 834.3 seconds (around 14 minutes) to solve these instances but as the instance size gets larger, it increases quickly to reach more than 1 hour for the largest small-sized instance. Thus it would not be practical for a company to use it every week for real instances. The average gaps (12.6, 0.2, and 6.6) for the heuristics are all quite small with Relax+Greedy being the smallest. However, when the running times are taken into account, we can see that the Basic Greedy and the Aggregated+Greedy are faster. So the trade-off of time and performance should be considered with respect to the original problem sizes. Note that there's an Upper Bound column in this table. This column will be of use when we explain the solutions for the medium- and large-sized instances in the next paragraph. In Figures 1 and 2 we can see that Relax+Greedy and Aggregate+Greedy both provide solutions with less than $< 15\%$ gap, but Basic-Greedy is less stable across instances, and Relax+Greedy tends to increase in computational time more quickly than others, although still doing much better (less than half on average) than the BIP.

During the execution of the BIP model using CPLEX, and the Relax+Greedy heuristic, the memory usage becomes more than 10GB with a CPU usage of around %100, while the memory usage was less than 300MB with a CPU usage of around

Table 2: Objective value gap (%) and time (sec) for instances that could be solved exactly.

Instance	Objective Value Gap (%) with the BIP				Time (sec.)			
	Basic Greedy	Relax+ Greedy	Aggregate+ Greedy	Upper bound	BIP	Basic Greedy	Relax+ Greedy	Aggregate+ Greedy
1	0.0	0.0	0.0	0	0.1	0.0	0.1	0.0
2	3.6	0.0	3.6	0	1.6	0.0	1.6	0.0
3	0.0	0.0	0.0	35	3.5	0.2	2.9	0.1
4	4.8	0.0	4.8	50	6.3	0.1	5.1	0.2
5	8.2	0.0	8.2	80	12.3	0.5	15.3	0.6
6	1.7	0.1	1.7	83	23.0	0.6	16.7	0.8
7	13.9	0.0	0.4	73	23.9	1.6	27.3	3.0
8	24.2	0.0	4.3	88	63.8	2.8	56.7	2.7
9	3.3	0.6	3.3	92	95.8	7.7	106.9	6.9
10	22.5	1.1	6.9	93	123.1	6.0	183.1	6.9
11	29.5	0.0	11.9	85	216.9	23.1	219.8	16.9
12	26.2	0.7	14.7	93	326.2	23.3	238.7	14.7
13	28.0	0.1	14.1	92	547.8	72.7	476.2	81.9
14	17.9	0.0	12.4	91	2062.3	163.7	821.6	193.6
15	13.6	0.2	12.2	89	2337.4	248.1	1174.8	190.7
16	8.2	0.0	5.1	93	3581.3	390.0	1400.9	393.6
17	8.6	0.1	8.6	93	4757.9	514.7	1667.6	597.0
Average	12.6	0.2	6.6	82	834.3	85.6	377.4	88.8

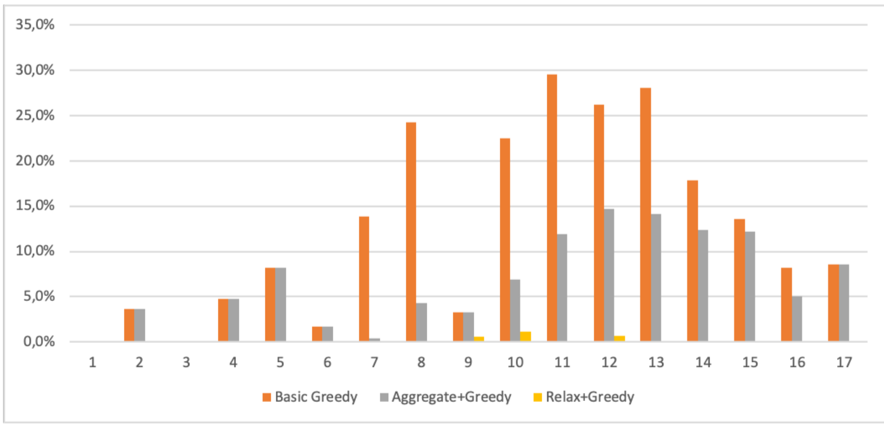

Figure 1: Objective value gap (%) for the instances that could be solved exactly.

%20 in both Basic Greedy and the Aggregate+Greedy heuristics.

For medium and large-sized instances (Instances 17-45), even loading the input data takes longer than an hour, i.e., they were not practical to be solved using CPLEX even with a time limit. For such cases, we find an upper bound to compare our heuristic results. To calculate the upper bound, we use the following method. Looking at the small-sized instances, we see that the BIP can be solved in around 5 minutes for a customer size of 5000. So, we consider customer subgroups of

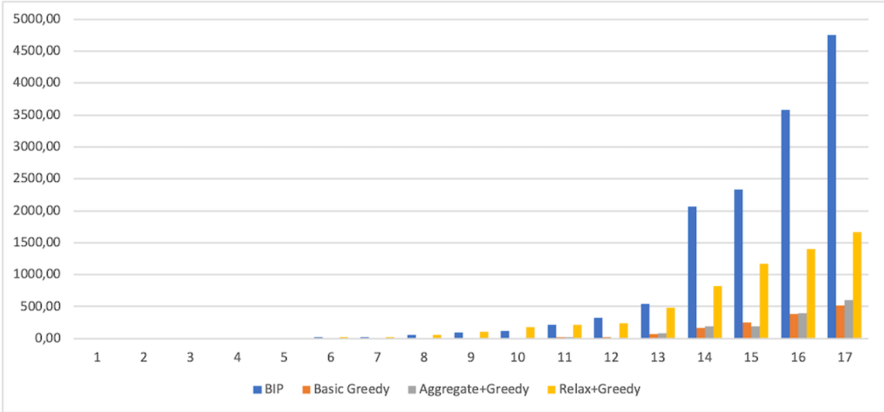

Figure 2: Time (sec) for the instances that could be solved exactly.

size at most 5000 and solve the BIP for each subgroup to optimality and then sum up the resulting objective function values. For each instance, the number of subproblems to be solved changes depending on the total customer size. We use parallel processing to solve 8 problems at the same time. Although this provides some time efficiency, this would not be a good way to solve the original problem due to the required computer power and run time. Note that dividing the customers into groups may violate the constraints on the capacity of each channel on each day. Thus, we obtain an upper bound for the original problem. In Table 3, we compare Basic Greedy and Aggregate+Greedy solutions with this upper bound. The LP-relaxation phase of the Relax+Greedy heuristic could not be run for these medium- and large-sized instances because of computer memory problems. The computational time for both algorithms has an average of just above 2 hours. For the largest customer sizes, this increases to less than 5 hours. The percentage difference is calculated by $[(UB - Z_h)/UB] \times 100$, where UB gives the upper bound value and Z_h gives the result for the corresponding heuristic. The average percentage difference is 36.5% and 29.7% for Basic Greedy and Aggregate+Greedy, respectively. Since the difference from the upper bound is large, we test the behavior of the upper bound with the small-sized instances where we can compare with the exact results. The Upper Bound column in Table 2 shows the percentage difference between the upper bound and the exact solution ($[(UB - Z_e)/UB] \times 100$). The average difference is 82%, suggesting that the upper bound is not tight. Figures 3 and 4 illustrate the percentage differences between the objective values of the Basic Greedy and Aggregate+Greedy and the upper bound and the computational time required for each instance, respectively. Overall, we can say that the Aggregate+Greedy performs better than the Basic Greedy heuristic in terms of both quality and time on average.

Table 3: Difference (%) with the UB and time (sec) for the medium- and large-sized instances.

Instance	Difference (%) with the Upper Bound		Time (sec.)	
	Basic Greedy	Aggregate+Greedy	Basic Greedy	Aggregate+Greedy
18	18.8	10.5	821.0	806.8
19	26.7	20.0	1146.8	946.4
20	41.6	35.5	1405.8	1130.5
21	56.8	50.2	1716.5	1214.8
22	47.9	46.0	2198.0	1589.9
23	35.0	26.6	2662.0	1943.5
24	46.7	38.0	2893.7	2495.7
25	43.6	35.5	3058.6	2908.5
26	40.3	15.3	3471.3	3239.0
27	45.2	38.6	3748.2	3381.9
28	40.4	35.2	4326.1	3821.7
29	38.8	34.3	4900.4	4109.8
30	54.5	47.2	5119.2	5087.0
31	20.9	16.0	6648.4	5334.0
32	42.6	36.0	7592.5	6070.9
33	23.3	20.1	7971.3	6849.5
34	25.2	18.9	8257.3	7988.5
35	25.9	24.5	9738.9	8639.4
36	44.4	38.6	11858.8	9209.3
37	44.8	33.2	12023.5	10806.4
38	38.1	29.8	12260.4	12865.2
39	20.5	19.7	13711.8	12139.4
40	17.1	17.0	14013.5	13295.5
41	34.0	21.6	14241.6	14883.0
42	29.2	23.9	14894.8	15845.3
43	29.2	22.1	15338.4	15975.7
44	37.0	30.0	16377.3	15850.0
45	52.3	48.1	17626.0	16546.5
Average	36.5	29.7	7857.9	7320.5

3.3.3 Special Case: Rich Channel Capacity

In the telecommunications sector, the availability of channel capacity is typically high, resulting in fewer restrictions on the volume of messages that can be sent across different communication channels. This subsection focuses on this scenario,

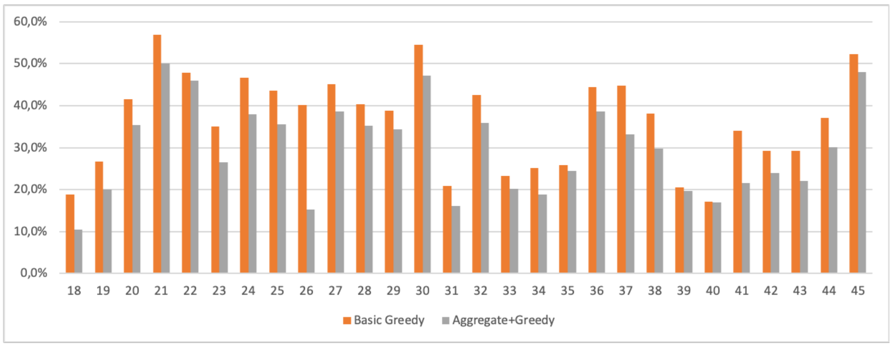

Figure 3: Difference (%) with the UB for the medium- and large-sized instances.

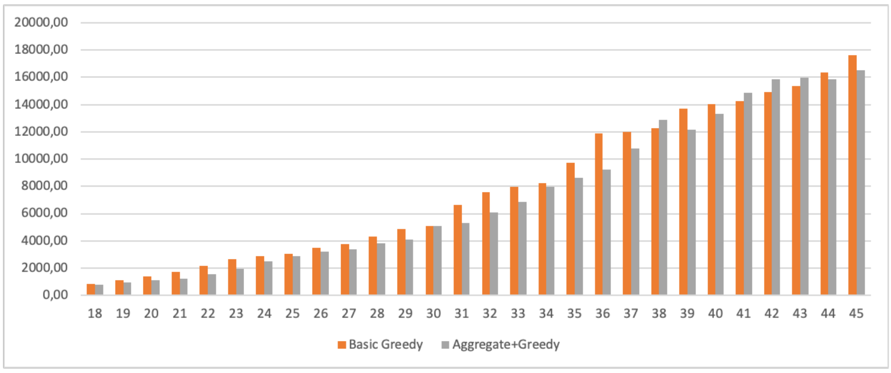

Figure 4: Time (sec) for the medium- and large-sized instances.

which we refer to as the *Rich Channel Capacity* case, as a specific instance of our broader problem.

For computational experimentation, we create a channel capacity matrix of size $|H| \times |D|$ by assigning random values from the set $\{0.5|U|, 0.6|U|, 0.7|U|\}$, where $|U|$ denotes the total number of customers, instead of the lower capacity set $\{0.01|U|, 0.02|U|, 0.03|U|\}$. We repeat our tests using 45 newly generated instances to reflect the rich channel capacity conditions.

For smaller problem instances (Instances 1–15), the exact solutions can still be obtained using the CPLEX solver. However, for the larger instances (Instances 16–45), we continue using a strategy that partitions customers into smaller groups to make the problem more manageable. Given the significantly larger channel capacity in this case, constraint (12) becomes effectively relaxed, simplifying the problem structure. Since this constraint is the only one applied at an aggregate level rather than to individual customers, its relaxation enables us to optimally solve the model using a decomposition approach. By splitting the customer base into smaller groups, we can again employ CPLEX to find optimal solutions for each subproblem.

Unlike the scenario with tighter capacity constraints, where the partitioned subproblems yield upper bounds, in this rich capacity setting, the sum of the subproblem solutions provides the exact solution to the original binary integer programming (BIP) model. Nevertheless, despite this simplification, computational time may still be considerable due to the large scale of the customer base involved in these tests.

Table 4: Objective Value Gap (%) and time (sec) for the rich channel capacity small-sized instances.

Instance	Objective Value Gap (%) with the BIP			Time (sec.)			
	Basic Greedy	Relax+ Greedy	Aggregate+ Greedy	BIP	Basic Greedy	Relax+ Greedy	Aggregate+ Greedy
1	0.0	0.0	0.0	0.1	0.0	0.1	0.0
2	0.0	0.0	0.0	1.1	0.0	1.0	0.0
3	2.4	0.0	0.0	2.3	0.1	2.1	0.1
4	0.0	0.0	0.0	4.0	0.1	3.7	0.1
5	7.5	0.0	0.0	10.4	0.4	11.1	0.4
6	13.4	0.0	0.0	18.2	0.6	15.7	0.8
7	28.4	0.0	0.0	23.1	1.0	21.6	2.0
8	5.1	0.0	0.4	44.7	1.9	38.4	2.6
9	6.7	0.1	1.2	94.8	4.8	105.7	6.1
10	13.6	0.0	0.0	128.7	5.3	145.2	7.3
11	26.1	0.1	0.2	223.2	14.3	205.1	11.7
12	12.7	0.2	0.7	244.3	17.2	246.8	17.3
13	3.4	0.0	0.3	465.3	60.1	450.1	58.0
14	13.8	0.1	1.0	1721.7	176.7	727.8	134.6
15	39.5	0.0	0.0	2033.9	233.7	1143.2	184.4
Average	11.5	0.0	0.3	334.4	34.4	207.8	28.4

Tables 4, 5, and 6 present the results for small, medium, and large instances, respectively, comparing the objective value gap and running times for each algorithm. The Relax+Greedy heuristic continues to provide superior objective values but requires longer computation times. In contrast, the Aggregate+Greedy heuristic demonstrates a balanced performance by yielding good solutions within significantly shorter timeframes. For example, across all instances, the Aggregate+Greedy method achieved an average gap of 0.6% and an average runtime of around 4245.53 seconds (approximately 1.18 hours). For the largest, most realistic instance, this approach finds a feasible solution in under 4.5 hours, offering a practical balance between computational efficiency and solution quality.

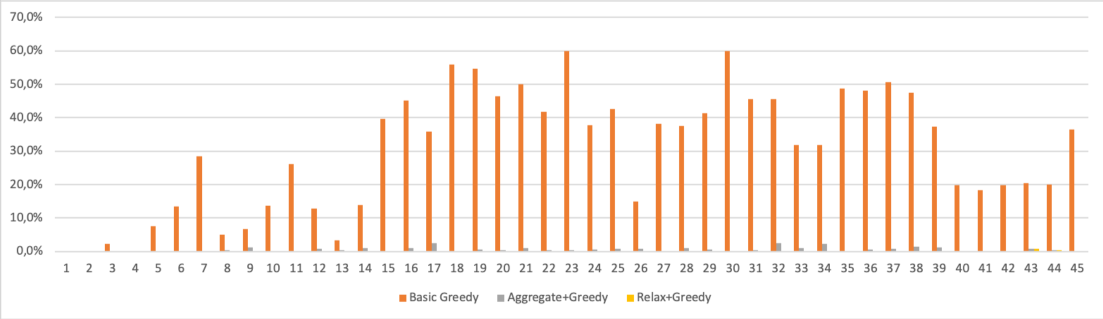

Figure 5: Objective value gap (%) for the rich channel capacity instances.

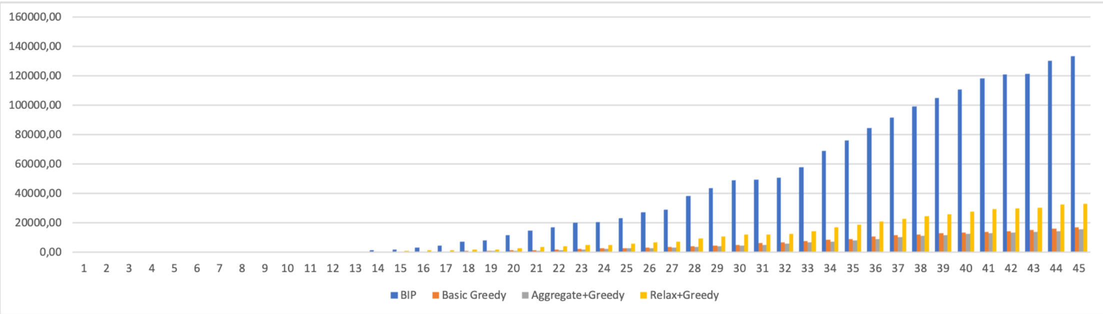

Figure 6: Time (sec) for the rich channel capacity instances.

Figure 5 and Figure 6 visually depict the differences in objective value gaps and computation times for each heuristic, underscoring the increasing computational demands of exact solutions as problem size scales up. For larger problems, where exact solutions become computationally prohibitive, the Aggregate+Greedy approach provides a viable alternative by delivering near-optimal solutions within a practical time frame.

The results presented in these tables highlight the trade-offs between solution precision and computational effort, with the Relax+Greedy approach excelling in accuracy, while the Aggregate+Greedy method remains competitive in terms of both speed and solution quality.

In the next chapter, we develop network construction methodologies and customer prioritization strategies that will form the foundation for network-aware campaign optimization. These methods enable the identification of influential customers whose targeting can amplify campaign reach through social network

Table 5: Objective value gap (%) and time (sec) for the rich channel capacity medium-sized instances.

Instance	Objective Value Gap (%) with the BIP			Time (sec.)			
	Basic Greedy	Relax+ Greedy	Aggregate+ Greedy	BIP	Basic Greedy	Relax+ Greedy	Aggregate+ Greedy
16	45.0	0.1	0.9	3348.7	370.3	1389.0	384.4
17	35.9	0.0	2.4	4728.5	563.0	1601.1	584.2
18	55.8	0.1	0.2	7368.0	931.8	1842.0	794.1
19	54.6	0.0	0.5	8126.4	1062.1	2024.9	891.7
20	46.3	0.0	0.4	11567.5	1341.2	2885.6	1014.0
21	49.9	0.0	1.0	14750.2	1617.7	3669.2	1291.6
22	41.7	0.0	0.4	16961.6	2007.9	4231.4	1564.8
23	59.9	0.2	0.4	20276.9	2408.8	5047.3	1930.8
24	37.8	0.0	0.6	20687.2	2781.8	5146.5	2315.6
25	42.7	0.0	0.8	23317.3	3053.1	5819.4	2650.8
26	15.0	0.0	0.9	27289.3	3337.2	6793.2	2914.5
27	38.1	0.0	0.1	29017.8	3722.8	7253.8	3201.2
28	37.6	0.2	0.9	38582.6	4306.8	9607.5	3558.9
29	41.3	0.0	0.5	43735.1	4848.3	10921.4	4159.9
30	59.9	0.0	0.0	48914.3	5267.0	12174.3	4685.6
Average	44.1	0.0	0.7	21244.8	2508.0	5360.4	2129.5

Table 6: Objective value gap (%) and time (sec) for the rich channel capacity large-sized instances.

Instance	Objective Value Gap (%) with the BIP			Time (sec.)			
	Basic Greedy	Relax+ Greedy	Aggregate+ Greedy	BIP	Basic Greedy	Relax+ Greedy	Aggregate+ Greedy
31	45.5	0.0	0.5	49561.8	6257.3	12356.5	5102.2
32	45.5	0.1	2.5	51034.4	7009.4	12727.0	5988.4
33	31.8	0.0	1.0	57914.9	7813.5	14468.2	6669.9
34	31.8	0.1	2.3	68891.4	8411.0	17178.9	7443.9
35	48.7	0.0	0.3	76391.5	9221.1	19035.9	8069.1
36	48.2	0.1	0.7	84451.2	11020.5	21093.5	8853.6
37	50.7	0.1	0.9	91784.4	11828.9	22945.5	10522.8
38	47.6	0.0	1.4	99314.4	12270.3	24791.4	11304.6
39	37.4	0.1	1.3	104934.7	13185.3	26109.4	11761.4
40	19.9	0.0	0.0	110981.6	13489.4	27672.6	12623.2
41	18.4	0.0	0.0	118392.3	13929.7	29455.2	12945.4
42	19.8	0.0	0.0	120934.0	14446.1	30130.4	13400.2
43	20.5	0.9	0.9	121304.8	15187.7	30234.7	13881.2
44	20.0	0.4	0.4	130217.1	16257.3	32471.3	14543.9
45	36.5	0.2	0.2	133405.4	17148.9	33238.8	15571.3
Average	34.8	0.1	0.8	94634.3	11831.8	23594.0	10578.7

effects.

CHAPTER IV

NETWORK CONSTRUCTION AND CUSTOMER RANKING METHODS

4.1 Introduction and Motivation

The campaign optimization model presented in Chapter 3 treats customers as independent entities, where the value of contacting a customer depends solely on their individual eligibility and the campaign's priority. However, in practice, customer decisions are often influenced by their social connections. When a customer learns about a promotional offer, they may share this information with friends, family, or colleagues, thereby extending the campaign's reach beyond the directly contacted individuals.

This observation motivates the need for a network-aware approach to campaign optimization. By modeling the social relationships among customers, we can identify individuals whose positions in the network make them particularly valuable for campaign dissemination. Targeting such *influential* customers allows marketers to achieve broader campaign coverage without increasing the number of direct communications—a crucial consideration given the strict communication limits imposed by regulations and company policies.

The telecommunications sector provides a unique opportunity to construct customer social networks from operational data. Call Detail Records (CDRs), which log communication events between subscribers, offer a foundation for inferring social ties. However, CDR-based networks often include weak connections (e.g., one-time business calls) that may not reflect relationships where marketing influence propagates effectively. To address this limitation, we propose an enrichment methodology that incorporates behavioral signals—specifically, tariff-change patterns—to identify stronger, more influential connections.

The contributions of this chapter are threefold:

1. We present a methodology for constructing enriched customer networks by combining CDR data with tariff-change histories, capturing both communication patterns and behavioral similarity.

2. We develop three customer prioritization strategies—CustSortA, CustSortB, and CustSortC—that leverage network structure to rank customers based on their influence potential.

3. We provide formal theoretical analysis of the proposed algorithms, including complexity bounds and coverage guarantees.

The network construction and customer ranking methods developed in this chapter form the foundation for the network-aware campaign optimization model presented in Chapter 5. By separating these methodological contributions from the optimization model itself, we emphasize their independent value and applicability to related problems in social network analysis and targeted marketing.

The remainder of this chapter is organized as follows. Section 4.2 describes the methodology for building enriched customer networks from telecommunications data. Section 4.3 presents the three customer prioritization strategies. Section 4.4 provides formal theoretical analysis of the proposed algorithms. Section 4.5 reports preliminary computational results demonstrating the effectiveness of the prioritization strategies. Finally, Section 4.6 discusses practical implications and offers guidance for parameter selection and algorithm choice.

4.2 Building Social Networks from Telecommunications Data

4.2.1 CDR-Based Networks

Call Detail Records (CDRs) are transactional logs generated by telecommunications operators whenever a communication event occurs between subscribers. A typical CDR entry includes the calling party, the called party, the timestamp, the duration, and the type of communication (voice call, SMS, etc.). These records provide a rich source of information for inferring social relationships among customers.

Formally, let U denote the set of customers. From CDR data, we can construct a social network $G = (U, E)$ where an edge $(u, v) \in E$ exists if customers u and v have communicated at least once during the observation period. The edge may be weighted by the frequency or duration of communications, though for the purposes of this study, we consider unweighted graphs.

While CDR-based networks capture communication patterns, they have several limitations for campaign optimization:

1. **Inclusion of weak ties:** CDR networks include all communication events, regardless of the underlying relationship strength. A single business call or wrong-number attempt creates an edge that may not represent a meaningful social connection.

2. **Limited predictive power:** Communication frequency alone does not necessarily indicate influence potential. Two individuals may communicate frequently without influencing each other's purchasing decisions.

3. **Network density:** Large-scale CDR networks tend to be sparse overall but may contain dense clusters of weakly connected individuals, complicating the identification of truly influential nodes.

These limitations motivate the need for network enrichment techniques that filter the CDR-based network to retain only the connections most likely to facilitate marketing influence propagation.

4.2.2 Enriching Networks with Behavioral Data

To construct a more accurate representation of influential relationships, we propose enriching the CDR-based network with behavioral signals derived from customers' tariff-change histories. The key insight is that customers who are both socially connected (as evidenced by CDR data) and behaviorally similar (as evidenced by correlated purchasing decisions) are more likely to influence each other's choices.

Specifically, we hypothesize that when two connected customers change their tariff plans within a short time window, this temporal co-occurrence suggests a stronger relationship where one customer's decision may have influenced the other. This hypothesis is grounded in the literature on social contagion and peer effects in consumer behavior [28].

4.2.2.1 Network Enrichment Parameters

The enrichment process is governed by two key parameters:

- **Temporal threshold (T):** The maximum number of days between two customers' tariff changes for them to be considered co-occurring. In our implementation, we set $T = 7$ days, reflecting the weekly nature of telecommunications billing cycles and promotional campaigns.

- **Co-occurrence threshold (CO):** The minimum number of products for which two customers must exhibit co-occurring tariff changes to retain their connection. Higher values of CO yield sparser networks with stronger ties.

4.2.2.2 Enrichment Algorithm

Algorithm 4 presents the pseudocode for the network enrichment process.

The algorithm iterates through all edges in the original CDR network. For each edge (u, v), it counts the number of products for which both customers made tariff changes within the temporal threshold T. If this count meets or exceeds the co-occurrence threshold CO, the edge is retained in the enriched network; otherwise, it is removed.

4.2.2.3 Complexity Analysis

Let $|E|$ denote the number of edges in the original network and $|P|$ denote the number of products. The time complexity of Algorithm 4 is $O(|E| \cdot |P|)$, as each

Algorithm 4 Network Enrichment Algorithm

Input: $G = (U, E)$: CDR-based network, DF: Tariff-change dataset, P: Set of products, T: Temporal threshold, CO: Co-occurrence threshold

Output: $G' = (U, E')$: Enriched network

$E' \leftarrow \emptyset$
for each edge $(u, v) \in E$ **do**
    $CoOccurrenceCount \leftarrow 0$
    for each product $p \in P$ **do**
        $t_u \leftarrow$ purchase date of product p by customer u (or ∞ if not purchased)
        $t_v \leftarrow$ purchase date of product p by customer v (or ∞ if not purchased)
        if $|t_u - t_v| \leq T$ **then**
            $CoOccurrenceCount \leftarrow CoOccurrenceCount + 1$
        end if
    end for
    if $CoOccurrenceCount \geq CO$ **then**
        $E' \leftarrow E' \cup \{(u, v)\}$
    end if
end for
return $G' = (U, E')$

edge requires examining all products. In practice, $|P|$ is typically small (single digits), making the algorithm efficient even for large networks.

4.2.3 Connection Retention Rate

To quantify the effect of network enrichment, we introduce the Connection Retention Rate (CRR), defined as the proportion of original connections retained after enrichment:

$$CRR = \frac{|E'|}{|E|} \tag{21}$$

where $|E|$ is the number of edges in the original CDR network and $|E'|$ is the number of edges in the enriched network.

A lower CRR indicates more aggressive filtering, yielding a sparser network of stronger ties. A higher CRR preserves more connections but may include weaker relationships. The choice of CO directly affects the CRR: increasing CO typically decreases CRR, as more stringent co-occurrence requirements eliminate additional edges.

4.2.4 An Illustrative Example

To illustrate the network enrichment process, consider the example depicted in Figure 7. The original CDR network (left) contains eight customers and their communication relationships. The tariff-change history is given in Table 7.

Figure 7: A sample social network modified with tariff-change data. Left: Original CDR network. Right: Enriched network after applying temporal threshold $T = 7$ days and co-occurrence threshold $CO = 1$.

Table 7: An example of tariff-change history

Day	Customer	New Tariff
1	1	F
1	4	F
1	2	H
1	3	G
6	5	H
6	6	F
9	3	F

Using temporal threshold $T = 7$ days and co-occurrence threshold $CO = 1$, we analyze each connection:

- **Customers 1–4:** Both switched to tariff F on Day 1. The time difference is 0 days (≤ 7), so the co-occurrence count is 1 ($\geq CO = 1$). **Connection retained.**
- **Customers 2–5:** Customer 2 switched to tariff H on Day 1; Customer 5 switched to tariff H on Day 6. Time difference is 5 days (≤ 7), co-occurrence count is 1. **Connection retained.**
- **Customers 1–3:** Customer 1 switched to tariff F on Day 1; Customer 3 switched to tariff G on Day 1 and later to tariff F on Day 9. For tariff

F, the time difference is 8 days (> 7). No co-occurring purchases exist. **Connection removed.**

- **Customers 1–2, 2–3, 3–4, 3–5, 5–6:** These pairs either have different tariffs or time gaps exceeding 7 days. **Connections removed.**
- **Customers 7, 8:** No tariff-change data available. **Nodes isolated and removed from the network.**

The enriched network (right side of Figure 7) retains only the connections between customers 1–4 and 2–5, representing the pairs with demonstrated behavioral similarity.

4.2.5 Analysis of Co-occurrence Thresholds and Connection Retention Rates

To understand the practical impact of the co-occurrence threshold, we applied the enrichment methodology to a real-world CDR dataset from a telecommunications operator. The dataset comprises 412,126 individuals with purchase records for 6 distinct products over 202 days. The original CDR network contained 114,902 connections.

We set the temporal threshold $T = 7$ days and varied the co-occurrence threshold CO from 1 to 5. Table 8 presents the resulting Connection Retention Rates.

Table 8: Connection Retention Rates for varying co-occurrence thresholds ($T = 7$ days)

Co-occurrence Threshold (CO)	Retention Rate (%)
1	14.890
2	0.530
3	0.040
4	0.010
5	0.003

The results reveal a steep decline in retention rates as CO increases. At $CO = 1$, approximately 14.89% of original connections are retained, indicating that most edges in the CDR network do not exhibit even a single co-occurring purchase behavior. At $CO = 2$, only 0.53% of connections remain, and for $CO \geq 3$, retention rates approach zero.

This rapid decline has important implications for campaign optimization:

1. **Sparse co-purchase events:** The low retention rates suggest that shared purchasing behaviors are rare, even among socially connected customers. Marketing influence may propagate through weak ties that do not exhibit behavioral similarity.

2. **Trade-off between precision and coverage:** A low CO value (e.g., $CO = 1$) preserves more connections, potentially capturing valuable weak ties that facilitate information spread. A high CO value yields a sparser network of stronger ties but risks excluding influential pathways.

3. **Practical recommendation:** For most campaign optimization scenarios, we recommend $CO = 1$ as a starting point, balancing network density with the requirement for behavioral evidence of influence.

4.3 Customer Prioritization Strategies

Given an enriched social network $G = (U, E)$ and campaign eligibility information, the customer prioritization problem is to rank customers in an order that maximizes the expected network influence when customers are contacted sequentially according to this ranking. In this section, we present three prioritization strategies that leverage different aspects of network structure.

4.3.1 Problem Setting

4.3.1.1 Input

- Enriched customer network $G = (U, E)$, where U is the set of customers and E is the set of edges representing influential relationships.

- Eligibility parameter $e_{cu} \in \{0, 1\}$ indicating whether customer u is eligible for campaign c.

- Adjacency parameter $a_{uv} \in \{0, 1\}$ indicating whether customers u and v are connected in G.

4.3.1.2 Output

An ordered list $(u_1, u_2, \ldots, u_{|U|})$ of customers, where customers appearing earlier in the list are prioritized for direct communication.

4.3.1.3 Objective

Maximize the network coverage achieved when a limited number of customers (determined by communication constraints) are contacted in the order specified by the prioritization.

We now present three strategies with increasing levels of sophistication.

4.3.2 Degree-Based Prioritization (CustSortA)

4.3.2.1 Motivation

The simplest approach to identifying influential customers is to prioritize those with the most connections. In network terminology, this corresponds to ranking customers by their *degree*—the number of adjacent nodes. High-degree customers have more neighbors who can learn about a campaign indirectly, making them natural candidates for direct communication.

4.3.2.2 Algorithm Description

Algorithm 5 presents the CustSortA prioritization strategy. The algorithm iteratively selects the customer with the highest degree in the remaining network, adds them to the sorted list, and removes them from the network. This process continues until all customers are ranked.

4.3.2.3 Analysis

Advantages:

- Simple to implement and understand.
- Computationally efficient.

Algorithm 5 CustSortA: Degree-Based Prioritization

Input: $G = (U, E)$: Customer network graph
Output: $SortedCustomers$: List of customers in prioritized order

$SortedCustomers \leftarrow []$
for $i \leftarrow 1$ **to** $|U|$ **do**
  $SelectedCustomer \leftarrow \arg\max_{u \in V(G)} \text{degree}(u)$ ▷ Break ties randomly
  $SortedCustomers.\text{append}(SelectedCustomer)$
  $G \leftarrow G \setminus \{SelectedCustomer\}$
end for
return $SortedCustomers$

- Effective when network influence correlates strongly with node degree.

Disadvantages:

- Does not account for overlap among high-degree customers' neighborhoods. Two high-degree customers may share many neighbors, leading to redundant coverage.

- Ignores the global network structure beyond immediate neighbors.

4.3.3 Coverage-Aware Prioritization (CustSortB)

4.3.3.1 Motivation

CustSortA's weakness lies in its disregard for coverage overlap. If the top-ranked customers share many neighbors, selecting them provides diminishing returns in terms of network coverage. CustSortB addresses this by tracking which customers have already been *covered* (i.e., reached directly or indirectly) and prioritizing customers who extend coverage to new regions of the network.

4.3.3.2 Algorithm Description

Algorithm 6 introduces a $CoveredSet$ that tracks customers who are either selected or adjacent to a selected customer. At each iteration, the algorithm preferentially selects high-degree customers from outside the $CoveredSet$, ensuring that new selections maximize marginal coverage.

4.3.3.3 Analysis

Advantages:

Algorithm 6 CustSortB: Coverage-Aware Prioritization

Input: $G = (U, E)$: Customer network graph
Output: $SortedCustomers$: List of customers in prioritized order

$SortedCustomers \leftarrow []$
$CoveredSet \leftarrow \emptyset$
for $i \leftarrow 1$ **to** $|U|$ **do**
    $MaxDegree \leftarrow \max_{u \in V(G)} \text{degree}(u)$
    $MaxDegreeCustomers \leftarrow \{u \in V(G) \mid \text{degree}(u) = MaxDegree\}$
    $UncoveredSet \leftarrow MaxDegreeCustomers \setminus CoveredSet$
    if $UncoveredSet \neq \emptyset$ **then**
        $SelectedCustomer \leftarrow$ first element of $UncoveredSet$
    else
        $SelectedCustomer \leftarrow$ first element of $MaxDegreeCustomers$
    end if
    $SortedCustomers.\text{append}(SelectedCustomer)$
    $CoveredSet \leftarrow CoveredSet \cup \{SelectedCustomer\} \cup \text{neighbors}(SelectedCustomer)$
    $G \leftarrow G \setminus \{SelectedCustomer\}$
end for
return $SortedCustomers$

- Reduces redundant coverage by tracking already-reached customers.
- Dynamically adapts to the evolving network structure as customers are selected.
- Generally achieves better coverage than CustSortA for the same number of selections.

Disadvantages:

- Still relies on degree as the primary selection criterion.
- Does not guarantee optimal coverage—greedy selection may miss globally better solutions.

4.3.4 Minimum Vertex Cover Based Prioritization (CustSortC)

4.3.4.1 Motivation

The most sophisticated approach recognizes that the customer prioritization problem is fundamentally about selecting a small set of influential customers whose direct contact enables broad one-hop reach. This perspective connects our problem to the classical Minimum Vertex Cover (MVC) problem in graph theory.

A vertex cover of a graph $G = (U, E)$ is a subset $S \subseteq U$ such that every edge in E has at least one endpoint in S. The Minimum Vertex Cover problem seeks the smallest such subset. In our context, if we focus on the subgraph induced by eligible customers, selecting a vertex cover provides a principled way to choose customers whose direct contact can cover all eligible connections, supporting one-hop propagation through the network.

4.3.4.2 Mathematical Formulation

We formulate the problem of deciding which customers should receive direct messages as a minimum vertex cover model on the subgraph induced by eligible customers. Let $e_u = 1$ if customer u is eligible for the campaign, and 0 otherwise. Let E_e denote the set of edges in this induced subgraph.

Decision Variables

$W_u = 1$, if customer $u \in U$ should receive a direct message, and 0 otherwise.

$$\begin{aligned} \text{Minimize} \quad & \sum_{u \in U} W_u && (22) \\ \text{subject to} \quad & W_u + W_v \geq 1, \quad \forall (u, v) \in E_e && (23) \\ & W_u \in \{0, 1\}, \quad \forall u \in U && (24) \end{aligned}$$

The objective (22) minimizes the number of customers contacted directly. Constraints (23) ensure that every eligible edge is covered by at least one contacted customer. Constraints (24) enforce binary decisions.

4.3.4.3 Algorithm Description

CustSortC combines the MVC solution with the coverage-aware approach of CustSortB. Customers identified by the MVC model are prioritized first, as they collectively ensure complete network coverage. Within this priority set, selection proceeds by degree and coverage status, as in CustSortB.

Algorithm 7 CustSortC: Minimum Vertex Cover Based Prioritization

Input: $G = (U, E)$: Customer network graph, e: Eligibility vector
Output: $SortedCustomers$: List of customers in prioritized order

Solve MVC model (22)–(24) to obtain W
$SortedCustomers \leftarrow []$
$CoveredSet \leftarrow \emptyset$
for $i \leftarrow 1$ **to** $|U|$ **do**
    $MaxDegree \leftarrow \max_{u \in V(G)} \text{degree}(u)$
    $MaxDegreeCustomers \leftarrow \{u \in V(G) \mid \text{degree}(u) = MaxDegree\}$
    $PriorityCustomers \leftarrow \{u \in V(G) \mid W_u = 1\}$
    $UncoveredPriority \leftarrow PriorityCustomers \setminus CoveredSet$
    if $UncoveredPriority \neq \emptyset$ **then**
        $SelectedCustomer \leftarrow$ first element of $UncoveredPriority$
    else
        $UncoveredSet \leftarrow MaxDegreeCustomers \setminus CoveredSet$
        if $UncoveredSet \neq \emptyset$ **then**
            $SelectedCustomer \leftarrow$ first element of $UncoveredSet$
        else
            $SelectedCustomer \leftarrow$ first element of $MaxDegreeCustomers$
        end if
    end if
    $SortedCustomers.\text{append}(SelectedCustomer)$
    $CoveredSet \leftarrow CoveredSet \cup \{SelectedCustomer\} \cup \text{neighbors}(SelectedCustomer)$
    $G \leftarrow G \setminus \{SelectedCustomer\}$
end for
return $SortedCustomers$

4.3.4.4 Key Property

Unlike CustSortA and CustSortB, the customer ordering produced by CustSortC varies across campaigns because the MVC solution W depends on the campaign-specific eligibility constraints. This campaign-dependent prioritization allows CustSortC to adapt to different target populations.

4.4 Theoretical Analysis

In this section, we provide formal analysis of the proposed customer prioritization algorithms, including time complexity bounds, coverage guarantees, and approximation properties.

4.4.1 Time Complexity Analysis

Theorem 4.1 (Complexity of CustSortA)**.** *Algorithm 5 (CustSortA) runs in* $O(|U|^2)$ *time, or* $O(|U|\log|U|+|E|)$ *time with appropriate data structures.*

Proof. The algorithm performs $|U|$ iterations, each requiring identification of the maximum-degree vertex. A naive implementation examines all remaining vertices, yielding $O(|U|)$ per iteration and $O(|U|^2)$ overall.

Using a max-heap to maintain vertex degrees, we can identify the maximum in $O(\log|U|)$ time. Removing a vertex requires updating the degrees of its neighbors, with total updates across all iterations bounded by $O(|E|)$. Each degree update involves a heap operation costing $O(\log|U|)$, but the total number of updates is $O(|E|)$. Thus, the heap-based implementation achieves $O(|U|\log|U|+|E|\log|U|)=O((|U|+|E|)\log|U|)$ time.

For sparse graphs where $|E|=O(|U|)$, this simplifies to $O(|U|\log|U|)$. □

Theorem 4.2 (Complexity of CustSortB)**.** *Algorithm 6 (CustSortB) runs in* $O(|U|^2+|U|\cdot|E|)$ *time, or* $O(|U|\log|U|+|E|)$ *time with appropriate data structures and amortized analysis.*

Proof. Beyond the operations in CustSortA, CustSortB maintains a *CoveredSet* that grows as customers are selected. At each iteration, we must:

1. Identify max-degree customers: $O(|U|)$ naive, $O(\log|U|)$ with heap.

2. Check membership in *CoveredSet*: $O(1)$ with hash set.

3. Update *CoveredSet* with new neighbors: $O(\text{degree}(u))$ per selection.

The total time for updating *CoveredSet* across all iterations is $O(\sum_{u\in U}\text{degree}(u))=O(|E|)$.

With heap-based degree maintenance and hash-based set operations, the overall complexity is $O(|U|\log|U|+|E|)$. □

Theorem 4.3 (Complexity of CustSortC)**.** *Algorithm 7 (CustSortC) requires solving the Minimum Vertex Cover problem, which is NP-hard in general. Given an MVC solution, the remaining operations run in* $O(|U| \log |U| + |E|)$ *time.*

Proof. The MVC problem is a well-known NP-complete problem [29]. In the worst case, finding an optimal MVC requires exponential time in $|U|$.

However, in practice:

1. Our enriched networks are typically sparse (low edge density), making integer programming solvers effective.
2. Polynomial-time 2-approximation algorithms exist (see Theorem 4.5).
3. For campaign-specific subproblems, the effective network size may be substantially smaller than $|U|$.

Once W is obtained, the prioritization loop is identical to CustSortB with an additional constant-time check of W_u values, preserving the $O(|U| \log |U| + |E|)$ bound. □

4.4.2 Coverage Guarantees

Proposition 4.4 (Network Coverage of MVC Solution)**.** *If* $S = \{u \in U \mid W_u = 1\}$ *is a vertex cover of the subgraph induced by eligible customers, then contacting all customers in* S *ensures that every eligible customer either receives a direct message or has at least one neighbor who receives a direct message.*

Proof. By the definition of vertex cover, every edge (u, v) in the graph has at least one endpoint in S. Consider any eligible customer u:

Case 1: $u \in S$. Then u receives a direct message.

Case 2: $u \notin S$. If u has no edges, constraints (23) require $W_u \geq e_u = 1$, contradicting $u \notin S$. Thus u has at least one neighbor v. The edge (u, v) must have an endpoint in S; since $u \notin S$, we have $v \in S$. Therefore, u's neighbor v receives a direct message.

In both cases, u learns about the campaign (directly or indirectly). □

Theorem 4.5 (Approximation Bound for MVC)**.** *There exists a polynomial-time algorithm that finds a vertex cover of size at most $2 \cdot OPT$, where OPT is the size of the minimum vertex cover.*

Proof. The standard matching-based algorithm achieves a 2-approximation. The algorithm repeatedly selects an arbitrary uncovered edge (u, v), adds both u and v to the cover, and removes all edges incident to u or v. This continues until no edges remain.

Each selected edge contributes exactly two vertices to the cover. Moreover, every selected edge must be covered by at least one vertex in an optimal solution, so the number of selected edges is at most OPT. Therefore, the algorithm produces a vertex cover of size at most $2 \cdot OPT$. □

Corollary 4.6 (Approximation Guarantee for CustSortC)**.** *When using a 2-approximation algorithm for the MVC component, CustSortC identifies a priority set W such that $|W| \leq 2 \cdot OPT_{MVC}$, where OPT_{MVC} is the size of the minimum vertex cover.*

4.4.3 Comparative Analysis

Proposition 4.7 (Ordering of Coverage)**.** *For any graph G and selection budget k, let $Cov_A(k)$, $Cov_B(k)$, and $Cov_C(k)$ denote the number of customers covered (directly or indirectly) by selecting the top k customers according to CustSortA, CustSortB, and CustSortC, respectively. Then:*

1. *$Cov_B(k) \geq Cov_A(k)$ in general, with equality when high-degree nodes have disjoint neighborhoods.*

2. *$Cov_C(k) \geq Cov_B(k)$ when k equals the MVC size, with CustSortC achieving complete coverage.*

Proof. **Part 1:** CustSortB explicitly avoids selecting customers whose neighborhoods overlap with already-selected customers. This prevents "wasted" coverage where a new selection reaches customers already covered. CustSortA makes no such distinction, potentially selecting high-degree customers with substantial neighborhood overlap. Equality occurs when all high-degree customers have disjoint neighborhoods, making the overlap-avoidance mechanism irrelevant.

Part 2: By Proposition 4.4, selecting all customers in a vertex cover guarantees complete edge coverage; under the one-hop propagation model, this implies that every eligible customer either is contacted directly or has a contacted neighbor. CustSortC prioritizes customers in the MVC set, so when k equals the MVC size, all MVC customers are selected (since they appear first in the prioritization), achieving complete coverage. CustSortB does not have this guarantee, as its greedy coverage-aware approach may not identify a minimum vertex cover. □

Table 9 summarizes the key properties of the three algorithms.

Table 9: Comparison of customer prioritization algorithms

Property	CustSortA	CustSortB	CustSortC
Time Complexity	$O(\|U\| \log \|U\| + \|E\|)$	$O(\|U\| \log \|U\| + \|E\|)$	NP-hard + $O(\|U\| \log \|U\| + \|E\|)$
Overlap Avoidance	No	Yes	Yes
Guaranteed Coverage	No	No	Yes (at MVC size)
Campaign-Specific	No	No	Yes

4.5 Preliminary Computational Study

Before applying the customer prioritization strategies to full-scale campaign optimization instances, we evaluate their effectiveness on a controlled test case where the optimal solution can be computed exactly.

4.5.1 Miniworld Test Instance

We construct a "miniworld" test instance with the following characteristics:

- **Customers:** 50 customers (nodes), all eligible for the campaign.
- **Network:** 19 edges representing the enriched social network.
- **Optimization problem:** Select at most 15 customers for direct communication to maximize the total number of customers reached (directly or indirectly).

- **Constraints:** Single campaign, single channel, single day—isolating the effect of customer prioritization.

Figure 8 shows the network structure. The edge list is: (3, 6), (4, 7), (4, 28), (5, 8), (6, 45), (7, 8), (7, 22), (9, 11), (9, 20), (9, 43), (10, 29), (10, 35), (10, 48), (11, 19), (13, 20), (15, 17), (21, 28), (22, 38), (31, 45).

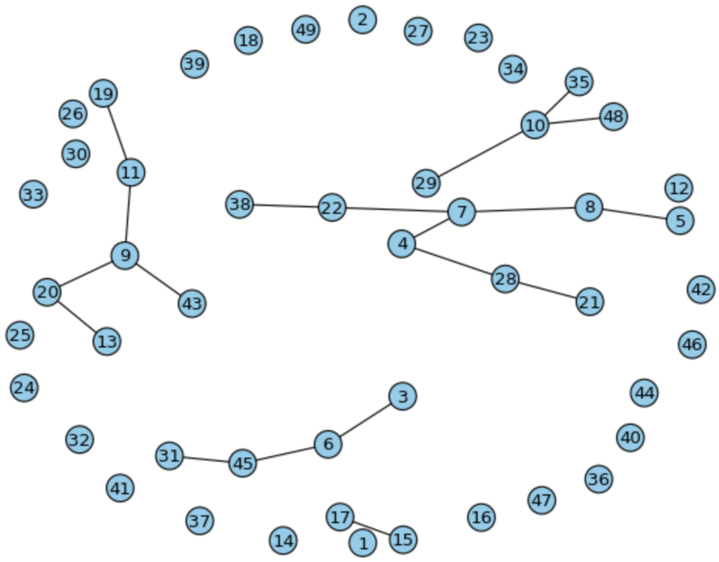

Figure 8: Miniworld sales-enriched customer network (50 customers, 19 edges)

4.5.2 Results

We solved the instance optimally using CPLEX and compared the results with the three prioritization strategies. Table 10 summarizes the findings.

Table 10: Results for the miniworld test instance (selection budget = 15)

Method	Objective Value	Gap (%)	First 15 Selected Customers
Optimal (CPLEX)	348	—	(varies with solver)
CustSortA	324	6.9	7, 9, 10, 6, 28, 5, 11, 13, 15, 22, 31, 0, 1, 2, 3
CustSortB	336	3.4	7, 9, 10, 6, 28, 5, 13, 15, 19, 31, 38, 0, 1, 2, 12
CustSortC	348	0.0	10, 6, 8, 20, 22, 28, 15, 19, 43, 45, 0, 1, 2, 12, 14

4.5.3 Analysis

1. **CustSortA (Gap: 6.9%):** The degree-based approach selects high-degree nodes (7, 9, 10) first but fails to account for neighborhood overlap. Nodes 7 and 9, for instance, share neighbors, leading to redundant coverage.

2. **CustSortB (Gap: 3.4%):** The coverage-aware approach improves upon CustSortA by avoiding the selection of nodes whose neighborhoods are already covered. However, it still relies on degree as the primary criterion and does not guarantee optimal coverage.

3. **CustSortC (Gap: 0.0%):** The MVC-based approach achieves the optimal objective value. By first solving the minimum vertex cover problem, CustSortC identifies the set of nodes that collectively cover all edges, ensuring maximum network reach with the given selection budget.

These results validate the theoretical advantages of CustSortC and demonstrate that the additional computational cost of solving the MVC problem is justified by improved solution quality.

4.6 Discussion and Practical Implications

4.6.1 Threshold Selection Guidelines

Based on the analysis presented in Section 4.2.5, we offer the following guidelines for selecting the enrichment parameters:

Table 11: Guidelines for threshold selection

Scenario	Recommended CO	Rationale
Broad reach campaigns	1	Preserves more connections, maximizing potential reach through weak ties.
Targeted influence campaigns	2–3	Focuses on stronger behavioral relationships, identifying core influencers.
Premium/loyalty campaigns	3+	Emphasizes only the strongest demonstrated correlations, targeting tight-knit groups.

The temporal threshold $T = 7$ days is recommended as a default, aligning with weekly billing cycles and campaign planning horizons common in telecommunications.

4.6.2 Algorithm Selection Criteria

The choice among CustSortA, CustSortB, and CustSortC depends on several factors:

Table 12: Algorithm selection decision matrix

Factor	CustSortA	CustSortB	CustSortC
Computational budget	Low	Low	Medium–High
Solution quality priority	Low	Medium	High
Network density	Any	Any	Sparse preferred
Campaign frequency	High (daily)	Medium (weekly)	Low (strategic)

For operational campaigns requiring rapid turnaround, CustSortA or CustSortB provide efficient solutions with acceptable quality. For strategic campaigns where maximizing network reach justifies additional computation, CustSortC is recommended.

4.6.3 Integration with Campaign Optimization

The customer prioritization strategies developed in this chapter serve as inputs to the network-aware campaign optimization model presented in Chapter 5. Specifically:

1. The enriched network G' constructed in Section 4.2 defines the adjacency parameters a_{uv} used in the COP-LN model.

2. The customer rankings produced by CustSortA, CustSortB, or CustSortC determine the order in which customers are considered during the greedy assignment phase of the solution heuristics.

3. The MVC solution from CustSortC provides initialization for the HANSA metaheuristic and guides its neighborhood search operators.

By separating network construction and customer prioritization from the optimization model itself, this chapter establishes a modular framework where improvements to either component can be developed and evaluated independently.

4.6.4 Limitations and Future Directions

Several limitations of the current approach suggest directions for future research:

1. **Static network assumption:** The enriched network is constructed once and assumed fixed during campaign planning. In reality, social relationships evolve over time. Dynamic network models that update with new CDR and purchase data could improve accuracy.

2. **Homogeneous influence:** Our model assumes uniform influence strength across all edges. Incorporating edge weights based on communication frequency or recency could capture heterogeneous influence effects.

3. **Single-hop propagation:** We consider only direct neighbors in the influence model. Multi-hop propagation models, while computationally more expensive, may better capture cascade effects in dense networks.

4. **MVC scalability:** Although our networks are typically sparse, the NP-hardness of MVC may limit CustSortC's applicability to very large instances. Investigating more scalable approximation algorithms is a promising direction.

Despite these limitations, the framework presented in this chapter provides a principled and empirically effective approach to leveraging network structure for campaign optimization in telecommunications.

CHAPTER V

CAMPAIGN OPTIMIZATION PROBLEM WITH COMMUNICATION LIMITATIONS AND NETWORK EFFECTS

5.1 Problem Definition

In this chapter, we extend the Campaign Optimization Problem with Communication Limitations (COPL) by incorporating network effects, resulting in the Campaign Optimization Problem with Communication Limitations and Network Effect (COP-LN). This enhanced model not only maximizes the number of prioritized campaigns delivered to customers but also leverages social interactions among customers to amplify campaign reach through indirect communications. The objective is to optimize both direct and indirect campaign communications while adhering to the communication constraints mandated by the company.

In the telecommunication industry, effective marketing strategies are crucial for promoting a diverse range of offerings, including tariffs, products, and services. While direct marketing communications (e.g., SMS, email, push notifications) target specific customer segments, social interactions among customers present an opportunity to enhance campaign effectiveness through network effects. In this thesis, this mechanism is modeled as a one-hop network externality: the adoption or awareness of a product or service by one customer can influence directly connected neighbors in the social network.

However, integrating network effects into campaign optimization introduces additional complexity. Marketers must balance the benefits of indirect communications against communication constraints to prevent customer fatigue and ensure compliance with predefined communication rules. Thus, the COP-LN model aims to harness network effects to maximize campaign influence while maintaining customer satisfaction and adhering to operational constraints.

The primary objective of this study is to optimize the marketing communication process by maximizing the influence of high-priority campaigns through both direct messaging and network effects. This involves:

- **Maximizing Direct Communications:** Ensuring that prioritized campaigns reach as many eligible customers as possible within communication constraints.
- **Leveraging Network Effects:** Facilitating one-hop propagation of campaign awareness through the customer network to amplify reach without additional direct communications.
- **Adhering to Communication Constraints:** Maintaining compliance with daily and weekly communication limits per customer, per campaign, and per communication channel to prevent over-contacting and ensure customer satisfaction.

To formalize the COP-LN, we introduce additional sets, variables, and constraints to capture network effects alongside the existing communication constraints.

Decision Variables

$Z_{cu} = 1$, if customer $u \in U$ learns about campaign $c \in C$, either directly from the marketer (indicated by X_{cuhd}) or indirectly through a neighboring customer, and 0 otherwise.

Parameters

$a_{uv} = 1$, if customers $u \in U$ and $v \in U$ are adjacent (i.e., connected in the social network), and 0 otherwise.

$$\textbf{COP-LN}: \qquad \text{Max} \sum_{p \in \mathcal{P}} \sum_{c \in \mathcal{C}} \sum_{u \in \mathcal{U}} Z_{cu} r_{cp} \tag{25}$$

subject to

$$Z_{cu} \leq \sum_{h \in \mathcal{H}} \sum_{d \in \mathcal{D}} (X_{cuhd} + \sum_{v \in \mathcal{U}} X_{cvhd} a_{uv}) \qquad \forall c \in \mathcal{C}, \forall u \in \mathcal{U}, \tag{26}$$

$$Z_{cu} \in \{0, 1\} \qquad \forall c \in C, \forall u \in U, . \tag{27}$$

The objective function (25) seeks to maximize the total weighted influence of campaigns by summing the priority values of campaigns delivered to customers, considering both direct and indirect communications. Network coverage constraints (26) define that a customer learns about a campaign either directly through a message from the marketer or indirectly through messages received by their neighbors in the social network. Here, Z_{cu} is set to 1 if at least one such path exists. Integrity constraints (27) determine the domain of the Z_{cu} variables.

5.2 Solution Methods

In this section, we present solution approaches for the COP-LN that leverage the network construction methodologies and customer prioritization strategies developed in Chapter 4. The enriched customer network, constructed by combining CDR data with tariff-change histories as described in §4.2, provides the adjacency parameters a_{uv} used in the COP-LN model. The customer prioritization strategies CustSortA, CustSortB, and CustSortC, detailed in §4.3, determine the order in which customers are considered during the optimization process.

Decomposition-based exact approaches, such as Benders decomposition, are not well suited for the COP-LN. In addition to inheriting the fully binary assignment structure of COPL (with $X_{cuhd} \in \{0, 1\}$), COP-LN introduces network-coverage variables Z_{cu} and the network-coverage constraints (26), which couple decisions across neighboring customers through the adjacency parameters a_{uv}. This cross-customer coupling breaks the separability that Benders decomposition typically exploits, and fixing a subset of variables does not yield a tractable continuous subproblem whose dual can generate effective cuts. Consequently, any Benders subproblem would remain combinatorial and would likely produce weak or highly problem-specific cuts, motivating our focus on constructive heuristics and metaheuristics.

We begin with an Aggregate+Greedy matheuristic in §5.2.1, which combines campaign aggregation with network-aware customer ranking. The greedy solution then serves as the initial point for the HANSA (Hybrid Adaptive Network-Aware Simulated Annealing) metaheuristic, described in §5.2.1.4.

5.2.1 Proposed Heuristics

In this section, we present our solution approaches for the campaign optimization problem. We begin with an aggregate greedy heuristic: (i) the first step involves determining the total number of campaign messages to be sent, without yet specifying the recipients. This step aims to maximize overall message allocation and coverage. (ii) In the second step, we apply three distinct methods to prioritize the recipients. Then, the outcome of the aggregate greedy algorithm, utilizing one of these methods, provides the initial solution for the simulated annealing algorithm. The following sections provide further details on each component.

5.2.1.1 Aggregate+Greedy Matheuristic to Solve the COP-LN

Our solution approaches begin with an Aggregate+Greedy matheuristic. The pseudocode steps are outlined in Algorithm 8. First, we solve a linear programming (LP) model that approximates the COP-LN by aggregating key campaign attributes, such as customers and channels. This solution prioritizes campaigns based on their aggregated value, generating a ranked list according to their importance. In Step 1 of Algorithm 8, we use the following mathematical model to estimate the number of possible messages for each campaign and derive the corresponding ranking.

As a next step, to include the network effects, we should find the customers whose network precedence is greater. We use three different methods to sort the customers, as will be explained in §5.2.1.2.

Once we have ranked both the campaigns and customers, we apply a naive greedy algorithm to assign messages to customers and channels, considering both campaign relevance and potential customer impact.

5.2.1.2 Customer Prioritization Strategies

The customer prioritization strategies used in the COP-LN are the CustSortA, CustSortB, and CustSortC algorithms developed in Chapter 4. To briefly summarize:

- **CustSortA** (Algorithm 5): Ranks customers by their network degree, selecting high-degree nodes first.
- **CustSortB** (Algorithm 6): Extends CustSortA by tracking covered customers to reduce neighborhood overlap.
- **CustSortC** (Algorithm 7): Uses a minimum vertex cover formulation to identify the most influential customers for complete network coverage.

As demonstrated in the preliminary computational study (§4.5), CustSortC consistently achieves the best solution quality by leveraging the theoretical guarantees of minimum vertex cover. The sorted customer lists produced by these algorithms determine the order in which customers are considered during the greedy assignment phase.

5.2.1.3 Greedy Heuristic

Once both campaigns and customers are ranked, a naive greedy algorithm is applied to assign messages to customers, channels, and days, considering both campaign relevance and potential customer impact. For each day d, all campaigns c are sorted in descending order of their prioritization scores, computed as $(Y_{cd} \times r_{cp})$.

Campaigns with higher scores are thus given precedence in the assignment process. For each campaign and channel, customers are processed according to their prioritization order, and campaign c is assigned to customer u on day d through channel h by setting $X_{cuhd} = 1$. After each assignment, all constraints of the COP-LN are checked, including channel restrictions, customer eligibility, and daily communication limits. If any constraint is violated, the assignment is reverted by setting $X_{cuhd} = 0$, ensuring that the resulting campaign plan remains feasible and adheres to all operational constraints. Integrating all three components, Algorithm 8 presents the pseudocode for the Aggregate+Greedy matheuristic.

Algorithm 8 Aggregate+Greedy Matheuristic

Input: $c \in C$, $u \in U$, $h \in H$, $d \in D$, constraints (2) to (13) of COP-LN
Output: X_{cuhd}, $c \in C$, $u \in U$, $h \in H$, $d \in D$ feasible for constraints (2) to (13)

Solve the LP Model to obtain Y_{cd} $c \in C$ $d \in D$ (14) to (20)
for $d \leftarrow D$ **do**
    Sort campaigns ($c \in C$) such that ($Y_{c_i d} r_{c_i p} \geq Y_{c_{i+1} d} r_{c_{i+1} p}$)
    for $c \leftarrow C$ **do**
        for $h \leftarrow H$ **do**
            for $u \leftarrow$ CustSortA/B/C(U) **do**
                $X_{cuhd} = 1$
                if any one of the constraints (2) to (13) are violated **then**
                    $X_{cuhd} = 0$
                end if
            end for
        end for
    end for
end for

5.2.1.4 Hybrid Adaptive Network-Aware Simulated Annealing (HANSA) for the COP-LN

HANSA (Hybrid Adaptive Network-Aware Simulated Annealing) is a framework designed to exploit network effects in campaign optimization. It uses the customer prioritization from CustSort-C, which has proven to be the most effective sorting strategy (as demonstrated in §5.3.2), as the initial solution. The CustSort-C prioritization identifies key influencers through a minimum cover set formulation and balances network coverage, providing a high-quality starting point for the simulated annealing search. Algorithm 9 and Figure 9 summarize the HANSA procedure, providing both a stepwise description and a flowchart illustrating the integration of CustSort-C, adaptive search, and network-aware optimization. HANSA extends classical simulated annealing through adaptive mechanisms, specialized neighborhood structures, and exact optimization components to efficiently explore the COP-LN solution space.

Algorithm 9 HANSA: Hybrid Adaptive Network-Aware Simulated Annealing

Input: $c \in C$, $u \in U$, $h \in H$, $d \in D$, Constraints (2) to (13) of COP-LN
Output: X_{cuhd}, $c \in C$, $u \in U$, $h \in H$, $d \in D$ feasible for constraints (2) to (13) of COP-LN

Initialize with the solution obtained using CustSort-C (Algorithm 8)
$S \leftarrow$ Initial solution
$S_{best} \leftarrow S$
$T \leftarrow$ Calculate dynamic initial temperature based on instance characteristics
Initialize neighborhood performance statistics NS_{perf}
Initialize tabu list $TL \leftarrow \emptyset$
Initialize stagnation counter $SC \leftarrow 0$
while $T > T_{\text{final}}$ **and** not reached max iterations **do**
  for $i \leftarrow 1$ **to** $N_{\text{iterations}}$ **do**
    Select neighborhood structure N_k based on NS_{perf}
    Generate candidate neighbor solution S' using N_k
    if $S' \in TL$ **then**
      Continue
    end if
    Apply problem-specific local improvement to S'
    Solve network influence subproblem for S' using CPLEX
    Calculate the objective function value $f(S')$
    $\Delta f \leftarrow f(S') - f(S)$
    Calculate adaptive acceptance probability $p = e^{-\frac{\Delta f}{T \cdot \alpha(i)}}$
    Generate a random number r in $[0, 1]$
    if $\Delta f < 0$ **or** $r < p$ **then**
      $S \leftarrow S'$
      Add S' to TL (with limited tenure)
      Update NS_{perf} for neighborhood N_k
      if $f(S) < f(S_{best})$ **then**
        $S_{best} \leftarrow S$
        $SC \leftarrow 0$
      else
        $SC \leftarrow SC + 1$
      end if
    else
      $SC \leftarrow SC + 1$
    end if
    if $SC > \eta_{stag}$ **then**
      Apply diversification mechanism
      if $SC > \eta_{severe}$ **then**
        Apply reheating: $T \leftarrow T \cdot r_{heat}$
        $SC \leftarrow 0$
      end if
    end if
  end for
  Adaptively adjust cooling rate based on search progress
  $T \leftarrow T \cdot \lambda(T, f(S_{best}), SC)$
end while
return S_{best}

The performance and convergence behavior of HANSA are influenced by several adaptive mechanisms and carefully tuned parameters. The following describes

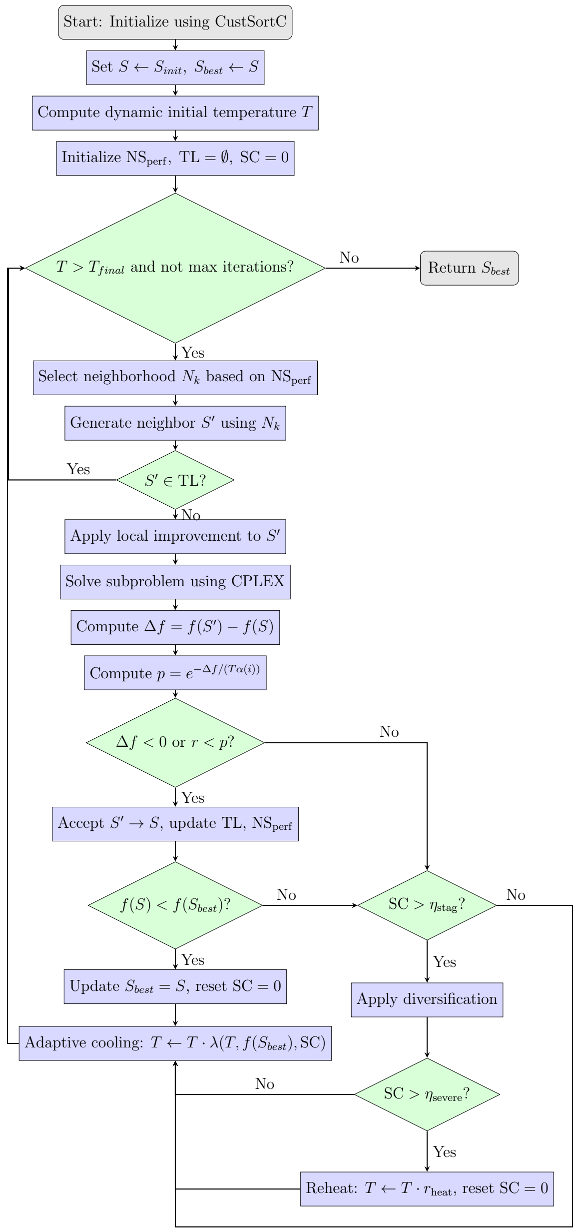

Figure 9: Flowchart for the HANSA Algorithm

the key components that distinguish HANSA from classical simulated annealing approaches.

Dynamic Initial Temperature: Instead of using a fixed starting temperature, the initial temperature is dynamically determined based on instance-specific characteristics as follows:

$$T_{\text{initial}} = \delta \cdot \sigma(f) \tag{28}$$

where $\sigma(f)$ denotes the standard deviation of objective values obtained from random perturbations of the initial solution, and $\delta = 1.5$ is a scaling parameter. This value was determined through preliminary tuning to ensure the initial temperature is sufficiently high to allow exploration but low enough to prevent random walk behavior. This approach ensures that the temperature reflects the complexity of the objective landscape for each problem instance.

Adaptive Cooling Schedule: A non-monotonic cooling schedule is employed to dynamically adjust the cooling rate according to the search progress and stagnation counter:

$$\lambda(T, f, SC) = \begin{cases} 0.98, & \text{if improvement occurs in the last } \eta_{\text{recent}} \text{ iterations} \\ 0.95, & \text{if } SC < \eta_{\text{stag}}/2 \\ 0.90, & \text{if } SC \geq \eta_{\text{stag}}/2 \end{cases} \tag{29}$$

This mechanism enables faster cooling during periods of limited improvement and slower cooling when the algorithm explores promising regions of the solution space, where $\eta_{\text{recent}} = 10$ and $\eta_{\text{stag}} = 50$.

Specialized Neighborhood Structures: Three complementary neighborhood operators are designed to effectively explore the solution space while preserving the underlying network characteristics:

- N_1 *(Customer-Pair Campaign Swap):* Exchanges campaign assignments between customers while maintaining the network's influence relationships.
- N_2 *(Channel-Day Reallocation):* Adjusts the communication channel and timing of campaign messages without modifying customer assignments.
- N_3 *(Minimum Vertex Cover Perturbation):* Introduces controlled modifications to the influential customers identified by the CustSort-C algorithm to diversify the search.

In our computational experiments, the adaptive cooling mechanism and the Vertex Cover–based neighborhood (N_3) were found to be particularly useful for enhancing solution quality.

Reactive Tabu Mechanism: A short-term memory structure is incorporated to prevent cycling by temporarily prohibiting the revisiting of recently explored solutions. The tabu duration is dynamically adjusted based on search progress:

$$\text{Tenure} = \text{base\_tenure} + \lfloor SC/10 \rfloor \tag{30}$$

where base_tenure = 7 iterations. This adaptive approach increases the prohibition duration as stagnation persists, forcing more aggressive diversification.

Matheuristic Integration: An exact optimization step using CPLEX is embedded within the heuristic framework for the critical customer sorting subproblem as follows:

1. The CustSort-C algorithm generates an initial high-quality customer prioritization.
2. CPLEX validates network influence constraints and ensures solution feasibility.
3. The solution is refined when necessary to strengthen network propagation performance.

This hybrid integration balances computational efficiency and solution accuracy, allowing the framework to handle the scalability and complexity of large-scale campaign optimization problems with network effects.

Reheating and Diversification: When search stagnation is detected ($SC > \eta_{stag}$), strategic diversification is applied through:

1. Random reassignment of a subset of campaign messages across different channels and days.
2. Preservation of the core customer assignments determined by CustSort-C.

For severe stagnation ($SC > \eta_{severe} = 100$), a reheating mechanism is activated, increasing the temperature by a factor of $r_{heat} = 1.5$ to escape local optima and restore search diversification.

Learning Components: During the search, the algorithm continuously monitors the performance of each neighborhood operator. The performance score of operator k is computed as:

$$NS_{\text{perf}}(k) = \frac{\alpha \cdot \text{ImprovementCount}(k) + \beta \cdot \text{ImprovementMagnitude}(k)}{\text{UsageCount}(k)} \tag{31}$$

This adaptive scoring mechanism enables dynamic selection of neighborhood operators, favoring those that contribute more effectively to solution improvement, where $\alpha = 0.7$ and $\beta = 0.3$ are weighting factors. These parameters were selected empirically to prioritize operators that consistently produce improvements (α) while also valuing the magnitude of those improvements (β). This scoring function is inspired by reinforcement learning principles, where actions (in this case, neighborhood operators) are evaluated based on both the frequency and magnitude of their contributions to solution improvement.

Early Stopping Criterion: To improve computational efficiency without compromising solution quality, an early stopping criterion is incorporated based on the observed convergence pattern. The algorithm terminates when no improvement greater than the threshold $\epsilon = 0.001$ is achieved in the objective function for $\eta_{\text{early}} = 5$ consecutive temperature levels.

Table 13: HANSA Algorithm Tuning Parameters.

Parameter	**Value**	**Description**
Initial Temperature (T_{init})	Dynamic	Calculated as $1.5 \times \sigma(f)$, where $\sigma(f)$ is the stdev of initial random perturbations.
Cooling Rate (α)	Adaptive	0.98 (if improving), 0.95 (normal), 0.90 (stagnation)
Tabu Tenure	$7 + \lfloor SC/10 \rfloor$	Adaptive tenure based on Stagnation Counter (SC).
Stagnation Thresholds	$\eta_{stag} = 50$, $\eta_{severe} = 100$	Triggers for diversification and reheating phases.
Learning Rates	$\alpha = 0.7, \beta = 0.3$	Weights for evaluating neighborhood operator performance (NS_perf).
Stopping Criterion	$\epsilon < 0.001$	Stop if improvement is negligible for 5 consecutive temperature levels.

By building on initial solutions from CustSort-C and refining them through adaptive search mechanisms, HANSA outperforms conventional heuristics and exact methods, particularly for large-scale instances where exact methods are computationally infeasible. These parameter choices and adaptive mechanisms reflect a balance between exploration, exploitation, and computational efficiency, specifically tailored to handle the complexity of the campaign optimization problem with network effects.

5.3 Computational Study

In this section, we present the computational study conducted to evaluate the performance of our solution methods for the COP-LN model. The network construction methodology and co-occurrence threshold analysis were presented in Chapter 4 (see §4.2.5 for detailed results on how co-occurrence thresholds affect network connectivity). We describe our test instances in §5.3.1, evaluate the efficiency and performance of our matheuristic strategies in §5.3.2, and analyze network effects in §5.3.3.

All computations were performed on a computer with a 64-bit Windows 10 operating system with Intel(R) Core(TM) i7-3630QM CPU 16 GB RAM, and CPLEX 20.1 was used in Python to solve the BIP problem exactly. The Aggregate+Greedy heuristic algorithm is also coded in Python using the Numpy and Numba libraries for fast and parallel mathematical operations.

5.3.1 Test Instances

The telecommunication company launches around 10 campaigns targeting around 500,000 customers through 3 communication channels each day, and the planning is done weekly. Following the provided specifications, we have generated a set of test instances tailored for this study. A summary of these instances is presented in Table 14.

The columns represent the number of campaigns (c), the number of customers (u), and the number of edges (a) in the sales-enriched customer network, respectively. We set the number of communication channels (h) as 3, and the planning

Table 14: Test instances for the campaign optimization problem

Small				Medium				Large			
Instance	c	u	a	Instance	c	u	a	Instance	c	u	a
1	1	100	0	8	10	2000	56	15	10	25000	9472
2	2	100	0	9	10	3000	122	16	10	30000	13646
3	5	200	0	10	10	4000	244	17	10	35000	18388
4	5	500	2	11	10	5000	392	18	10	40000	23990
5	5	700	4	12	10	10000	1606	19	10	45000	30476
6	5	1000	12	13	10	15000	3548	20	10	50000	37502
7	10	1000	12	14	10	20000	6100	21	10	55000	44992

horizon (d) as 7 days.

Our dataset encompasses a total of 21 instances: small-sized (Instances 1-7), medium-sized (Instances 8-14), and large-sized (Instances 15-21). In a real scenario, there would be around 500,000 customers, however generating such a network would require 100s GB of memory.

For each instance, we generated the additional parameters according to the following guidelines:

- The eligibility matrix e_{cu} of size $|C| \times |U|$ was populated with random binary values to indicate the eligibility of customers for each campaign.
- The campaign category matrix q_{ic} of size $|I| \times |C|$ was randomly generated using binary values to specify the categories for each campaign.
- The campaign priority array r_p of size $|P|$ was generated with random integer values ranging from 0 to 100, indicating the priority level of each category.
- The matrix r_{pc} of size $|P| \times |C|$ was created, with each campaign assigned to one of the $|P|$ priority values using random binary values. Each campaign's individual priority r_c was calculated by multiplying r_p and r_{pc}.
- We set the weekly limit b to 7 (representing 7 days in a week).
- The daily limit k was set to 3 (representing the maximum number of communications per customer per day).

- The campaign limit array l_c of size $|C|$ was populated with randomly selected integer values from the set $\{2, 3, 4\}$, indicating the maximum number of communications for each campaign.
- The campaign category limit per week array m_i of size $|I|$ was generated using randomly selected integer values from $\{3, 4, 5\}$, representing the maximum number of communications for each campaign category per week.
- The campaign category limit per day array n_i of size $|I|$ was generated with randomly selected integer values from $\{1, 2, 3\}$, indicating the maximum number of communications for each campaign category per day.
- The channel capacity matrix of size $|H| \times |D|$ was populated with random values from the set $\{0.01|U|, 0.02|U|, 0.03|U|\}$.
- The solution from previous weeks of size $|C| \times |U| \times |H| \times |D|$ was generated for s_{cuhd} with random binary values.

The generated values for these parameters are given in Appendix A.3. These test instances allow us to evaluate the performance of the algorithms under different conditions, providing insights into the behavior of the proposed heuristics and their ability to handle real-world scenarios.

Before proceeding with the full computational study, we validated our customer prioritization algorithms using a miniworld example in Chapter 4 (see §4.5). That preliminary study demonstrated that CustSort-C achieves optimal solutions in small network instances, while CustSort-A and CustSort-B provide efficient approximations suitable for time-constrained scenarios.

5.3.2 Evaluating Efficiency and Performance of Matheuristic Strategies

We assess the performance of the proposed heuristics in Table-15 and Table-16. For instances 1 to 15, exact solutions are obtainable using the CPLEX solver. The objective value gap (%) (calculated by $[(Z_e - Z_h)/Z_e] \times 100$, where Z_e gives the exact objective value found by CPLEX solver and Z_h gives the result for the corresponding heuristic) and time required for each instance are given in Table-15. Notably, the duration for solving these instances using CPLEX averages around 4143 seconds (approximately 69 minutes), but escalates considerably for larger instances, exceeding 14 hours (50792 seconds) for the largest instance. Such extensive computational time renders frequent application impractical for real-world scenarios.

Table-15 presents the objective value gap and computational time for instances solvable by the CPLEX solver. The average gaps for the Aggregate+Greedy heuristic are 62% for CustSort-A, 59% for CustSort-B, and 20% for CustSort-C. When HANSA is applied, the gap is further reduced to 5%, demonstrating its ability to enhance solution quality. However, the computational runtime varies significantly. CustSort-A and CustSort-B consistently outperform CustSort-C in terms of speed, with CustSort-C taking considerably longer to execute. Moreover, when HANSA is applied, the runtime increases drastically compared to other configurations. This highlights the need to carefully balance computational efficiency with performance, particularly for larger instances where runtime becomes a critical consideration.

Table 15: Objective value gap (%) and time (sec) for instances that could be solved exactly.

Instance	Objective Value Gap (%)				Time (sec)				
	Agg+ Grdy(A)	Agg+ Grdy(B)	Agg+ Grdy(C)	HANSA	Exact (BIP)	Agg+ Grdy(A)	Agg+ Grdy(B)	Agg+ Grdy(C)	HANSA
1	24.5	24.5	24.5	0.0	0.0	0.0	0.0	0.0	3.4
2	60.4	60.4	19.2	1.9	0.1	0.0	0.0	0.1	3.7
3	51.9	40.2	14.4	4.8	0.1	0.0	0.0	0.1	3.9
4	62.9	46.4	26.3	4.1	0.3	0.0	0.1	0.3	5.0
5	68.2	59.5	17.9	6.5	0.4	0.1	0.1	0.6	5.4
6	31.4	31.3	19.3	7.2	0.6	0.1	0.2	1.1	10.2
7	70.1	52.6	18.2	3.5	1.3	0.2	0.2	2.2	21.8
8	70.0	68.4	24.3	8.3	4.3	0.6	1.0	8.8	79.1
9	66.4	66.4	17.0	5.2	11.3	1.4	2.2	20.1	189.8
10	63.7	63.8	20.9	7.6	24.0	2.4	3.9	37.3	331.8
11	70.7	70.8	15.7	5.4	43.9	3.8	6.1	60.4	541.7
12	75.6	75.7	20.1	2.9	310.0	15.5	25.1	290.1	2601.6
13	73.2	73.3	23.9	4.2	1732.3	35.7	57.4	749.6	6781.3
14	76.9	76.1	26.3	9.4	9234.9	66.4	105.0	1209.1	11609.8
15	68.9	69.0	19.1	4.7	50792.0	109.2	168.0	1668.6	14829.9
Average	62.3	58.6	20.5	5.0	4143.7	15.7	24.6	269.9	2467.9

For large instances (Instances 16-21), the time required only for loading the input data exceeds about 20 hours, rendering exact solutions using CPLEX impractical even with constrained time limits. This limitation requires the adoption of alternative methods for these large-scale scenarios. To address this, a comparative evaluation was conducted among CustSort-A, CustSort-B, CustSort-C, and HANSA. HANSA demonstrated superior optimization performance compared to CustSort-A, CustSort-B, and CustSort-C. Specifically, it achieved an average improvement of 60% over CustSort-A, 56% over CustSort-B, and 15% over CustSort-C across the initial 15 instances. This trend continued in the final seven instances, where it showed a 65% improvement over CustSort-A, 61% over CustSort-B, and 15% over CustSort-C.

These results are particularly notable given the constraints on CPLEX for large instances. While CustSort-A and CustSort-B were more efficient in terms of execution duration, HANSA provided a favorable balance of solution quality and computational time. The execution durations of CustSort-A and CustSort-B, though efficient, exhibited a slower escalation compared to CustSort-C and HANSA. This indicates that, while CustSort-A and CustSort-B are faster for smaller instances, HANSA remains effective even as the problem scales, delivering higher-quality solutions without a proportionate increase in computation time.

Overall, despite the inability to obtain exact solutions for larger instances using CPLEX, the comparative results validate the effectiveness of HANSA. It consistently provides high-quality solutions with a reasonable computational effort, making it a practical choice for addressing large-scale optimization problems where exact methods fall short.

If the telecommunication company opts to integrate the sales-enriched customer network into its campaign optimization while encountering computational constraints and temporal restrictions, it may choose either CustSort-A or CustSort-B. Conversely, should the company choose to invest in enhanced computational resources to attain greater precision in its outcomes, HANSA represents a viable option.

Table 16: Obj. value gap (%) and time (sec) for comparison between each algorithm.

Instance	Obj. Value Gap To HANSA(%)			Time (sec)			
	Agg+ Grdy(A)	Agg+ Grdy(B)	Agg+ Grdy(C)	Agg+ Grdy(A)	Agg+ Grdy(B)	Agg+ Grdy(C)	HANSA
1	23.1	23.1	0.0	0.0	0.0	0.0	3.4
2	58.4	58.4	23.1	0.0	0.0	0.1	3.7
3	49.8	37.6	15.1	0.0	0.0	0.1	3.9
4	60.4	42.7	10.8	0.0	0.1	0.3	5.0
5	65.7	56.4	21.2	0.1	0.1	0.6	5.4
6	28.9	28.9	11.5	0.1	0.2	1.1	10.2
7	67.3	48.3	16.4	0.2	0.2	2.2	21.8
8	68.4	66.7	10.8	0.6	1.0	8.8	79.1
9	63.6	63.6	20.2	1.4	2.2	20.1	189.8
10	61.6	61.7	10.2	2.4	3.9	37.3	331.8
11	69.8	69.9	16.4	3.8	6.1	60.4	541.7
12	74.5	74.6	13.2	15.5	25.1	290.1	2601.6
13	70.4	70.5	16.6	35.7	57.4	749.6	6781.3
14	75.7	74.9	16.0	66.4	105.0	1209.1	11609.8
15	67.0	67.1	22.7	109.2	168.0	1668.6	14829.9
Average	60.3	56.3	14.9	15.69	24.62	269.90	2467.86
16	60.4	58.7	16.0	163.2	253.6	2128.1	18913.7
17	65.5	64.0	18.7	232.0	357.8	2587.6	25256.1
18	66.8	65.4	22.6	316.5	496.7	3047.1	28532.2
19	67.3	53.0	17.4	415.8	653.1	3506.6	33112.9
20	67.4	66.1	16.3	530.9	860.2	3966.1	39025.6
21	63.2	61.7	12.3	672.8	1067.2	4425.6	41087.9
Average	65.1	61.5	17.2	388.51	614.77	3276.89	30988.05

5.3.3 Evaluating Network Effect

Table 17 presents the objective value gained per direct message (OV/DM), providing a comparative analysis of various solution methods, including the Exact approach (utilizing the CPLEX Solver) and heuristic methods (Agg+Grdy with different sorting strategies: CustSort-A, CustSort-B, CustSort-C). Additionally, the performance of HANSA is examined. The OV/DM metric is used to evaluate the efficiency of direct messages in achieving the objective function as described in Section 3.1. This metric serves as an indicator of network utilization efficiency across varying levels of problem complexity. The Exact solution consistently delivers high OV/DM values, setting a benchmark for comparison. This method, however, is applicable only to problem instances where computational resources allow for optimal solutions. The Agg+Grdy heuristic methods show a progressive improvement in OV/DM across CustSort-A, CustSort-B, and CustSort-C, with CustSort-C yielding the most significant gains. This improvement reflects the increasing effectiveness of CustSort-C in handling complex problems due to its refined problem-solving capabilities. HANSA outperforms other heuristic methods in almost all instances. This method, combining adaptive simulated annealing with network-aware optimization strategies, demonstrates substantial enhancements in OV/DM, particularly in more complex instances. The increase in the OV/DM value across instances is attributed to the increasing complexity of the network. As the network complexity grows, more sophisticated methods like CustSort-C and HANSA become increasingly effective, leveraging the additional complexity to yield higher objective values relative to the number of direct messages.

- **Instances 1 to 7**: In these simpler instances, the difference in OV/DM between the Exact method and the heuristic approaches is minimal. The results indicate that even basic heuristic methods can achieve near-optimal performance for straightforward cases.

- **Instances 8 to 15**: As the complexity of the problem increases, CustSort-C begins to show a notable advantage over CustSort-A and CustSort-B. In these instances, HANSA closely follows or matches the performance of the Exact method, indicating its robustness and efficiency.

- **Instances 16 to 22**: For the most complex problem instances, HANSA exhibits a marked improvement in OV/DM, significantly outperforming CustSort-A and CustSort-B, and even surpassing the Exact method in certain cases. This demonstrates the scalability and adaptability of HANSA in handling large-scale, complex problems.

Table 17: Objective value gained per direct message

Instance	Exact (CPLEX Solver)	Agg+ Grdy(A)	Agg+ Grdy(B)	Agg+ Grdy(C)	HANSA
1	55.57	55.71	55.71	55.71	55.68
2	28.59	28.61	28.61	28.59	28.68
3	49.04	49.09	49.10	49.04	49.09
4	24.08	24.07	24.06	24.09	24.10
5	57.92	57.91	57.90	57.92	57.94
6	13.79	13.79	13. 79	13.79	13.79
7	23.73	23.73	23.73	23.73	23.73
8	39.39	29.08	29.48	30.75	39.39
9	59.16	28.17	28.45	50.20	59.16
10	52.60	35.76	35.73	45.98	52.60
11	50.42	30.80	30.76	47.33	50.42
12	88.59	33.49	34.78	49.64	88.59
13	102.40	43.16	44.64	53.67	102.40
14	115.67	43.06	46.14	69.61	115.67
15	116.88	47.54	48.05	80.70	116.88
16	-	48.32	50.65	83.75	103.56
17	-	50.39	52.74	98.95	132.85
18	-	54.70	56.06	139.03	181.57
19	-	60.31	58.44	217.36	282.84
20	-	58.33	58.76	267.35	272.98
21	-	57.05	58.24	206.95	291.45
22	-	55.19	57.79	213.20	303.55

5.3.4 Impact of Incorporating Network Effects in Campaign Planning

To assess the impact of incorporating customer network effects into the campaign optimization framework, we conducted a comparative analysis of two scenarios. In the first scenario, network propagation was modeled explicitly using the CustSortC strategy (Algorithm 3, Section 4.2.2). In the second scenario, network effects were excluded by setting the adjacency matrix a_{uv}, which represents the connections between customers u and v to a zero matrix in the network coverage constraint (Constraint 3) during the initial optimization phase. This effectively removed the influence of network propagation from the model. After obtaining the optimal solution under the configuration excluding network influence, we re-evaluated the objective function (1) by recalculating the network coverage constraint (26) using the actual, non-zero values of the adjacency parameter a_{uv} from the empirical social network. This enabled us to estimate the actual reach and impact of the campaign within the true network structure, based on targeting decisions made without considering network effects during planning.

Following the acquisition of an optimal solution under this "network-agnostic" configuration, a subsequent evaluation of the objective function (1) was performed. This re-evaluation entailed a recalculation of the network coverage constraint (26), this time employing the authentic, non-zero values of the adjacency parameter a_{uv} derived from the empirical social network. This step facilitated the determination of the potential campaign reach and impact within the existing network structure, predicated on the targeting decisions made without explicit network optimization. The efficacy of network optimization, specifically through the **CustSortC** strategy, was then assessed by computing the ratio of the objective value obtained from the original problem formulation (incorporating network optimization) to this post-optimization re-computed objective value (derived from the network-agnostic solution but evaluated within the true network). This ratio serves as a direct metric of the incremental benefit accrued from the network optimization component of the proposed model.

More specifically, we compared the objective value from the network-aware solution to the re-evaluated objective value of the solution that excluded network influence. The resulting ratio provides a direct measure of the performance improvement attributable to modeling network effects within the optimization process.

Let's consider a customer base containing c_1 and c_2, and assume we have two messages with priorities p_1 and p_2. If the network effect is ignored, it does not matter which customer receives which message, as long as the total number of customers receiving messages is maximized. Therefore, when customer c_1 receives the message with priority p_1, and customer c_2 receives the message with priority p_2, the total priority score for these two customers will be $p_1 + p_2$.

However, as the network effect takes place (even if we ignore it in optimization), the total priority score will be $[p_1 \cdot (\deg_1 + 1)] + [p_2 \cdot (\deg_2 + 1)]$, where $\deg_1$ is the degree of customer c_1 and $\deg_2$ is the degree of customer c_2.

If we consider the network effect in optimization, then the greater the degree of a customer, the higher the priority of the message they will receive. Consequently, if $\deg_1 < \deg_2$ and $p_1 > p_2$ the total priority score will be $[p_1 \cdot (\deg_2 + 1)] + [p_2 \cdot (\deg_1 + 1)]$.

The calculation of the Ratio of Objective Value would be $\frac{[p_1 \cdot (\deg_2+1)]+[p_2 \cdot (\deg_1+1)]}{[p_1 \cdot (\deg_1+1)]+[p_2 \cdot (\deg_2+1)]}$. Assume we have the following values:

- $p_1 = 5$
- $p_2 = 3$
- $\deg_1 = 4$
- $\deg_2 = 6$

Without Network Optimization: When the network effect is ignored, the optimized priority score is simply the sum of the priorities:

$$\text{Optimized Priority Score} = p_1 + p_2 = 5 + 3 = 8$$

$$\begin{aligned} \text{Priority Score With Network Effect} &= [p_1 \cdot (\deg_1 + 1)] + [p_2 \cdot (\deg_2 + 1)] \\ &= [5 \cdot (4 + 1)] + [3 \cdot (6 + 1)] \\ &= 46 \end{aligned}$$

With Network Optimization: When considering the network effect, the total priority score is calculated as:

$$\begin{aligned}
\text{Network Optimized Priority Score} &= [p_1 \cdot (\deg_2 + 1)] + [p_2 \cdot (\deg_1 + 1)] \\
&= [5 \cdot (6 + 1)] + [3 \cdot (5 + 1)] \\
&= 50
\end{aligned}$$

Ratio Calculation: Now, let's calculate the ratio of the objective value with network optimization to the re-computed objective value without network optimization.

$$\begin{aligned}
\text{Ratio} &= \frac{[p_1 \cdot (\deg_2 + 1)] + [p_2 \cdot (\deg_1 + 1)]}{[p_1 \cdot (\deg_1 + 1)] + [p_2 \cdot (\deg_2 + 1)]} \\
&= \frac{50}{46} = 1.09
\end{aligned}$$

However, when the differences between priority values, and degree values increases the ratio will be increased as well, see the following case:

- $p_1 = 90$
- $p_2 = 10$
- $\deg_1 = 2$
- $\deg_2 = 40$

$$\begin{aligned}
\text{Priority Score With Network Effect} &= [p_1 \cdot (\deg_1 + 1)] + [p_2 \cdot (\deg_2 + 1)] \\
&= [90 \cdot (2 + 1)] + [10 \cdot (40 + 1)] \\
&= 680
\end{aligned}$$

$$
\begin{aligned}
\text{Network Optimized Priority Score} &= [p_1 \cdot (\deg_2 + 1)] + [p_2 \cdot (\deg_1 + 1)] \\
&= [90 \cdot (40 + 1)] + [10 \cdot (2 + 1)] \\
&= 3720
\end{aligned}
$$

$$
\begin{aligned}
\text{Ratio} &= \frac{[p_1 \cdot (\deg_2 + 1)] + [p_2 \cdot (\deg_1 + 1)]}{[p_1 \cdot (\deg_1 + 1)] + [p_2 \cdot (\deg_2 + 1)]} \\
&= \frac{3720}{680} = 5.47
\end{aligned}
$$

Table 18 reports the resulting ratios for the subset of instances (Instances 1–15) where exact solutions could be obtained using CPLEX. The results show a clear and statistically significant improvement in the objective function when network effects are explicitly incorporated into the optimization, which leads to prioritization of influential customers. The ratio values range from 1.26 to 7.02, with a mean of 2.70. This implies that, on average, incorporating network effects during the optimization phase more than doubles the effective total weighted value of campaign deliveries in these instances, compared to solutions that initially ignore network influence (even when the network effects are accounted for retrospectively).

Table 18: Ratio of objective values from campaign plans with vs. without network effects during planning, obtained with CPLEX.

Instance	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**Avg.**
Ratio	1.26	2.22	1.99	1.67	2.30	1.40	1.91	2.19	1.67	2.20	2.43	4.00	5.19	7.02	3.07	2.70

Table 19 presents the corresponding ratios obtained using the HANSA heuristic for all problem instances (1–21). The same evaluation procedure was applied: initial optimization with a null adjacency matrix ($a_{uv} = 0$ in (26)), followed by re-evaluation of the objective function using the actual network structure compared to the HANSA heuristic that incorporates network effects. As with the exact solutions, incorporating network effects during optimization yields a significant performance gain. The resulting ratios range from 1.09 to 5.94, with an average of 2.90. These results indicate that HANSA, guided by the prioritization of influential customers, effectively leverages the social network to enhance campaign reach and impact. On average, network-aware optimization with HANSA results in nearly a threefold increase in the objective value compared to solutions that do not explicitly consider network influence.

Table 19: Ratio of objective values from campaign plans with vs. without network effects during planning, obtained with HANSA.

Instance	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**Avg.**
Ratio	1.09	2.19	3.06	3.30	2.83	1.80	4.73	3.21	3.25	2.24	1.91	3.43	2.14	2.33	4.35	2.53	3.12	1.88	5.94	3.43	2.13	2.90

The consistently high ratios observed for both the exact solutions and the HANSA heuristic (driven by the CustSortC customer prioritization strategy) provide strong empirical evidence of the value added by incorporating network optimization into telecommunication marketing campaigns. By strategically targeting influential customers identified through the network structure, the model significantly enhances campaign reach and effectiveness, yielding substantially higher total weighted values compared to approaches that ignore social connectivity during the planning phase. These results highlight both the practical relevance and theoretical contribution of the proposed model and heuristic in leveraging customer networks to improve communication strategies in the telecommunications sector.

CHAPTER VI

CONCLUSION

This thesis developed an optimization framework for outbound telecommunications campaigns under communication constraints and network effects. We addressed the problem of maximizing prioritized campaign reach while satisfying company and operational limits on communication frequency and channel capacity. The base model, *Campaign Optimization Problem with Communication Limitations (COPL)*, was extended with customer-network information from Call Data Records (CDRs) and tariff-change data to form *Campaign Optimization Problem with Communication Limitations and Network Effect (COP-LN).*

The first half of the thesis provided a detailed problem formulation, incorporating constraints and parameters that represent real-world limitations faced by telecommunication companies. Our mathematical model introduced binary decision variables to determine the allocation of campaign messages across customers, channels, and days, ensuring that communication thresholds were respected while prioritizing high-value campaigns. The model also accounted for rolling horizons to address long-term communication strategies and customer annoyance by regulating the frequency of messages over consecutive weeks.

Given the complexity of the proposed optimization problem, especially when considering large-scale instances with multiple campaigns, customers, and communication channels, we explored heuristic approaches to generate high-quality solutions in a reasonable timeframe. Three solution methods were presented: (1) the *Basic Greedy heuristic*, (2) the *Relax+Greedy matheuristic*, and (3) the *Aggregate+Greedy matheuristic*. Each method built upon the previous one, gradually incorporating more sophisticated techniques for balancing campaign priorities and customer eligibility while respecting the capacity and communication limits.

The *Basic Greedy heuristic* provided an initial solution by sequentially selecting campaigns based on their priority, but its simplicity often failed to fully utilize the available communication channels or customer capacity. To address this, the *Relax+Greedy matheuristic* employed a linear programming (LP) relaxation of the original binary integer program (BIP) model, improving solution quality by exploiting fractional allocations before rounding decisions. Finally, the *Aggregate+Greedy matheuristic* enhanced the solution further by aggregating the number of eligible customers per campaign and implementing a more structured approach to customer prioritization. This hybrid approach, combining aggregate analysis with a greedy allocation strategy, yielded significantly better solutions within practical time limits, especially for large-scale problem instances.

In the latter half of the thesis, we integrated one-hop network effects into the model. By extending the customer base with social network data—constructed from a refined CDR-based network and enriched with tariff-change information—we incorporated indirect communication benefits. This addition enabled campaign planning that considers both direct messages and one-hop neighbor awareness.

The *Hybrid Adaptive Network-Aware Simulated Annealing (HANSA)* heuristic, introduced in the final stages of the thesis, was a key development in refining the solution space exploration. By combining simulated annealing with adaptive mechanisms including dynamic temperature control, reactive tabu search, and diversification strategies, this approach balanced exploration and exploitation, preventing premature convergence to suboptimal solutions. It allowed us to fine-tune campaign communication strategies dynamically, accounting for changes in customer behavior and communication patterns over time.

6.0.1 Contribution Recap

The definitive contribution list is presented in Chapter 1. In summary, this thesis delivers: (i) COPL and COP-LN formulations for campaign planning under communication constraints and one-hop network effects, (ii) scalable heuristic and matheuristic solution methods, (iii) a practical network-construction and customer-prioritization pipeline (CustSortA/B/C), and (iv) the HANSA framework for higher-quality search on large instances.

6.0.2 Practical Implications and Future Work

The findings of this thesis have direct implications for marketing strategies in the telecommunications sector, where optimizing the timing, channel, and frequency of campaign communications is critical to both customer satisfaction and campaign success. The integration of social network data provides an innovative way to extend campaign reach beyond direct communication, opening new avenues for marketing optimization in other industries that rely on customer networks.

Future work could explore several extensions of this research:

- **Real-time Optimization**: Implementing a real-time version of the proposed model to dynamically adjust campaign strategies based on live customer interactions and responses.

- **Multi-Objective Optimization**: Expanding the model to incorporate multiple objectives, such as balancing customer churn rates with campaign success or minimizing marketing costs alongside maximizing campaign reach.

- **Advanced Social Network Models**: Further refinement of the network effect model, including the use of more sophisticated techniques such as machine learning, to predict customer influence more accurately.

- **Integration of Personalized Campaigns**: Extending the optimization to include personalized campaign offers and messages based on customer preferences and behaviors, thereby enhancing customer engagement and satisfaction.

In conclusion, this thesis presents a robust framework for optimizing outbound campaign communication in telecommunications under strict communication constraints. The combination of mathematical modeling, network-aware analysis, and scalable heuristics provides practical and transferable methods for campaign optimization in data-rich service industries.

APPENDIX A

PARAMETERS AND VALUES USED FOR TEST INSTANCES

A.1 Parameters and Values Used for Test Instances

Table 20: Parameters and values used in the test instances.

Instance	q_{ic}	r_c	b	k	l_c	m_i	n_i	t_{hd}
1	$\begin{bmatrix}0&1\\1&1\\0&0\end{bmatrix}$	$\begin{bmatrix}62&49\end{bmatrix}$	7	3	$\begin{bmatrix}3&3\end{bmatrix}$	$\begin{bmatrix}4&3&3\end{bmatrix}$	$\begin{bmatrix}3&2&1\end{bmatrix}$	$\begin{bmatrix}1&1&2&3&3&1&1\\1&2&1&2&3&2&1\\3&2&2&3&3&2&3\end{bmatrix}$
2	$\begin{bmatrix}0&1&0&1&1\\1&0&1&1&0\\1&1&1&1&1\end{bmatrix}$	$\begin{bmatrix}50&36&40&40&40\end{bmatrix}$	7	3	$\begin{bmatrix}2&4&3&4&2\end{bmatrix}$	$\begin{bmatrix}3&5&3\end{bmatrix}$	$\begin{bmatrix}2&2&3\end{bmatrix}$	$\begin{bmatrix}2&2&3&1&2&2&1\\3&1&3&3&3&3&2\\2&2&3&1&1&3&3\end{bmatrix}$
3	$\begin{bmatrix}0&0&0&1&0\\0&0&0&0&1\\0&0&1&0&1\end{bmatrix}$	$\begin{bmatrix}45&61&99&45&45\end{bmatrix}$	7	3	$\begin{bmatrix}3&2&2&2&4\end{bmatrix}$	$\begin{bmatrix}3&5&4\end{bmatrix}$	$\begin{bmatrix}1&1&1\end{bmatrix}$	$\begin{bmatrix}4&4&4&2&4&4&6\\2&4&2&6&6&4&4\\2&2&2&2&6&4&2\end{bmatrix}$
4	$\begin{bmatrix}0&0&1&0&1\\1&1&1&0&1\\1&0&0&0&1\end{bmatrix}$	$\begin{bmatrix}22&34&34&34&22\end{bmatrix}$	7	3	$\begin{bmatrix}3&3&3&2&4\end{bmatrix}$	$\begin{bmatrix}3&4&5\end{bmatrix}$	$\begin{bmatrix}2&2&2\end{bmatrix}$	$\begin{bmatrix}10&15&10&15&5&15&10\\15&15&5&15&10&10&15\\15&15&10&10&10&10&10\end{bmatrix}$
5	$\begin{bmatrix}1&1&0&0&1\\1&1&1&1&0\\1&1&1&0&1\end{bmatrix}$	$\begin{bmatrix}68&68&68&47&97\end{bmatrix}$	7	3	$\begin{bmatrix}3&2&2&2&3\end{bmatrix}$	$\begin{bmatrix}4&4&4\end{bmatrix}$	$\begin{bmatrix}2&2&2\end{bmatrix}$	$\begin{bmatrix}14&21&14&14&21&7&21\\21&21&21&14&21&21&21\\7&14&21&14&14&14&14\end{bmatrix}$
6	$\begin{bmatrix}0&1&1&1&1\\1&1&0&1&0\\0&1&0&1&0\end{bmatrix}$	$\begin{bmatrix}7&41&7&7&7\end{bmatrix}$	7	3	$\begin{bmatrix}3&3&4&4&4\end{bmatrix}$	$\begin{bmatrix}3&3&4\end{bmatrix}$	$\begin{bmatrix}2&1&3\end{bmatrix}$	$\begin{bmatrix}10&10&20&20&30&20&30\\30&30&20&30&10&20&30\\30&10&30&30&30&30&20\end{bmatrix}$
7	$\begin{bmatrix}1&0&1&0&0&0&0&1&0&0\\0&0&1&1&1&0&0&1&1&1\\0&0&1&0&1&0&0&0&1&1\end{bmatrix}$	$\begin{bmatrix}44&65&45&65&45&45&44&45&45&65\end{bmatrix}$	7	3	$\begin{bmatrix}3&2&2&2&2&3&3&2&2&2\end{bmatrix}$	$\begin{bmatrix}3&3&3\end{bmatrix}$	$\begin{bmatrix}1&1&2\end{bmatrix}$	$\begin{bmatrix}30&30&30&20&10&30&20\\10&20&10&30&20&10&30\\30&30&20&30&20&20&30\end{bmatrix}$
8	$\begin{bmatrix}1&0&1&1&0&0&1&1&1&1\\0&1&1&1&1&1&1&0&1&1\\1&1&0&1&0&0&0&1&0&0\end{bmatrix}$	$\begin{bmatrix}66&66&66&66&66&66&66&66&66&21\end{bmatrix}$	7	3	$\begin{bmatrix}4&2&2&4&2&4&4&4&4&4\end{bmatrix}$	$\begin{bmatrix}3&4&4\end{bmatrix}$	$\begin{bmatrix}1&1&2\end{bmatrix}$	$\begin{bmatrix}60&20&20&40&60&20&60\\60&40&60&40&20&40&40\\20&20&40&60&40&40&20\end{bmatrix}$
9	$\begin{bmatrix}1&0&1&0&0&1&1&0&1&1\\0&1&0&0&0&1&1&1&1&0\\0&1&0&1&0&0&0&1&1&1\end{bmatrix}$	$\begin{bmatrix}98&42&98&98&42&8&8&98&42&42\end{bmatrix}$	7	3	$\begin{bmatrix}4&3&3&3&4&3&2&3&4&4\end{bmatrix}$	$\begin{bmatrix}4&5&4\end{bmatrix}$	$\begin{bmatrix}1&2&3\end{bmatrix}$	$\begin{bmatrix}30&30&30&30&30&30&90\\90&60&90&90&30&90&60\\90&30&60&60&30&90&90\end{bmatrix}$
10	$\begin{bmatrix}0&1&1&1&1&0&1&0&0&1\\1&1&0&1&1&0&1&1&1&1\\0&0&1&1&0&1&1&1&1&0\end{bmatrix}$	$\begin{bmatrix}97&18&18&6&97&18&97&6&97&18\end{bmatrix}$	7	3	$\begin{bmatrix}4&2&4&3&2&3&2&3&2&3\end{bmatrix}$	$\begin{bmatrix}4&3&3\end{bmatrix}$	$\begin{bmatrix}2&2&1\end{bmatrix}$	$\begin{bmatrix}120&120&80&120&40&80&80\\80&40&40&120&120&80&120\\40&80&120&80&40&40&120\end{bmatrix}$
11	$\begin{bmatrix}0&1&0&0&0&0&1&1&1&1\\0&1&1&1&1&1&0&1&0&0\\1&1&0&1&1&0&0&1&0&0\end{bmatrix}$	$\begin{bmatrix}69&20&49&49&49&69&69&69&69&20\end{bmatrix}$	7	3	$\begin{bmatrix}2&2&2&3&3&2&3&2&2&4\end{bmatrix}$	$\begin{bmatrix}5&4&3\end{bmatrix}$	$\begin{bmatrix}1&2&1\end{bmatrix}$	$\begin{bmatrix}150&150&100&50&150&50&150\\150&100&50&50&100&50&100\\150&100&50&50&100&50&50\end{bmatrix}$
12	$\begin{bmatrix}1&1&0&0&0&1&1&0&0&0\\0&1&1&1&1&1&1&1&0&0\\0&0&1&0&0&1&1&1&0&1\end{bmatrix}$	$\begin{bmatrix}68&62&62&62&68&37&62&62&68&62\end{bmatrix}$	7	3	$\begin{bmatrix}4&2&4&4&2&4&3&3&4&2\end{bmatrix}$	$\begin{bmatrix}5&4&4\end{bmatrix}$	$\begin{bmatrix}2&3&1\end{bmatrix}$	$\begin{bmatrix}300&100&100&300&300&200&300\\100&200&300&200&300&300&200\\100&300&200&300&200&100&300\end{bmatrix}$
13	$\begin{bmatrix}1&0&0&0&1&1&1&0&1&0\\0&0&0&1&0&0&1&0&1&0\\0&0&1&0&0&0&1&1&1&1\end{bmatrix}$	$\begin{bmatrix}64&64&38&38&38&80&64&64&64&64\end{bmatrix}$	7	3	$\begin{bmatrix}3&4&2&4&4&3&4&2&4&2\end{bmatrix}$	$\begin{bmatrix}3&5&5\end{bmatrix}$	$\begin{bmatrix}3&1&3\end{bmatrix}$	$\begin{bmatrix}150&150&450&150&450&300&450\\300&450&150&450&450&300&450\\450&150&450&450&300&150&450\end{bmatrix}$
14	$\begin{bmatrix}0&1&0&1&1&1&0&1&1&1\\0&1&1&1&1&1&1&1&1&1\\0&0&0&1&0&0&1&1&0&1\end{bmatrix}$	$\begin{bmatrix}17&29&29&17&56&29&56&56&29&56\end{bmatrix}$	7	3	$\begin{bmatrix}2&2&4&4&4&3&3&2&4&4\end{bmatrix}$	$\begin{bmatrix}5&4&4\end{bmatrix}$	$\begin{bmatrix}3&3&1\end{bmatrix}$	$\begin{bmatrix}200&600&200&600&400&400&400\\600&400&200&400&600&600&600\\200&600&200&200&600&400&200\end{bmatrix}$
15	$\begin{bmatrix}1&0&0&1&0&0&1&1&1&0\\1&1&1&0&0&0&1&0&0&1\\1&1&0&0&1&1&0&1&1&0\end{bmatrix}$	$\begin{bmatrix}25&25&25&94&97&97&97&97&94&97\end{bmatrix}$	7	3	$\begin{bmatrix}2&4&2&2&4&2&2&4&4&3\end{bmatrix}$	$\begin{bmatrix}5&4&4\end{bmatrix}$	$\begin{bmatrix}1&1&1\end{bmatrix}$	$\begin{bmatrix}250&500&750&250&250&500&750\\500&500&250&500&750&500&250\\250&500&500&750&500&500&500\end{bmatrix}$
16	$\begin{bmatrix}1&1&1&1&0&1&1&0&0&0\\0&0&1&0&1&0&0&0&1&1\\1&1&1&0&0&1&1&0&1&0\end{bmatrix}$	$\begin{bmatrix}17&17&66&66&17&46&66&46&66&66\end{bmatrix}$	7	3	$\begin{bmatrix}2&2&4&2&3&2&3&3&4&3\end{bmatrix}$	$\begin{bmatrix}4&5&3\end{bmatrix}$	$\begin{bmatrix}1&2&1\end{bmatrix}$	$\begin{bmatrix}900&300&300&600&300&300&600\\600&900&900&600&600&300&600\\600&600&600&300&600&900&300\end{bmatrix}$
								Continues on the next page

Table 20 – Parameters and Values Used for Test Instance

Instance	q_{ic}	r_c	b	k	l_c	m_i	n_i	t_{hd}
17	$\begin{bmatrix} 0&1&0&0&1&1&0&0&1&1 \\ 0&1&1&0&1&0&1&1&0&1 \\ 0&1&1&1&1&0&1&0&1&0 \end{bmatrix}$	$\begin{bmatrix} 67&48&63&67&67&63&63&63&63&67 \end{bmatrix}$	7	3	$\begin{bmatrix} 2&4&2&2&2&4&4&2&4&3 \end{bmatrix}$	$\begin{bmatrix} 5&4&3 \end{bmatrix}$	$\begin{bmatrix} 2&3&2 \end{bmatrix}$	$\begin{bmatrix} 700&350&350&350&1050&350&350 \\ 350&1050&700&700&700&1050&350 \\ 700&1050&1050&350&700&1050&1050 \end{bmatrix}$
18	$\begin{bmatrix} 1&0&1&1&0&1&1&0&0&1 \\ 0&0&1&0&0&1&1&1&0&0 \\ 1&0&1&1&1&0&1&0&1&0 \end{bmatrix}$	$\begin{bmatrix} 73&73&73&73&26&58&58&73&73&58 \end{bmatrix}$	7	3	$\begin{bmatrix} 3&2&3&3&4&3&4&3&3&4 \end{bmatrix}$	$\begin{bmatrix} 4&5&4 \end{bmatrix}$	$\begin{bmatrix} 2&2&2 \end{bmatrix}$	$\begin{bmatrix} 1200&400&400&400&400&800&400 \\ 800&1200&1200&1200&400&800&400 \\ 400&800&400&1200&800&400&800 \end{bmatrix}$
19	$\begin{bmatrix} 1&1&0&1&0&0&0&0&0&1 \\ 0&1&0&1&1&0&0&1&1&0 \\ 0&0&0&1&0&1&1&1&0&0 \end{bmatrix}$	$\begin{bmatrix} 40&77&77&40&40&77&40&77&40&16 \end{bmatrix}$	7	3	$\begin{bmatrix} 2&2&2&4&3&3&4&4&4&3 \end{bmatrix}$	$\begin{bmatrix} 3&4&3 \end{bmatrix}$	$\begin{bmatrix} 1&3&2 \end{bmatrix}$	$\begin{bmatrix} 1350&1350&450&450&900&900&1350 \\ 900&1350&1350&450&900&900&1350 \\ 1350&900&1350&900&1350&900&900 \end{bmatrix}$
20	$\begin{bmatrix} 0&1&0&1&0&1&1&1&1&0 \\ 1&0&0&0&0&1&1&1&0&1 \\ 0&1&0&0&1&0&1&1&0&1 \end{bmatrix}$	$\begin{bmatrix} 22&22&22&10&52&22&22&22&10&52 \end{bmatrix}$	7	3	$\begin{bmatrix} 4&2&4&3&3&2&2&4&2&4 \end{bmatrix}$	$\begin{bmatrix} 5&5&3 \end{bmatrix}$	$\begin{bmatrix} 3&2&1 \end{bmatrix}$	$\begin{bmatrix} 1000&500&500&1000&1000&1000&1000 \\ 1000&1500&1500&500&1500&500&1500 \\ 1000&1500&1500&1500&500&1000&1000 \end{bmatrix}$
21	$\begin{bmatrix} 1&0&1&0&0&1&1&1&0&0 \\ 1&0&1&1&1&0&0&1&0&0 \\ 0&0&0&1&0&0&1&0&0&1 \end{bmatrix}$	$\begin{bmatrix} 22&41&21&41&41&41&41&21&21&21 \end{bmatrix}$	7	3	$\begin{bmatrix} 4&2&3&3&3&2&2&3&3&3 \end{bmatrix}$	$\begin{bmatrix} 3&5&5 \end{bmatrix}$	$\begin{bmatrix} 2&2&1 \end{bmatrix}$	$\begin{bmatrix} 1650&1650&1650&1100&1100&1650&550 \\ 1650&1100&1100&1100&550&550&1100 \\ 550&1650&550&1100&550&550&550 \end{bmatrix}$
22	$\begin{bmatrix} 0&0&1&1&1&1&1&0&1&1 \\ 1&1&1&0&0&1&1&1&0&1 \\ 0&1&0&1&1&0&1&0&0&1 \end{bmatrix}$	$\begin{bmatrix} 49&49&18&18&18&18&36&49&18&18 \end{bmatrix}$	7	3	$\begin{bmatrix} 4&2&2&4&2&4&2&4&2&2 \end{bmatrix}$	$\begin{bmatrix} 3&3&4 \end{bmatrix}$	$\begin{bmatrix} 2&3&3 \end{bmatrix}$	$\begin{bmatrix} 600&600&1200&1800&1800&1200&1200 \\ 1200&1200&1800&1800&1200&1800&1800 \\ 1200&1200&600&1800&1200&600&600 \end{bmatrix}$
23	$\begin{bmatrix} 0&0&1&1&1&0&0&0&0&1 \\ 0&0&0&1&1&0&0&1&0&1 \\ 1&1&1&1&1&0&1&1&0&1 \end{bmatrix}$	$\begin{bmatrix} 46&20&13&13&20&46&20&20&13&20 \end{bmatrix}$	7	3	$\begin{bmatrix} 4&2&2&2&2&4&3&2&4&3 \end{bmatrix}$	$\begin{bmatrix} 3&5&5 \end{bmatrix}$	$\begin{bmatrix} 2&2&1 \end{bmatrix}$	$\begin{bmatrix} 650&1300&650&1950&650&1950&650 \\ 1300&650&650&1950&650&1300&650 \\ 1950&650&1300&650&1300&1300&1950 \end{bmatrix}$
24	$\begin{bmatrix} 0&1&0&0&1&0&1&0&0&0 \\ 1&1&1&0&1&0&1&0&1&0 \\ 1&1&1&0&1&0&1&0&1&0 \end{bmatrix}$	$\begin{bmatrix} 78&35&80&78&35&78&35&80&78&80 \end{bmatrix}$	7	3	$\begin{bmatrix} 2&3&3&2&4&3&4&4&2&2 \end{bmatrix}$	$\begin{bmatrix} 3&5&3 \end{bmatrix}$	$\begin{bmatrix} 3&3&3 \end{bmatrix}$	$\begin{bmatrix} 700&700&700&700&2100&1400&1400 \\ 1400&700&2100&2100&1400&2100&2100 \\ 1400&1400&1400&2100&1400&2100&700 \end{bmatrix}$
25	$\begin{bmatrix} 1&0&1&0&0&1&0&1&0&1 \\ 0&1&1&1&0&1&0&0&1&1 \\ 0&0&0&1&1&1&0&1&0&0 \end{bmatrix}$	$\begin{bmatrix} 66&67&1&67&1&67&67&1&1&1 \end{bmatrix}$	7	3	$\begin{bmatrix} 4&2&4&2&2&2&2&3&4&3 \end{bmatrix}$	$\begin{bmatrix} 3&5&3 \end{bmatrix}$	$\begin{bmatrix} 1&2&3 \end{bmatrix}$	$\begin{bmatrix} 2250&2250&1500&750&2250&750&750 \\ 2250&750&1500&750&2250&2250&750 \\ 2250&750&2250&1500&2250&1500&750 \end{bmatrix}$
26	$\begin{bmatrix} 0&0&1&0&1&0&0&0&0&0 \\ 0&0&1&0&0&1&0&0&1&1 \\ 0&1&1&0&1&1&1&0&0&1 \end{bmatrix}$	$\begin{bmatrix} 92&0&92&0&92&92&92&92&0&92 \end{bmatrix}$	7	3	$\begin{bmatrix} 3&2&2&2&2&4&4&3&3&2 \end{bmatrix}$	$\begin{bmatrix} 5&5&5 \end{bmatrix}$	$\begin{bmatrix} 3&3&3 \end{bmatrix}$	$\begin{bmatrix} 800&2400&1600&2400&2400&1600&800 \\ 2400&800&1600&800&800&1600&1600 \\ 2400&1600&2400&1600&1600&800&1600 \end{bmatrix}$
27	$\begin{bmatrix} 1&1&1&0&0&0&1&1&0&1 \\ 0&0&0&0&1&0&1&1&0&1 \\ 1&1&0&1&1&0&0&0&1&1 \end{bmatrix}$	$\begin{bmatrix} 23&92&23&4&23&92&92&23&23&23 \end{bmatrix}$	7	3	$\begin{bmatrix} 2&3&3&2&3&2&4&4&4&3 \end{bmatrix}$	$\begin{bmatrix} 4&4&5 \end{bmatrix}$	$\begin{bmatrix} 2&1&1 \end{bmatrix}$	$\begin{bmatrix} 850&2550&1700&1700&1700&2550&1700 \\ 850&2550&1700&850&1700&1700&850 \\ 1700&850&2550&2550&1700&850&2550 \end{bmatrix}$
28	$\begin{bmatrix} 1&0&0&0&0&1&1&0&1&0 \\ 1&0&0&0&1&1&1&1&0&1 \\ 0&0&0&1&1&0&1&1&1&0 \end{bmatrix}$	$\begin{bmatrix} 71&17&71&45&17&45&17&71&17&17 \end{bmatrix}$	7	3	$\begin{bmatrix} 4&4&2&2&3&2&4&2&4&4 \end{bmatrix}$	$\begin{bmatrix} 3&3&5 \end{bmatrix}$	$\begin{bmatrix} 3&1&1 \end{bmatrix}$	$\begin{bmatrix} 900&2700&900&2700&1800&1800&900 \\ 900&2700&900&900&1800&1800&1800 \\ 2700&900&1800&900&900&1800&2700 \end{bmatrix}$
29	$\begin{bmatrix} 0&1&0&1&0&0&1&0&1&1 \\ 1&1&0&1&1&1&1&0&1&1 \\ 1&0&0&0&0&0&1&0&0&1 \end{bmatrix}$	$\begin{bmatrix} 56&14&14&49&56&14&14&14&56&56 \end{bmatrix}$	7	3	$\begin{bmatrix} 4&3&3&2&3&4&2&4&2&3 \end{bmatrix}$	$\begin{bmatrix} 4&3&5 \end{bmatrix}$	$\begin{bmatrix} 2&2&3 \end{bmatrix}$	$\begin{bmatrix} 950&2850&950&1900&950&950&2850 \\ 2850&1900&1900&950&950&1900&1900 \\ 950&2850&950&950&2850&950&950 \end{bmatrix}$
30	$\begin{bmatrix} 1&1&0&1&0&1&1&1&1&1 \\ 0&1&1&0&1&1&0&0&1&1 \\ 1&1&0&0&0&1&1&1&1&0 \end{bmatrix}$	$\begin{bmatrix} 76&76&33&76&33&33&18&33&18&76 \end{bmatrix}$	7	3	$\begin{bmatrix} 4&3&2&4&2&2&2&3&2&2 \end{bmatrix}$	$\begin{bmatrix} 5&5&3 \end{bmatrix}$	$\begin{bmatrix} 3&2&2 \end{bmatrix}$	$\begin{bmatrix} 3000&2000&2000&1000&3000&2000&3000 \\ 2000&1000&1000&3000&3000&2000&3000 \\ 3000&2000&2000&1000&3000&3000&2000 \end{bmatrix}$
31	$\begin{bmatrix} 1&1&0&1&0&1&1&1&1&1 \\ 0&1&1&0&1&1&0&0&1&1 \\ 1&1&0&0&0&1&1&1&1&0 \end{bmatrix}$	$\begin{bmatrix} 76&76&33&76&33&33&18&33&18&76 \end{bmatrix}$	7	3	$\begin{bmatrix} 4&3&2&4&2&2&2&3&2&2 \end{bmatrix}$	$\begin{bmatrix} 5&5&3 \end{bmatrix}$	$\begin{bmatrix} 3&2&2 \end{bmatrix}$	$\begin{bmatrix} 3000&2000&2000&1000&3000&2000&3000 \\ 2000&1000&1000&3000&3000&2000&3000 \\ 3000&2000&2000&1000&3000&3000&2000 \end{bmatrix}$
32	$\begin{bmatrix} 1&1&0&1&0&0&0&1&0&0 \\ 0&0&1&0&0&0&0&0&1&0 \\ 1&0&1&0&1&1&1&0&0&1 \end{bmatrix}$	$\begin{bmatrix} 12&24&0&12&12&24&24&24&12&0 \end{bmatrix}$	7	3	$\begin{bmatrix} 2&3&3&4&2&2&3&2&3&2 \end{bmatrix}$	$\begin{bmatrix} 3&4&5 \end{bmatrix}$	$\begin{bmatrix} 2&3&3 \end{bmatrix}$	$\begin{bmatrix} 4500&4500&4500&3000&1500&4500&3000 \\ 4500&3000&1500&3000&1500&4500&3000 \\ 3000&3000&1500&3000&4500&1500&1500 \end{bmatrix}$
33	$\begin{bmatrix} 1&1&1&0&0&1&1&0&1&1 \\ 1&1&1&0&0&0&1&0&1&0 \\ 1&1&0&0&1&0&1&0&0&1 \end{bmatrix}$	$\begin{bmatrix} 2&83&2&2&42&42&42&83&83&2 \end{bmatrix}$	7	3	$\begin{bmatrix} 2&4&3&3&2&2&3&3&2&2 \end{bmatrix}$	$\begin{bmatrix} 4&3&4 \end{bmatrix}$	$\begin{bmatrix} 1&3&2 \end{bmatrix}$	$\begin{bmatrix} 6000&2000&2000&4000&6000&4000&6000 \\ 4000&2000&4000&4000&4000&4000&6000 \\ 4000&2000&6000&2000&4000&2000&4000 \end{bmatrix}$
34	$\begin{bmatrix} 1&1&0&1&0&0&1&1&1&0 \\ 1&0&1&0&0&0&1&1&1&1 \\ 0&1&0&1&0&1&0&1&1&0 \end{bmatrix}$	$\begin{bmatrix} 30&15&15&15&15&15&15&30&83&15 \end{bmatrix}$	7	3	$\begin{bmatrix} 2&4&4&2&4&3&2&3&4&2 \end{bmatrix}$	$\begin{bmatrix} 4&3&5 \end{bmatrix}$	$\begin{bmatrix} 1&1&1 \end{bmatrix}$	$\begin{bmatrix} 7500&5000&5000&7500&2500&7500&7500 \\ 5000&2500&2500&7500&5000&5000&2500 \\ 2500&5000&5000&2500&7500&2500&5000 \end{bmatrix}$
35	$\begin{bmatrix} 1&0&1&0&1&0&1&0&0&1 \\ 1&1&1&1&0&0&0&0&0&1 \\ 0&0&1&1&1&0&0&0&0&0 \end{bmatrix}$	$\begin{bmatrix} 43&66&66&43&99&43&43&43&99&66 \end{bmatrix}$	7	3	$\begin{bmatrix} 4&4&4&2&4&4&2&3&2&4 \end{bmatrix}$	$\begin{bmatrix} 3&4&5 \end{bmatrix}$	$\begin{bmatrix} 2&3&2 \end{bmatrix}$	$\begin{bmatrix} 9000&3000&3000&9000&6000&9000&9000 \\ 3000&3000&9000&6000&6000&6000&3000 \\ 9000&9000&6000&6000&9000&3000&6000 \end{bmatrix}$
36	$\begin{bmatrix} 0&1&0&1&0&1&0&0&1&0 \\ 1&0&0&0&1&0&1&1&1&1 \\ 1&0&0&0&0&1&0&0&0&1 \end{bmatrix}$	$\begin{bmatrix} 37&37&58&35&35&37&37&58&37&58 \end{bmatrix}$	7	3	$\begin{bmatrix} 4&4&4&2&2&3&4&2&3&4 \end{bmatrix}$	$\begin{bmatrix} 5&5&5 \end{bmatrix}$	$\begin{bmatrix} 3&1&3 \end{bmatrix}$	$\begin{bmatrix} 3500&7000&7000&7000&10500&7000&3500 \\ 7000&10500&10500&7000&7000&3500&7000 \\ 10500&3500&3500&7000&10500&3500&3500 \end{bmatrix}$
37	$\begin{bmatrix} 1&1&0&0&0&1&0&0&1&0 \\ 1&0&1&1&1&1&1&1&1&1 \\ 1&0&1&0&1&0&1&1&1&0 \end{bmatrix}$	$\begin{bmatrix} 4&4&45&4&4&45&45&63&63&45 \end{bmatrix}$	7	3	$\begin{bmatrix} 3&3&3&4&2&2&4&4&4&2 \end{bmatrix}$	$\begin{bmatrix} 4&5&5 \end{bmatrix}$	$\begin{bmatrix} 1&1&2 \end{bmatrix}$	$\begin{bmatrix} 4000&4000&12000&8000&12000&8000&8000 \\ 8000&4000&4000&4000&4000&4000&8000 \\ 12000&8000&8000&8000&4000&12000&4000 \end{bmatrix}$
38	$\begin{bmatrix} 1&0&1&0&0&0&1&1&1&1 \\ 1&0&1&0&0&1&0&1&1&1 \\ 0&1&0&1&1&1&0&1&0&1 \end{bmatrix}$	$\begin{bmatrix} 77&43&2&2&43&43&2&43&2&43 \end{bmatrix}$	7	3	$\begin{bmatrix} 2&4&3&2&3&3&3&3&2&3 \end{bmatrix}$	$\begin{bmatrix} 5&5&5 \end{bmatrix}$	$\begin{bmatrix} 2&3&1 \end{bmatrix}$	$\begin{bmatrix} 9000&4500&9000&4500&13500&9000&9000 \\ 13500&4500&13500&13500&4500&9000&13500 \\ 13500&4500&9000&9000&13500&9000&13500 \end{bmatrix}$
39	$\begin{bmatrix} 1&1&1&0&0&0&1&0&1&0 \\ 0&0&0&1&0&1&1&1&0&0 \\ 0&0&1&0&0&1&0&0&1&1 \end{bmatrix}$	$\begin{bmatrix} 21&19&77&77&19&21&19&77&19&77 \end{bmatrix}$	7	3	$\begin{bmatrix} 3&2&2&4&3&2&3&2&3&2 \end{bmatrix}$	$\begin{bmatrix} 5&4&3 \end{bmatrix}$	$\begin{bmatrix} 2&2&2 \end{bmatrix}$	$\begin{bmatrix} 5000&15000&5000&15000&10000&5000&10000 \\ 15000&15000&15000&5000&10000&5000&5000 \\ 5000&15000&5000&5000&5000&10000&10000 \end{bmatrix}$

Continues on the next page

Table 20 – Parameters and Values Used for Test Instance

Instance	q_{ic}	r_c	b	k	l_c	m_i	n_i	t_{hd}
40	$\begin{bmatrix} 1&1&0&1&0&1&0&1&0&1 \\ 1&0&1&0&0&1&0&1&0&1 \\ 0&1&0&0&1&1&1&0&0&1 \end{bmatrix}$	$\begin{bmatrix} 37&60&37&37&60&67&60&67&67&60 \end{bmatrix}$	7	3	$\begin{bmatrix} 2&2&3&4&3&3&3&2&2&3 \end{bmatrix}$	$\begin{bmatrix} 3&4&4 \end{bmatrix}$	$\begin{bmatrix} 2&3&2 \end{bmatrix}$	$\begin{bmatrix} 11000&11000&11000&16500&5500&5500&11000 \\ 16500&11000&11000&5500&16500&5500&11000 \\ 11000&5500&11000&11000&5500&11000&16500 \end{bmatrix}$
41	$\begin{bmatrix} 0&1&1&0&0&0&0&0&0&1 \\ 1&1&1&1&1&1&0&1&1&0 \\ 0&1&1&1&1&1&0&1&0&1 \end{bmatrix}$	$\begin{bmatrix} 38&38&38&54&54&54&54&54&92&54 \end{bmatrix}$	7	3	$\begin{bmatrix} 3&3&4&3&4&3&3&3&3&4 \end{bmatrix}$	$\begin{bmatrix} 4&5&5 \end{bmatrix}$	$\begin{bmatrix} 3&2&2 \end{bmatrix}$	$\begin{bmatrix} 12000&12000&12000&18000&12000&6000&18000 \\ 12000&6000&18000&6000&6000&6000&18000 \\ 12000&12000&18000&6000&6000&12000&6000 \end{bmatrix}$
42	$\begin{bmatrix} 0&0&0&1&0&0&1&0&1&1 \\ 1&0&0&0&1&1&0&1&0&1 \\ 1&0&0&0&0&1&0&0&1&1 \end{bmatrix}$	$\begin{bmatrix} 42&42&42&76&42&42&71&71&42&71 \end{bmatrix}$	7	3	$\begin{bmatrix} 2&3&3&3&2&3&4&4&4&3 \end{bmatrix}$	$\begin{bmatrix} 3&3&4 \end{bmatrix}$	$\begin{bmatrix} 3&1&3 \end{bmatrix}$	$\begin{bmatrix} 19500&19500&19500&13000&6500&6500&19500 \\ 19500&13000&13000&13000&6500&19500&13000 \\ 6500&13000&6500&6500&13000&13000&6500 \end{bmatrix}$
43	$\begin{bmatrix} 1&0&0&0&0&1&0&1&0&0 \\ 0&1&1&1&0&1&0&1&0&1 \\ 0&0&0&0&0&1&1&1&0&0 \end{bmatrix}$	$\begin{bmatrix} 53&53&57&57&53&57&20&20&57&53 \end{bmatrix}$	7	3	$\begin{bmatrix} 4&2&2&4&2&2&4&4&4&3 \end{bmatrix}$	$\begin{bmatrix} 4&5&4 \end{bmatrix}$	$\begin{bmatrix} 1&3&2 \end{bmatrix}$	$\begin{bmatrix} 21000&14000&14000&7000&7000&21000&14000 \\ 21000&21000&14000&7000&14000&21000&14000 \\ 21000&21000&21000&7000&7000&7000&14000 \end{bmatrix}$
44	$\begin{bmatrix} 0&1&0&1&1&0&1&1&0&0 \\ 1&1&0&0&0&1&1&0&0&0 \\ 0&1&1&0&1&0&1&1&1&1 \end{bmatrix}$	$\begin{bmatrix} 39&39&2&2&97&2&39&97&2&97 \end{bmatrix}$	7	3	$\begin{bmatrix} 2&4&2&2&4&2&3&4&3&3 \end{bmatrix}$	$\begin{bmatrix} 5&3&4 \end{bmatrix}$	$\begin{bmatrix} 3&1&3 \end{bmatrix}$	$\begin{bmatrix} 22500&7500&15000&15000&7500&7500&15000 \\ 15000&7500&15000&7500&15000&22500&22500 \\ 7500&15000&15000&15000&7500&7500&15000 \end{bmatrix}$
45	$\begin{bmatrix} 1&0&0&0&1&0&1&1&0&1 \\ 0&1&0&1&0&0&1&1&1&1 \\ 1&1&0&1&0&0&1&0&1&0 \end{bmatrix}$	$\begin{bmatrix} 32&32&37&37&85&85&32&37&37&32 \end{bmatrix}$	7	3	$\begin{bmatrix} 3&2&2&3&4&4&4&3&3&3 \end{bmatrix}$	$\begin{bmatrix} 3&4&4 \end{bmatrix}$	$\begin{bmatrix} 1&3&3 \end{bmatrix}$	$\begin{bmatrix} 8000&8000&24000&16000&8000&16000&16000 \\ 16000&8000&16000&8000&8000&16000&8000 \\ 24000&24000&8000&16000&24000&24000&8000 \end{bmatrix}$

A.2 Parameter Values for the Rich Channel Capacity Test Instances.

Table 21: Parameter values for the rich channel capacity test instances.

Instance	q_{ic}	r_c	b	k	l_c	m_i	n_i	t_{hd}
1	$\begin{bmatrix}0&1\\1&1\\0&0\end{bmatrix}$	$\begin{bmatrix}62&49\end{bmatrix}$	7	3	$\begin{bmatrix}3&3\end{bmatrix}$	$\begin{bmatrix}4&3&3\end{bmatrix}$	$\begin{bmatrix}3&2&1\end{bmatrix}$	$\begin{bmatrix}70&70&60&50&50&70&70\\70&60&70&60&50&60&70\\50&60&60&50&50&60&50\end{bmatrix}$
2	$\begin{bmatrix}0&1&0&1&1\\1&0&1&1&0\\1&1&1&1&1\end{bmatrix}$	$\begin{bmatrix}50&36&40&40&40\end{bmatrix}$	7	3	$\begin{bmatrix}2&4&3&4&2\end{bmatrix}$	$\begin{bmatrix}3&5&3\end{bmatrix}$	$\begin{bmatrix}2&2&3\end{bmatrix}$	$\begin{bmatrix}60&60&50&70&60&60&70\\50&70&50&50&50&50&60\\60&60&50&70&70&50&50\end{bmatrix}$
3	$\begin{bmatrix}0&0&0&1&0\\0&0&0&0&1\\0&0&1&0&1\end{bmatrix}$	$\begin{bmatrix}45&61&99&45&45\end{bmatrix}$	7	3	$\begin{bmatrix}3&2&2&2&4\end{bmatrix}$	$\begin{bmatrix}3&5&4\end{bmatrix}$	$\begin{bmatrix}1&1&1\end{bmatrix}$	$\begin{bmatrix}120&120&120&140&120&120&100\\140&120&140&100&100&120&120\\140&140&140&140&100&120&140\end{bmatrix}$
4	$\begin{bmatrix}0&0&1&0&1\\1&1&1&0&1\\1&0&0&0&1\end{bmatrix}$	$\begin{bmatrix}22&34&34&34&22\end{bmatrix}$	7	3	$\begin{bmatrix}3&3&3&2&4\end{bmatrix}$	$\begin{bmatrix}3&4&5\end{bmatrix}$	$\begin{bmatrix}2&2&2\end{bmatrix}$	$\begin{bmatrix}300&250&300&250&350&250&300\\250&250&350&250&300&300&250\\250&250&300&300&300&300&300\end{bmatrix}$
5	$\begin{bmatrix}1&1&0&0&1\\1&1&1&1&0\\1&1&1&0&1\end{bmatrix}$	$\begin{bmatrix}68&68&68&47&97\end{bmatrix}$	7	3	$\begin{bmatrix}3&2&2&2&3\end{bmatrix}$	$\begin{bmatrix}4&4&4\end{bmatrix}$	$\begin{bmatrix}2&2&2\end{bmatrix}$	$\begin{bmatrix}420&350&420&420&350&490&350\\350&350&350&420&350&350&350\\490&420&350&420&420&420&420\end{bmatrix}$
6	$\begin{bmatrix}0&1&1&1&1\\1&1&0&1&0\\0&1&0&1&0\end{bmatrix}$	$\begin{bmatrix}7&41&7&7&7\end{bmatrix}$	7	3	$\begin{bmatrix}3&3&4&4&4\end{bmatrix}$	$\begin{bmatrix}3&3&4\end{bmatrix}$	$\begin{bmatrix}2&1&3\end{bmatrix}$	$\begin{bmatrix}700&700&600&600&500&600&500\\500&500&600&500&700&600&500\\500&700&500&500&500&500&600\end{bmatrix}$
7	$\begin{bmatrix}1&0&1&0&0&0&0&1&0&0\\0&0&1&1&1&0&0&1&1&1\\0&0&1&0&1&0&0&0&1&1\end{bmatrix}$	$\begin{bmatrix}44&65&45&65&45&45&44&45&45&65\end{bmatrix}$	7	3	$\begin{bmatrix}3&2&2&2&2&3&3&2&2&2\end{bmatrix}$	$\begin{bmatrix}3&3&3\end{bmatrix}$	$\begin{bmatrix}1&1&2\end{bmatrix}$	$\begin{bmatrix}500&500&500&600&700&500&600\\700&600&700&500&600&700&500\\500&500&600&500&600&600&500\end{bmatrix}$
8	$\begin{bmatrix}1&0&1&1&0&0&1&1&1&1\\0&1&1&1&1&1&1&0&1&1\\1&1&0&1&0&0&0&1&0&0\end{bmatrix}$	$\begin{bmatrix}66&66&66&66&66&66&66&66&66&21\end{bmatrix}$	7	3	$\begin{bmatrix}4&2&2&4&2&4&4&4&4&4\end{bmatrix}$	$\begin{bmatrix}3&4&4\end{bmatrix}$	$\begin{bmatrix}1&1&2\end{bmatrix}$	$\begin{bmatrix}1000&1400&1400&1200&1000&1400&1000\\1000&1200&1000&1200&1400&1200&1200\\1400&1400&1200&1000&1200&1200&1400\end{bmatrix}$
9	$\begin{bmatrix}1&0&1&0&0&1&1&0&1&1\\0&1&0&0&0&1&1&1&1&0\\0&1&0&1&0&0&0&1&1&1\end{bmatrix}$	$\begin{bmatrix}98&42&98&98&42&8&8&98&42&42\end{bmatrix}$	7	3	$\begin{bmatrix}4&3&3&3&4&3&2&3&4&4\end{bmatrix}$	$\begin{bmatrix}4&5&4\end{bmatrix}$	$\begin{bmatrix}1&2&3\end{bmatrix}$	$\begin{bmatrix}2100&2100&2100&2100&2100&2100&1500\\1500&1800&1500&1500&2100&1500&1800\\1500&2100&1800&1800&2100&1500&1500\end{bmatrix}$
10	$\begin{bmatrix}0&1&1&1&1&0&1&0&0&1\\1&1&0&1&1&0&1&1&1&1\\0&0&1&1&0&1&1&1&1&0\end{bmatrix}$	$\begin{bmatrix}97&18&18&6&97&18&97&6&97&18\end{bmatrix}$	7	3	$\begin{bmatrix}4&2&4&3&2&3&2&3&2&3\end{bmatrix}$	$\begin{bmatrix}4&3&3\end{bmatrix}$	$\begin{bmatrix}2&2&1\end{bmatrix}$	$\begin{bmatrix}2000&2000&2400&2000&2800&2400&2400\\2400&2800&2800&2000&2000&2400&2000\\2800&2400&2000&2400&2800&2800&2000\end{bmatrix}$
11	$\begin{bmatrix}0&1&0&0&0&0&1&1&1&1\\0&1&1&1&1&1&0&1&0&0\\1&1&0&1&1&0&0&1&0&0\end{bmatrix}$	$\begin{bmatrix}69&20&49&49&49&69&69&69&69&20\end{bmatrix}$	7	3	$\begin{bmatrix}2&2&2&3&3&2&3&2&2&4\end{bmatrix}$	$\begin{bmatrix}5&4&3\end{bmatrix}$	$\begin{bmatrix}1&2&1\end{bmatrix}$	$\begin{bmatrix}2500&2500&3000&3500&2500&3500&2500\\2500&3000&3500&3500&3000&3500&3000\\2500&3000&3500&3500&3000&3500&3500\end{bmatrix}$
12	$\begin{bmatrix}1&1&0&0&0&1&1&0&0&0\\0&1&1&1&1&1&1&1&0&0\\0&0&1&0&0&1&1&1&0&1\end{bmatrix}$	$\begin{bmatrix}68&62&62&62&68&37&62&62&68&62\end{bmatrix}$	7	3	$\begin{bmatrix}4&2&4&4&2&4&3&3&4&2\end{bmatrix}$	$\begin{bmatrix}5&4&4\end{bmatrix}$	$\begin{bmatrix}2&3&1\end{bmatrix}$	$\begin{bmatrix}5000&7000&7000&5000&5000&6000&5000\\7000&6000&5000&6000&5000&5000&6000\\7000&5000&6000&5000&6000&7000&5000\end{bmatrix}$
13	$\begin{bmatrix}1&0&0&0&1&1&1&0&1&0\\0&0&0&1&0&0&1&0&1&0\\0&0&1&0&0&0&1&1&1&1\end{bmatrix}$	$\begin{bmatrix}64&64&38&38&38&80&64&64&64&64\end{bmatrix}$	7	3	$\begin{bmatrix}3&4&2&4&4&3&4&2&4&2\end{bmatrix}$	$\begin{bmatrix}3&5&5\end{bmatrix}$	$\begin{bmatrix}3&1&3\end{bmatrix}$	$\begin{bmatrix}10500&10500&7500&10500&7500&9000&7500\\9000&7500&10500&7500&7500&9000&7500\\7500&10500&7500&7500&9000&10500&7500\end{bmatrix}$
14	$\begin{bmatrix}0&1&0&1&1&1&0&1&1&1\\0&1&1&1&1&1&1&1&1&1\\0&0&0&1&0&0&1&1&0&1\end{bmatrix}$	$\begin{bmatrix}17&29&29&17&56&29&56&56&29&56\end{bmatrix}$	7	3	$\begin{bmatrix}2&2&4&4&4&3&3&2&4&4\end{bmatrix}$	$\begin{bmatrix}5&4&4\end{bmatrix}$	$\begin{bmatrix}3&3&1\end{bmatrix}$	$\begin{bmatrix}14000&10000&14000&10000&12000&12000&12000\\10000&12000&14000&12000&10000&10000&10000\\14000&10000&14000&14000&10000&12000&14000\end{bmatrix}$
15	$\begin{bmatrix}1&0&0&1&0&0&1&1&1&0\\1&1&1&0&0&0&1&0&0&1\\1&1&0&0&1&1&0&1&1&0\end{bmatrix}$	$\begin{bmatrix}25&25&25&94&97&97&97&97&94&97\end{bmatrix}$	7	3	$\begin{bmatrix}2&4&2&2&4&2&2&4&4&3\end{bmatrix}$	$\begin{bmatrix}5&4&4\end{bmatrix}$	$\begin{bmatrix}1&1&1\end{bmatrix}$	$\begin{bmatrix}17500&15000&12500&17500&17500&15000&12500\\15000&15000&17500&15000&12500&15000&17500\\17500&15000&15000&12500&15000&15000&15000\end{bmatrix}$
16	$\begin{bmatrix}1&1&1&1&0&1&1&0&0&0\\0&0&1&0&1&0&0&0&1&1\\1&1&1&0&0&1&1&0&1&0\end{bmatrix}$	$\begin{bmatrix}17&17&66&66&17&46&66&46&66&66\end{bmatrix}$	7	3	$\begin{bmatrix}2&2&4&2&3&2&3&3&4&3\end{bmatrix}$	$\begin{bmatrix}4&5&3\end{bmatrix}$	$\begin{bmatrix}1&2&1\end{bmatrix}$	$\begin{bmatrix}15000&21000&21000&18000&21000&21000&18000\\18000&15000&15000&18000&18000&21000&18000\\18000&18000&18000&21000&18000&15000&21000\end{bmatrix}$
17	$\begin{bmatrix}0&1&0&0&1&1&0&0&1&1\\0&1&1&0&1&0&1&1&0&1\\0&1&1&1&1&0&1&0&1&0\end{bmatrix}$	$\begin{bmatrix}67&48&63&67&67&63&63&63&63&67\end{bmatrix}$	7	3	$\begin{bmatrix}2&4&2&2&2&4&4&2&4&3\end{bmatrix}$	$\begin{bmatrix}5&4&3\end{bmatrix}$	$\begin{bmatrix}2&3&2\end{bmatrix}$	$\begin{bmatrix}21000&24500&24500&24500&17500&24500&24500\\24500&17500&21000&21000&21000&17500&24500\\21000&17500&17500&24500&21000&17500&17500\end{bmatrix}$
18	$\begin{bmatrix}1&0&1&1&0&1&1&0&0&1\\0&0&1&0&0&1&1&1&0&0\\1&0&1&1&1&0&1&0&1&0\end{bmatrix}$	$\begin{bmatrix}73&73&73&73&26&58&58&73&73&58\end{bmatrix}$	7	3	$\begin{bmatrix}3&2&3&3&4&3&4&3&3&4\end{bmatrix}$	$\begin{bmatrix}4&5&4\end{bmatrix}$	$\begin{bmatrix}2&2&2\end{bmatrix}$	$\begin{bmatrix}20000&28000&28000&28000&28000&24000&28000\\24000&20000&20000&20000&28000&24000&28000\\28000&24000&28000&20000&24000&28000&24000\end{bmatrix}$
19	$\begin{bmatrix}1&1&0&1&0&0&0&0&0&1\\0&1&0&1&1&0&0&1&1&0\\0&0&0&1&0&1&1&1&0&0\end{bmatrix}$	$\begin{bmatrix}40&77&77&40&40&77&40&77&40&16\end{bmatrix}$	7	3	$\begin{bmatrix}2&2&2&4&3&3&4&4&4&3\end{bmatrix}$	$\begin{bmatrix}3&4&3\end{bmatrix}$	$\begin{bmatrix}1&3&2\end{bmatrix}$	$\begin{bmatrix}22500&22500&31500&31500&27000&27000&22500\\27000&22500&22500&31500&27000&27000&22500\\22500&27000&22500&27000&22500&27000&27000\end{bmatrix}$
20	$\begin{bmatrix}0&1&0&1&0&1&1&1&1&0\\1&0&0&0&0&1&1&1&0&1\\0&1&0&0&1&0&1&1&0&1\end{bmatrix}$	$\begin{bmatrix}22&22&22&10&52&22&22&22&10&52\end{bmatrix}$	7	3	$\begin{bmatrix}4&2&4&3&3&2&2&4&2&4\end{bmatrix}$	$\begin{bmatrix}5&5&3\end{bmatrix}$	$\begin{bmatrix}3&2&1\end{bmatrix}$	$\begin{bmatrix}30000&35000&35000&30000&30000&30000&30000\\30000&25000&25000&35000&25000&35000&25000\\30000&25000&25000&25000&35000&30000&30000\end{bmatrix}$
21	$\begin{bmatrix}1&0&1&0&0&1&1&1&0&0\\1&0&1&1&1&0&0&1&0&0\\0&0&0&1&0&0&1&0&0&1\end{bmatrix}$	$\begin{bmatrix}22&41&21&41&41&41&41&21&21&21\end{bmatrix}$	7	3	$\begin{bmatrix}4&2&3&3&3&2&2&3&3&3\end{bmatrix}$	$\begin{bmatrix}3&5&5\end{bmatrix}$	$\begin{bmatrix}2&2&1\end{bmatrix}$	$\begin{bmatrix}27500&27500&27500&33000&33000&27500&38500\\27500&33000&33000&33000&38500&38500&33000\\38500&27500&38500&33000&38500&38500&38500\end{bmatrix}$

Continues on the next page

Table 21 – Parameters and Values Used for Test Instance in Case of Rich Channel Capacity

Instance	q_{ic}	r_c	b	k	l_c	m_i	n_i	t_{hd}
22	$\begin{bmatrix}0&0&1&1&1&1&1&0&1&1\\1&1&1&0&0&1&1&1&0&1\\0&1&0&1&1&0&1&0&0&1\end{bmatrix}$	$\begin{bmatrix}49&49&18&18&18&18&36&49&18&18\end{bmatrix}$	7	3	$\begin{bmatrix}4&2&2&4&2&4&2&4&2&2\end{bmatrix}$	$\begin{bmatrix}3&3&4\end{bmatrix}$	$\begin{bmatrix}2&3&3\end{bmatrix}$	$\begin{bmatrix}42000&42000&36000&30000&30000&36000&36000\\36000&36000&30000&30000&36000&30000&30000\\36000&36000&42000&30000&36000&42000&42000\end{bmatrix}$
23	$\begin{bmatrix}0&0&1&1&1&0&0&0&0&1\\0&0&0&1&1&0&0&1&0&1\\1&1&1&1&1&0&1&1&0&1\end{bmatrix}$	$\begin{bmatrix}46&20&13&13&20&46&20&20&13&20\end{bmatrix}$	7	3	$\begin{bmatrix}4&2&2&2&2&4&3&2&4&3\end{bmatrix}$	$\begin{bmatrix}3&5&5\end{bmatrix}$	$\begin{bmatrix}2&2&1\end{bmatrix}$	$\begin{bmatrix}45500&39000&45500&32500&45500&32500&45500\\39000&45500&45500&32500&45500&39000&45500\\32500&45500&39000&45500&39000&39000&32500\end{bmatrix}$
24	$\begin{bmatrix}0&1&0&0&1&0&1&0&0&0\\1&1&1&0&1&0&1&0&1&0\\1&1&1&0&1&0&1&0&1&0\end{bmatrix}$	$\begin{bmatrix}78&35&80&78&35&78&35&80&78&80\end{bmatrix}$	7	3	$\begin{bmatrix}2&3&3&2&4&3&4&4&2&2\end{bmatrix}$	$\begin{bmatrix}3&5&3\end{bmatrix}$	$\begin{bmatrix}3&3&3\end{bmatrix}$	$\begin{bmatrix}49000&49000&49000&49000&35000&42000&42000\\42000&49000&35000&35000&42000&35000&35000\\42000&42000&42000&35000&42000&35000&49000\end{bmatrix}$
25	$\begin{bmatrix}1&0&1&0&0&1&0&1&0&1\\0&1&1&1&0&1&0&0&1&1\\0&0&0&1&1&1&0&1&0&0\end{bmatrix}$	$\begin{bmatrix}66&67&1&67&1&67&67&1&1&1\end{bmatrix}$	7	3	$\begin{bmatrix}4&2&4&2&2&2&2&3&4&3\end{bmatrix}$	$\begin{bmatrix}3&5&3\end{bmatrix}$	$\begin{bmatrix}1&2&3\end{bmatrix}$	$\begin{bmatrix}37500&37500&45000&52500&37500&52500&52500\\37500&52500&45000&52500&37500&37500&52500\\37500&52500&37500&45000&37500&45000&52500\end{bmatrix}$
26	$\begin{bmatrix}0&0&1&0&1&0&0&0&0&0\\0&0&1&0&0&1&0&0&1&1\\0&1&1&0&1&1&1&0&0&1\end{bmatrix}$	$\begin{bmatrix}92&0&92&0&92&92&92&92&0&92\end{bmatrix}$	7	3	$\begin{bmatrix}3&2&2&2&2&4&4&3&3&2\end{bmatrix}$	$\begin{bmatrix}5&5&5\end{bmatrix}$	$\begin{bmatrix}3&3&3\end{bmatrix}$	$\begin{bmatrix}56000&40000&48000&40000&40000&48000&56000\\40000&56000&48000&56000&56000&48000&48000\\40000&48000&40000&48000&48000&56000&48000\end{bmatrix}$
27	$\begin{bmatrix}1&1&1&0&0&0&1&1&0&1\\0&0&0&0&1&0&1&1&0&1\\1&1&0&1&1&0&0&0&1&1\end{bmatrix}$	$\begin{bmatrix}23&92&23&4&23&92&92&23&23&23\end{bmatrix}$	7	3	$\begin{bmatrix}2&3&3&2&3&2&4&4&4&3\end{bmatrix}$	$\begin{bmatrix}4&4&5\end{bmatrix}$	$\begin{bmatrix}2&1&1\end{bmatrix}$	$\begin{bmatrix}59500&42500&51000&51000&51000&42500&51000\\59500&42500&51000&59500&51000&51000&59500\\51000&59500&42500&42500&51000&59500&42500\end{bmatrix}$
28	$\begin{bmatrix}1&0&0&0&0&1&1&0&1&0\\1&0&0&0&1&1&1&1&0&1\\0&0&0&1&1&0&1&1&1&0\end{bmatrix}$	$\begin{bmatrix}71&17&71&45&17&45&17&71&17&17\end{bmatrix}$	7	3	$\begin{bmatrix}4&4&2&2&3&2&4&2&4&4\end{bmatrix}$	$\begin{bmatrix}3&3&5\end{bmatrix}$	$\begin{bmatrix}3&1&1\end{bmatrix}$	$\begin{bmatrix}63000&45000&63000&45000&54000&54000&63000\\63000&45000&63000&63000&54000&54000&54000\\45000&63000&54000&63000&63000&54000&45000\end{bmatrix}$
29	$\begin{bmatrix}0&1&0&1&0&0&1&0&1&1\\1&1&0&1&1&1&1&0&1&1\\1&0&0&0&0&0&1&0&0&1\end{bmatrix}$	$\begin{bmatrix}56&14&14&49&56&14&14&14&56&56\end{bmatrix}$	7	3	$\begin{bmatrix}4&3&3&2&3&4&2&4&2&3\end{bmatrix}$	$\begin{bmatrix}4&3&5\end{bmatrix}$	$\begin{bmatrix}2&2&3\end{bmatrix}$	$\begin{bmatrix}66500&47500&66500&57000&66500&66500&47500\\47500&57000&57000&66500&66500&57000&57000\\66500&47500&66500&66500&47500&66500&66500\end{bmatrix}$
30	$\begin{bmatrix}1&1&0&1&0&1&1&1&1&1\\0&1&1&0&1&1&0&0&1&1\\1&1&0&0&0&1&1&1&1&0\end{bmatrix}$	$\begin{bmatrix}76&76&33&76&33&33&18&33&18&76\end{bmatrix}$	7	3	$\begin{bmatrix}4&3&2&4&2&2&2&3&2&2\end{bmatrix}$	$\begin{bmatrix}5&5&3\end{bmatrix}$	$\begin{bmatrix}3&2&2\end{bmatrix}$	$\begin{bmatrix}50000&60000&60000&70000&50000&60000&50000\\60000&70000&70000&50000&50000&60000&50000\\50000&60000&60000&70000&50000&50000&60000\end{bmatrix}$
31	$\begin{bmatrix}1&1&0&1&0&1&1&1&1&1\\0&1&1&0&1&1&0&0&1&1\\1&1&0&0&0&1&1&1&1&0\end{bmatrix}$	$\begin{bmatrix}76&76&33&76&33&33&18&33&18&76\end{bmatrix}$	7	3	$\begin{bmatrix}4&3&2&4&2&2&2&3&2&2\end{bmatrix}$	$\begin{bmatrix}5&5&3\end{bmatrix}$	$\begin{bmatrix}3&2&2\end{bmatrix}$	$\begin{bmatrix}50000&60000&60000&70000&50000&60000&50000\\60000&70000&70000&50000&50000&60000&50000\\50000&60000&60000&70000&50000&50000&60000\end{bmatrix}$
32	$\begin{bmatrix}1&1&0&1&0&0&0&1&0&0\\0&0&1&0&0&0&0&0&1&0\\1&0&1&0&1&1&1&0&0&1\end{bmatrix}$	$\begin{bmatrix}12&24&0&12&12&24&24&24&12&0\end{bmatrix}$	7	3	$\begin{bmatrix}2&3&3&4&2&2&3&2&3&2\end{bmatrix}$	$\begin{bmatrix}3&4&5\end{bmatrix}$	$\begin{bmatrix}2&3&3\end{bmatrix}$	$\begin{bmatrix}75000&75000&75000&90000&105000&75000&90000\\75000&90000&105000&90000&105000&75000&90000\\90000&90000&105000&90000&75000&105000&105000\end{bmatrix}$
33	$\begin{bmatrix}1&1&1&0&0&1&1&0&1&1\\1&1&1&0&0&0&1&0&1&0\\1&1&0&0&1&0&1&0&0&1\end{bmatrix}$	$\begin{bmatrix}2&83&2&2&42&42&42&83&83&2\end{bmatrix}$	7	3	$\begin{bmatrix}2&4&3&3&2&2&3&3&2&2\end{bmatrix}$	$\begin{bmatrix}4&3&4\end{bmatrix}$	$\begin{bmatrix}1&3&2\end{bmatrix}$	$\begin{bmatrix}100000&140000&140000&120000&100000&120000&100000\\120000&140000&120000&120000&120000&120000&100000\\120000&140000&100000&140000&120000&140000&120000\end{bmatrix}$
34	$\begin{bmatrix}1&1&0&1&0&0&1&1&1&0\\1&0&1&0&0&0&1&1&1&1\\0&1&0&1&0&1&0&1&1&0\end{bmatrix}$	$\begin{bmatrix}30&15&15&15&15&15&15&30&83&15\end{bmatrix}$	7	3	$\begin{bmatrix}2&4&4&2&4&3&2&3&4&2\end{bmatrix}$	$\begin{bmatrix}4&3&5\end{bmatrix}$	$\begin{bmatrix}1&1&1\end{bmatrix}$	$\begin{bmatrix}125000&150000&150000&125000&175000&125000&125000\\150000&175000&175000&125000&150000&150000&175000\\175000&150000&150000&175000&125000&175000&150000\end{bmatrix}$
35	$\begin{bmatrix}1&0&1&0&1&0&1&0&0&1\\1&1&1&1&0&0&0&0&0&1\\0&0&1&1&1&0&0&0&0&0\end{bmatrix}$	$\begin{bmatrix}43&66&66&43&99&43&43&43&99&66\end{bmatrix}$	7	3	$\begin{bmatrix}4&4&4&2&4&4&2&3&2&4\end{bmatrix}$	$\begin{bmatrix}3&4&5\end{bmatrix}$	$\begin{bmatrix}2&3&2\end{bmatrix}$	$\begin{bmatrix}150000&210000&210000&150000&180000&150000&150000\\210000&210000&150000&180000&180000&180000&210000\\150000&150000&180000&180000&150000&210000&180000\end{bmatrix}$
36	$\begin{bmatrix}0&1&0&1&0&1&0&0&1&0\\1&0&0&0&1&0&1&1&1&1\\1&0&0&0&0&1&0&0&0&1\end{bmatrix}$	$\begin{bmatrix}37&37&58&35&35&37&37&58&37&58\end{bmatrix}$	7	3	$\begin{bmatrix}4&4&4&2&2&3&4&2&3&4\end{bmatrix}$	$\begin{bmatrix}5&5&5\end{bmatrix}$	$\begin{bmatrix}3&1&3\end{bmatrix}$	$\begin{bmatrix}245000&210000&210000&210000&175000&210000&245000\\210000&175000&175000&210000&210000&245000&210000\\175000&245000&245000&210000&175000&245000&245000\end{bmatrix}$
37	$\begin{bmatrix}1&1&0&0&0&1&0&0&1&0\\1&0&1&1&1&1&1&1&1&1\\1&0&1&0&1&0&1&1&1&0\end{bmatrix}$	$\begin{bmatrix}4&4&45&4&4&45&45&63&63&45\end{bmatrix}$	7	3	$\begin{bmatrix}3&3&3&4&2&2&4&4&4&2\end{bmatrix}$	$\begin{bmatrix}4&5&5\end{bmatrix}$	$\begin{bmatrix}1&1&2\end{bmatrix}$	$\begin{bmatrix}280000&280000&200000&240000&200000&240000&240000\\240000&280000&280000&280000&280000&280000&240000\\200000&240000&240000&240000&280000&200000&280000\end{bmatrix}$
38	$\begin{bmatrix}1&0&1&0&0&0&1&1&1&1\\1&0&1&0&0&1&0&1&1&1\\0&1&0&1&1&1&0&1&0&1\end{bmatrix}$	$\begin{bmatrix}77&43&2&2&43&43&2&43&2&43\end{bmatrix}$	7	3	$\begin{bmatrix}2&4&3&2&3&3&3&3&2&3\end{bmatrix}$	$\begin{bmatrix}5&5&5\end{bmatrix}$	$\begin{bmatrix}2&3&1\end{bmatrix}$	$\begin{bmatrix}270000&315000&270000&315000&225000&270000&270000\\225000&315000&225000&225000&315000&270000&225000\\225000&315000&270000&270000&225000&270000&225000\end{bmatrix}$
39	$\begin{bmatrix}1&1&1&0&0&0&1&0&1&0\\0&0&0&1&0&1&1&1&0&0\\0&0&1&0&0&1&0&0&1&1\end{bmatrix}$	$\begin{bmatrix}21&19&77&77&19&21&19&77&19&77\end{bmatrix}$	7	3	$\begin{bmatrix}3&2&2&4&3&2&3&2&3&2\end{bmatrix}$	$\begin{bmatrix}5&4&3\end{bmatrix}$	$\begin{bmatrix}2&2&2\end{bmatrix}$	$\begin{bmatrix}350000&250000&350000&250000&300000&350000&300000\\250000&250000&250000&350000&300000&350000&350000\\350000&250000&350000&350000&350000&300000&300000\end{bmatrix}$
40	$\begin{bmatrix}1&1&0&1&0&1&0&1&0&1\\1&0&1&0&0&1&0&1&0&1\\0&1&0&0&1&1&1&0&0&1\end{bmatrix}$	$\begin{bmatrix}37&60&37&37&60&67&60&67&67&60\end{bmatrix}$	7	3	$\begin{bmatrix}2&2&3&4&3&3&3&2&2&3\end{bmatrix}$	$\begin{bmatrix}3&4&4\end{bmatrix}$	$\begin{bmatrix}2&3&2\end{bmatrix}$	$\begin{bmatrix}330000&330000&330000&275000&385000&385000&330000\\275000&330000&330000&385000&275000&385000&330000\\330000&385000&330000&330000&385000&330000&275000\end{bmatrix}$
41	$\begin{bmatrix}0&1&1&0&0&0&0&0&0&1\\1&1&1&1&1&1&0&1&1&0\\0&1&1&1&1&1&0&1&0&1\end{bmatrix}$	$\begin{bmatrix}38&38&38&54&54&54&54&54&92&54\end{bmatrix}$	7	3	$\begin{bmatrix}3&3&4&3&4&3&3&3&3&4\end{bmatrix}$	$\begin{bmatrix}4&5&5\end{bmatrix}$	$\begin{bmatrix}3&2&2\end{bmatrix}$	$\begin{bmatrix}360000&360000&360000&300000&360000&420000&300000\\360000&420000&300000&420000&420000&420000&300000\\360000&360000&300000&420000&420000&360000&420000\end{bmatrix}$
42	$\begin{bmatrix}0&0&0&1&0&0&1&0&1&1\\1&0&0&0&1&1&0&1&0&1\\1&0&0&0&0&1&0&0&1&1\end{bmatrix}$	$\begin{bmatrix}42&42&42&76&42&42&71&71&42&71\end{bmatrix}$	7	3	$\begin{bmatrix}2&3&3&3&2&3&4&4&4&3\end{bmatrix}$	$\begin{bmatrix}3&3&4\end{bmatrix}$	$\begin{bmatrix}3&1&3\end{bmatrix}$	$\begin{bmatrix}325000&325000&325000&390000&455000&455000&325000\\325000&390000&390000&390000&455000&325000&390000\\455000&390000&455000&455000&390000&390000&455000\end{bmatrix}$
43	$\begin{bmatrix}1&0&0&0&0&1&0&1&0&0\\0&1&1&1&0&1&0&1&0&1\\0&0&0&0&0&1&1&1&0&0\end{bmatrix}$	$\begin{bmatrix}53&53&57&57&53&57&20&20&57&53\end{bmatrix}$	7	3	$\begin{bmatrix}4&2&2&4&2&2&4&4&4&3\end{bmatrix}$	$\begin{bmatrix}4&5&4\end{bmatrix}$	$\begin{bmatrix}1&3&2\end{bmatrix}$	$\begin{bmatrix}350000&420000&420000&490000&490000&350000&420000\\350000&350000&420000&490000&420000&350000&420000\\350000&350000&350000&490000&490000&490000&420000\end{bmatrix}$
44	$\begin{bmatrix}0&1&0&1&1&0&1&1&0&0\\1&1&0&0&0&1&1&0&0&0\\0&1&1&0&1&0&1&1&1&1\end{bmatrix}$	$\begin{bmatrix}39&39&2&2&97&2&39&97&2&97\end{bmatrix}$	7	3	$\begin{bmatrix}2&4&2&2&4&2&3&4&3&3\end{bmatrix}$	$\begin{bmatrix}5&3&4\end{bmatrix}$	$\begin{bmatrix}3&1&3\end{bmatrix}$	$\begin{bmatrix}375000&525000&450000&450000&525000&525000&450000\\450000&525000&450000&525000&450000&375000&375000\\525000&450000&450000&450000&525000&525000&450000\end{bmatrix}$

Continues on the next page

Table 21 – Parameters and Values Used for Test Instance in Case of Rich Channel Capacity

Instance	q_{ic}	r_c	b	k	l_c	m_i	n_i	t_{hd}
45	$\begin{bmatrix} 1\ 0\ 0\ 0\ 1\ 0\ 1\ 1\ 0\ 1 \\ 0\ 1\ 0\ 1\ 0\ 0\ 1\ 1\ 1\ 1 \\ 1\ 1\ 0\ 1\ 0\ 0\ 1\ 0\ 1\ 0 \end{bmatrix}$	$\begin{bmatrix} 32\ 32\ 37\ 37\ 85\ 85\ 32\ 37\ 37\ 32 \end{bmatrix}$	7	3	$\begin{bmatrix} 3\ 2\ 2\ 3\ 4\ 4\ 4\ 3\ 3\ 3 \end{bmatrix}$	$\begin{bmatrix} 3\ 4\ 4 \end{bmatrix}$	$\begin{bmatrix} 1\ 3\ 3 \end{bmatrix}$	$\begin{bmatrix} 560000\ 560000\ 400000\ 480000\ 560000\ 480000\ 480000 \\ 480000\ 560000\ 480000\ 560000\ 560000\ 480000\ 560000 \\ 400000\ 400000\ 560000\ 480000\ 400000\ 400000\ 560000 \end{bmatrix}$

A.3 Parameter Values for the Test Instances with Network Effect

Table 22: Parameters and values used in the test instances with network effect.

Instance	q_{ic}	r_c	b	k	l_c	m_i	n_i	t_{hd}
1	$\begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 62 & 49 \end{bmatrix}$	7	3	$\begin{bmatrix} 3 & 3 \end{bmatrix}$	$\begin{bmatrix} 4 & 3 & 3 \end{bmatrix}$	$\begin{bmatrix} 3 & 2 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 & 2 & 3 & 3 & 1 & 1 \\ 1 & 2 & 1 & 2 & 3 & 2 & 1 \\ 3 & 2 & 2 & 3 & 3 & 2 & 3 \end{bmatrix}$
2	$\begin{bmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$	$\begin{bmatrix} 50 & 36 & 40 & 40 & 40 \end{bmatrix}$	7	3	$\begin{bmatrix} 2 & 4 & 3 & 4 & 2 \end{bmatrix}$	$\begin{bmatrix} 3 & 5 & 3 \end{bmatrix}$	$\begin{bmatrix} 2 & 2 & 3 \end{bmatrix}$	$\begin{bmatrix} 2 & 2 & 3 & 1 & 2 & 2 & 1 \\ 3 & 1 & 3 & 3 & 3 & 3 & 2 \\ 2 & 2 & 3 & 1 & 1 & 3 & 3 \end{bmatrix}$
3	$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 45 & 61 & 99 & 45 & 45 \end{bmatrix}$	7	3	$\begin{bmatrix} 3 & 2 & 2 & 2 & 4 \end{bmatrix}$	$\begin{bmatrix} 3 & 5 & 4 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$	$\begin{bmatrix} 4 & 4 & 4 & 2 & 4 & 4 & 6 \\ 2 & 4 & 2 & 6 & 6 & 4 & 4 \\ 2 & 2 & 2 & 2 & 6 & 4 & 2 \end{bmatrix}$
4	$\begin{bmatrix} 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 22 & 34 & 34 & 34 & 22 \end{bmatrix}$	7	3	$\begin{bmatrix} 3 & 3 & 3 & 2 & 4 \end{bmatrix}$	$\begin{bmatrix} 3 & 4 & 5 \end{bmatrix}$	$\begin{bmatrix} 2 & 2 & 2 \end{bmatrix}$	$\begin{bmatrix} 10 & 15 & 10 & 15 & 5 & 15 & 10 \\ 15 & 15 & 5 & 15 & 10 & 10 & 15 \\ 15 & 15 & 10 & 10 & 10 & 10 & 10 \end{bmatrix}$
5	$\begin{bmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 68 & 68 & 68 & 47 & 97 \end{bmatrix}$	7	3	$\begin{bmatrix} 3 & 2 & 2 & 2 & 3 \end{bmatrix}$	$\begin{bmatrix} 4 & 4 & 4 \end{bmatrix}$	$\begin{bmatrix} 2 & 2 & 2 \end{bmatrix}$	$\begin{bmatrix} 14 & 21 & 14 & 14 & 21 & 7 & 21 \\ 21 & 21 & 21 & 14 & 21 & 21 & 21 \\ 7 & 14 & 21 & 14 & 14 & 14 & 14 \end{bmatrix}$
6	$\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 7 & 41 & 7 & 7 & 7 \end{bmatrix}$	7	3	$\begin{bmatrix} 3 & 3 & 4 & 4 & 4 \end{bmatrix}$	$\begin{bmatrix} 3 & 3 & 4 \end{bmatrix}$	$\begin{bmatrix} 2 & 1 & 3 \end{bmatrix}$	$\begin{bmatrix} 10 & 10 & 20 & 20 & 30 & 20 & 30 \\ 30 & 30 & 20 & 30 & 10 & 20 & 30 \\ 30 & 10 & 30 & 30 & 30 & 30 & 20 \end{bmatrix}$
7	$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}$	$\begin{bmatrix} 44 & 65 & 45 & 65 & 45 & 45 & 44 & 45 & 45 & 65 \end{bmatrix}$	7	3	$\begin{bmatrix} 3 & 2 & 2 & 2 & 2 & 3 & 3 & 2 & 2 & 2 \end{bmatrix}$	$\begin{bmatrix} 3 & 3 & 3 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 & 2 \end{bmatrix}$	$\begin{bmatrix} 30 & 30 & 30 & 20 & 10 & 30 & 20 \\ 10 & 20 & 10 & 30 & 20 & 10 & 30 \\ 30 & 30 & 20 & 30 & 20 & 20 & 30 \end{bmatrix}$
8	$\begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 66 & 66 & 66 & 66 & 66 & 66 & 66 & 66 & 66 & 21 \end{bmatrix}$	7	3	$\begin{bmatrix} 4 & 2 & 2 & 4 & 2 & 4 & 4 & 4 & 4 & 4 \end{bmatrix}$	$\begin{bmatrix} 3 & 4 & 4 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 & 2 \end{bmatrix}$	$\begin{bmatrix} 60 & 20 & 20 & 40 & 60 & 20 & 60 \\ 60 & 40 & 60 & 40 & 20 & 40 & 40 \\ 20 & 20 & 40 & 60 & 40 & 40 & 20 \end{bmatrix}$
9	$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$	$\begin{bmatrix} 98 & 42 & 98 & 98 & 42 & 8 & 8 & 98 & 42 & 42 \end{bmatrix}$	7	3	$\begin{bmatrix} 4 & 3 & 3 & 3 & 4 & 3 & 2 & 3 & 4 & 4 \end{bmatrix}$	$\begin{bmatrix} 4 & 5 & 4 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$	$\begin{bmatrix} 30 & 30 & 30 & 30 & 30 & 30 & 90 \\ 90 & 60 & 90 & 90 & 30 & 90 & 60 \\ 90 & 30 & 60 & 60 & 30 & 90 & 90 \end{bmatrix}$
10	$\begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 97 & 18 & 18 & 6 & 97 & 18 & 97 & 6 & 97 & 18 \end{bmatrix}$	7	3	$\begin{bmatrix} 4 & 2 & 4 & 3 & 2 & 3 & 2 & 3 & 2 & 3 \end{bmatrix}$	$\begin{bmatrix} 4 & 3 & 3 \end{bmatrix}$	$\begin{bmatrix} 2 & 2 & 1 \end{bmatrix}$	$\begin{bmatrix} 120 & 120 & 80 & 120 & 40 & 80 & 80 \\ 80 & 40 & 40 & 120 & 120 & 80 & 120 \\ 40 & 80 & 120 & 80 & 40 & 40 & 120 \end{bmatrix}$
11	$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 69 & 20 & 49 & 49 & 49 & 69 & 69 & 69 & 69 & 20 \end{bmatrix}$	7	3	$\begin{bmatrix} 2 & 2 & 2 & 3 & 3 & 2 & 3 & 2 & 2 & 4 \end{bmatrix}$	$\begin{bmatrix} 5 & 4 & 3 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$	$\begin{bmatrix} 150 & 150 & 100 & 50 & 150 & 50 & 150 \\ 150 & 100 & 50 & 50 & 100 & 50 & 100 \\ 150 & 100 & 50 & 50 & 100 & 50 & 50 \end{bmatrix}$
12	$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 68 & 62 & 62 & 62 & 68 & 37 & 62 & 62 & 68 & 62 \end{bmatrix}$	7	3	$\begin{bmatrix} 4 & 2 & 4 & 4 & 2 & 4 & 3 & 3 & 4 & 2 \end{bmatrix}$	$\begin{bmatrix} 5 & 4 & 4 \end{bmatrix}$	$\begin{bmatrix} 2 & 3 & 1 \end{bmatrix}$	$\begin{bmatrix} 300 & 100 & 100 & 300 & 300 & 200 & 300 \\ 100 & 200 & 300 & 200 & 300 & 300 & 200 \\ 100 & 300 & 200 & 300 & 200 & 100 & 300 \end{bmatrix}$
13	$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$	$\begin{bmatrix} 64 & 64 & 38 & 38 & 38 & 80 & 64 & 64 & 64 & 64 \end{bmatrix}$	7	3	$\begin{bmatrix} 3 & 4 & 2 & 4 & 4 & 3 & 4 & 2 & 4 & 2 \end{bmatrix}$	$\begin{bmatrix} 3 & 5 & 5 \end{bmatrix}$	$\begin{bmatrix} 3 & 1 & 3 \end{bmatrix}$	$\begin{bmatrix} 150 & 150 & 450 & 150 & 450 & 300 & 450 \\ 300 & 450 & 150 & 450 & 450 & 300 & 450 \\ 450 & 150 & 450 & 450 & 300 & 150 & 450 \end{bmatrix}$
14	$\begin{bmatrix} 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 17 & 29 & 29 & 17 & 56 & 29 & 56 & 56 & 29 & 56 \end{bmatrix}$	7	3	$\begin{bmatrix} 2 & 2 & 4 & 4 & 4 & 3 & 3 & 2 & 4 & 4 \end{bmatrix}$	$\begin{bmatrix} 5 & 4 & 4 \end{bmatrix}$	$\begin{bmatrix} 3 & 3 & 1 \end{bmatrix}$	$\begin{bmatrix} 200 & 600 & 200 & 600 & 400 & 400 & 400 \\ 600 & 400 & 200 & 400 & 600 & 600 & 600 \\ 200 & 600 & 200 & 200 & 600 & 400 & 200 \end{bmatrix}$
15	$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 25 & 25 & 25 & 94 & 97 & 97 & 97 & 97 & 94 & 97 \end{bmatrix}$	7	3	$\begin{bmatrix} 2 & 4 & 2 & 2 & 4 & 2 & 2 & 4 & 4 & 3 \end{bmatrix}$	$\begin{bmatrix} 5 & 4 & 4 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$	$\begin{bmatrix} 250 & 500 & 750 & 250 & 250 & 500 & 750 \\ 500 & 500 & 250 & 500 & 750 & 500 & 250 \\ 250 & 500 & 500 & 750 & 500 & 500 & 500 \end{bmatrix}$
16	$\begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 17 & 17 & 66 & 66 & 17 & 46 & 66 & 46 & 66 & 66 \end{bmatrix}$	7	3	$\begin{bmatrix} 2 & 2 & 4 & 2 & 3 & 2 & 3 & 3 & 4 & 3 \end{bmatrix}$	$\begin{bmatrix} 4 & 5 & 3 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$	$\begin{bmatrix} 900 & 300 & 300 & 600 & 300 & 300 & 600 \\ 600 & 900 & 900 & 600 & 600 & 300 & 600 \\ 600 & 600 & 600 & 300 & 600 & 900 & 300 \end{bmatrix}$
17	$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 67 & 48 & 63 & 67 & 67 & 63 & 63 & 63 & 63 & 67 \end{bmatrix}$	7	3	$\begin{bmatrix} 2 & 4 & 2 & 2 & 2 & 4 & 4 & 2 & 4 & 3 \end{bmatrix}$	$\begin{bmatrix} 5 & 4 & 3 \end{bmatrix}$	$\begin{bmatrix} 2 & 3 & 2 \end{bmatrix}$	$\begin{bmatrix} 700 & 350 & 350 & 350 & 1050 & 350 & 350 \\ 350 & 1050 & 700 & 700 & 700 & 1050 & 350 \\ 700 & 1050 & 1050 & 350 & 700 & 1050 & 1050 \end{bmatrix}$
18	$\begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 73 & 73 & 73 & 73 & 26 & 58 & 58 & 73 & 73 & 58 \end{bmatrix}$	7	3	$\begin{bmatrix} 3 & 2 & 3 & 3 & 4 & 3 & 4 & 3 & 3 & 4 \end{bmatrix}$	$\begin{bmatrix} 4 & 5 & 4 \end{bmatrix}$	$\begin{bmatrix} 2 & 2 & 2 \end{bmatrix}$	$\begin{bmatrix} 1200 & 400 & 400 & 400 & 400 & 800 & 400 \\ 800 & 1200 & 1200 & 1200 & 400 & 800 & 400 \\ 400 & 800 & 400 & 1200 & 800 & 400 & 800 \end{bmatrix}$
19	$\begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 40 & 77 & 77 & 40 & 40 & 77 & 40 & 77 & 40 & 16 \end{bmatrix}$	7	3	$\begin{bmatrix} 2 & 2 & 2 & 4 & 3 & 3 & 4 & 4 & 4 & 3 \end{bmatrix}$	$\begin{bmatrix} 3 & 4 & 3 \end{bmatrix}$	$\begin{bmatrix} 1 & 3 & 2 \end{bmatrix}$	$\begin{bmatrix} 1350 & 1350 & 450 & 450 & 900 & 900 & 1350 \\ 900 & 1350 & 1350 & 450 & 900 & 900 & 1350 \\ 1350 & 900 & 1350 & 900 & 1350 & 900 & 900 \end{bmatrix}$
20	$\begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 22 & 22 & 22 & 10 & 52 & 22 & 22 & 22 & 10 & 52 \end{bmatrix}$	7	3	$\begin{bmatrix} 4 & 2 & 4 & 3 & 3 & 2 & 2 & 4 & 2 & 4 \end{bmatrix}$	$\begin{bmatrix} 5 & 5 & 3 \end{bmatrix}$	$\begin{bmatrix} 3 & 2 & 1 \end{bmatrix}$	$\begin{bmatrix} 1000 & 500 & 500 & 1000 & 1000 & 1000 & 1000 \\ 1000 & 1500 & 1500 & 500 & 1500 & 500 & 1500 \\ 1000 & 1500 & 1500 & 1500 & 500 & 1000 & 1000 \end{bmatrix}$
21	$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 22 & 41 & 21 & 41 & 41 & 41 & 41 & 21 & 21 & 21 \end{bmatrix}$	7	3	$\begin{bmatrix} 4 & 2 & 3 & 3 & 3 & 2 & 2 & 3 & 3 & 3 \end{bmatrix}$	$\begin{bmatrix} 3 & 5 & 5 \end{bmatrix}$	$\begin{bmatrix} 2 & 2 & 1 \end{bmatrix}$	$\begin{bmatrix} 1650 & 1650 & 1650 & 1100 & 1100 & 1650 & 550 \\ 1650 & 1100 & 1100 & 1100 & 550 & 550 & 1100 \\ 550 & 1650 & 550 & 1100 & 550 & 550 & 550 \end{bmatrix}$

APPENDIX B

MINIWORLD MATHMETICAL MODEL

B.1 Mathmetical Model and Solution for Mini Campaign Optimization Problem with Network Effect

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Model:

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Max :

$12Y[0,0] + 12Y[0,1] + 12Y[0,2] + 12Y[0,3] + 12Y[0,4] + 12Y[0,5] + 12Y[0,6] + 12Y[0,7] + 12Y[0,8] + 12Y[0,9] + 12Y[0,10] + 12Y[0,11] + 12Y[0,12] + 12Y[0,13] + 12Y[0,14] + 12Y[0,15] + 12Y[0,16] + 12Y[0,17] + 12Y[0,18] + 12Y[0,19] + 12Y[0,20] + 12Y[0,21] + 12Y[0,22] + 12Y[0,23] + 12Y[0,24] + 12Y[0,25] + 12Y[0,26] + 12Y[0,27] + 12Y[0,28] + 12Y[0,29] + 12Y[0,30] + 12Y[0,31] + 12Y[0,32] + 12Y[0,33] + 12Y[0,34] + 12Y[0,35] + 12Y[0,36] + 12Y[0,37] + 12Y[0,38] + 12Y[0,39] + 12Y[0,40] + 12Y[0,41] + 12Y[0,42] + 12Y[0,43] + 12Y[0,44] + 12Y[0,45] + 12Y[0,46] + 12Y[0,47] + 12Y[0,48] + 12Y[0,49]$

$subject\ to$:

$X[0,0,0,0] \leq 1.0$

$X[0,1,0,0] \leq 1.0$

$X[0,2,0,0] \leq 1.0$

$X[0,3,0,0] \leq 1.0$

$X[0,4,0,0] \leq 1.0$

$X[0,5,0,0] \leq 1.0$

$X[0,6,0,0] \leq 1.0$

$X[0,7,0,0] \leq 1.0$

$X[0,8,0,0] \leq 1.0$

$X[0,9,0,0] \leq 1.0$

$X[0,10,0,0] \leq 1.0$

$X[0,11,0,0] \leq 1.0$

$X[0,12,0,0] \leq 1.0$

$X[0,13,0,0] \leq 1.0$

$X[0,14,0,0] \leq 1.0$

$X[0,15,0,0] \leq 1.0$

$X[0,16,0,0] \leq 1.0$

$X[0,17,0,0] \leq 1.0$

$X[0,18,0,0] \leq 1.0$

$X[0,19,0,0] \leq 1.0$

$X[0,20,0,0] \leq 1.0$

$X[0,21,0,0] \leq 1.0$

$X[0,22,0,0] \leq 1.0$

$X[0,23,0,0] \leq 1.0$

$X[0,24,0,0] \leq 1.0$

$X[0,25,0,0] \leq 1.0$

$X[0,26,0,0] \leq 1.0$

$X[0,27,0,0] \leq 1.0$

$X[0,28,0,0] \leq 1.0$

$X[0,29,0,0] \leq 1.0$

$X[0,30,0,0] \leq 1.0$

$X[0,31,0,0] \leq 1.0$

$X[0,32,0,0] \leq 1.0$

$X[0,33,0,0] \leq 1.0$

$X[0,34,0,0] \leq 1.0$

$X[0,35,0,0] \leq 1.0$

$X[0,36,0,0] \leq 1.0$

$X[0, 37, 0, 0] \le 1.0$
$X[0, 38, 0, 0] \le 1.0$
$X[0, 39, 0, 0] \le 1.0$
$X[0, 40, 0, 0] \le 1.0$
$X[0, 41, 0, 0] \le 1.0$
$X[0, 42, 0, 0] \le 1.0$
$X[0, 43, 0, 0] \le 1.0$
$X[0, 44, 0, 0] \le 1.0$
$X[0, 45, 0, 0] \le 1.0$
$X[0, 46, 0, 0] \le 1.0$
$X[0, 47, 0, 0] \le 1.0$
$X[0, 48, 0, 0] \le 1.0$
$X[0, 49, 0, 0] \le 1.0$
$X[0, 0, 0, 0]+X[0, 1, 0, 0]+X[0, 2, 0, 0]+X[0, 3, 0, 0]+X[0, 4, 0, 0]+X[0, 5, 0, 0]+X[0, 6, 0, 0]+X[0, 7, 0, 0]+X[0, 8, 0, 0]+X[0, 9, 0, 0]+ X[0, 10, 0, 0]+X[0, 11, 0, 0]+X[0, 12, 0, 0]+X[0, 13, 0, 0]+X[0, 14, 0, 0]+X[0, 15, 0, 0]+X[0, 16, 0, 0]+X[0, 17, 0, 0]+X[0, 18, 0, 0]+ X[0, 19, 0, 0]+X[0, 20, 0, 0]+X[0, 21, 0, 0]+X[0, 22, 0, 0]+X[0, 23, 0, 0]+X[0, 24, 0, 0]+X[0, 25, 0, 0]+X[0, 26, 0, 0]+X[0, 27, 0, 0]+ X[0, 28, 0, 0]+X[0, 29, 0, 0]+X[0, 30, 0, 0]+X[0, 31, 0, 0]+X[0, 32, 0, 0]+X[0, 33, 0, 0]+X[0, 34, 0, 0]+X[0, 35, 0, 0]+X[0, 36, 0, 0]+ X[0, 37, 0, 0]+X[0, 38, 0, 0]+X[0, 39, 0, 0]+X[0, 40, 0, 0]+X[0, 41, 0, 0]+X[0, 42, 0, 0]+X[0, 43, 0, 0]+X[0, 44, 0, 0]+X[0, 45, 0, 0]+ X[0, 46, 0, 0] + X[0, 47, 0, 0] + X[0, 48, 0, 0] + X[0, 49, 0, 0] \le 15.0$
$Y[0, 0] \le X[0, 0, 0, 0]$
$Y[0, 1] \le X[0, 1, 0, 0]$
$Y[0, 2] \le X[0, 2, 0, 0]$
$Y[0, 3] \le X[0, 3, 0, 0] + X[0, 6, 0, 0]$
$Y[0, 4] \le X[0, 4, 0, 0] + X[0, 7, 0, 0] + X[0, 28, 0, 0]$
$Y[0, 5] \le X[0, 5, 0, 0] + X[0, 8, 0, 0]$
$Y[0, 6] \le X[0, 3, 0, 0] + X[0, 6, 0, 0] + X[0, 45, 0, 0]$
$Y[0, 7] \le X[0, 4, 0, 0] + X[0, 7, 0, 0] + X[0, 8, 0, 0] + X[0, 22, 0, 0]$
$Y[0, 8] \le X[0, 5, 0, 0] + X[0, 7, 0, 0] + X[0, 8, 0, 0]$
$Y[0, 9] \le X[0, 9, 0, 0] + X[0, 11, 0, 0] + X[0, 20, 0, 0] + X[0, 43, 0, 0]$
$Y[0, 10] \le X[0, 10, 0, 0] + X[0, 29, 0, 0] + X[0, 35, 0, 0] + X[0, 48, 0, 0]$
$Y[0, 11] \le X[0, 9, 0, 0] + X[0, 11, 0, 0] + X[0, 19, 0, 0]$
$Y[0, 12] \le X[0, 12, 0, 0]$
$Y[0, 13] \le X[0, 13, 0, 0] + X[0, 20, 0, 0]$
$Y[0, 14] \le X[0, 14, 0, 0]$
$Y[0, 15] \le X[0, 15, 0, 0] + X[0, 17, 0, 0]$
$Y[0, 16] \le X[0, 16, 0, 0]$
$Y[0, 17] \le X[0, 15, 0, 0] + X[0, 17, 0, 0]$
$Y[0, 18] \le X[0, 18, 0, 0]$
$Y[0, 19] \le X[0, 11, 0, 0] + X[0, 19, 0, 0]$
$Y[0, 20] \le X[0, 9, 0, 0] + X[0, 13, 0, 0] + X[0, 20, 0, 0]$
$Y[0, 21] \le X[0, 21, 0, 0] + X[0, 28, 0, 0]$
$Y[0, 22] \le X[0, 7, 0, 0] + X[0, 22, 0, 0] + X[0, 38, 0, 0]$
$Y[0, 23] \le X[0, 23, 0, 0]$
$Y[0, 24] \le X[0, 24, 0, 0]$
$Y[0, 25] \le X[0, 25, 0, 0]$
$Y[0, 26] \le X[0, 26, 0, 0]$
$Y[0, 27] \le X[0, 27, 0, 0]$
$Y[0, 28] \le X[0, 4, 0, 0] + X[0, 21, 0, 0] + X[0, 28, 0, 0]$
$Y[0, 29] \le X[0, 10, 0, 0] + X[0, 29, 0, 0]$
$Y[0, 30] \le X[0, 30, 0, 0]$
$Y[0, 31] \le X[0, 31, 0, 0] + X[0, 45, 0, 0]$
$Y[0, 32] \le X[0, 32, 0, 0]$
$Y[0, 33] \le X[0, 33, 0, 0]$
$Y[0, 34] \le X[0, 34, 0, 0]$
$Y[0, 35] \le X[0, 10, 0, 0] + X[0, 35, 0, 0]$
$Y[0, 36] \le X[0, 36, 0, 0]$
$Y[0, 37] \le X[0, 37, 0, 0]$
$Y[0, 38] \le X[0, 22, 0, 0] + X[0, 38, 0, 0]$
$Y[0, 39] \le X[0, 39, 0, 0]$
$Y[0, 40] \le X[0, 40, 0, 0]$

$Y[0,41] \leq X[0,41,0,0]$

$Y[0,42] \leq X[0,42,0,0]$

$Y[0,43] \leq X[0,9,0,0] + X[0,43,0,0]$

$Y[0,44] \leq X[0,44,0,0]$

$Y[0,45] \leq X[0,6,0,0] + X[0,31,0,0] + X[0,45,0,0]$

$Y[0,46] \leq X[0,46,0,0]$

$Y[0,47] \leq X[0,47,0,0]$

$Y[0,48] \leq X[0,10,0,0] + X[0,48,0,0]$

$Y[0,49] \leq X[0,49,0,0]$

$========$

$Solution:$

$obj: 348.0$

$========$

$X[0,0,0,0] = 1.0$

$X[0,1,0,0] = 1.0$

$X[0,2,0,0] = 1.0$

$X[0,6,0,0] = 1.0$

$X[0,8,0,0] = 1.0$

$X[0,9,0,0] = 1.0$

$X[0,10,0,0] = 1.0$

$X[0,11,0,0] = 1.0$

$X[0,12,0,0] = 1.0$

$X[0,15,0,0] = 1.0$

$X[0,20,0,0] = 1.0$

$X[0,22,0,0] = 1.0$

$X[0,28,0,0] = 1.0$

$X[0,34,0,0] = 1.0$

$X[0,45,0,0] = 1.0$

$Y[0,0] = 1.0$

$Y[0,1] = 1.0$

$Y[0,2] = 1.0$

$Y[0,3] = 1.0$

$Y[0,4] = 1.0$

$Y[0,5] = 1.0$

$Y[0,6] = 1.0$

$Y[0,7] = 1.0$

$Y[0,8] = 1.0$

$Y[0,9] = 1.0$

$Y[0,10] = 1.0$

$Y[0,11] = 1.0$

$Y[0,12] = 1.0$

$Y[0,13] = 1.0$

$Y[0,15] = 1.0$

$Y[0,17] = 1.0$

$Y[0,19] = 1.0$

$Y[0,20] = 1.0$

$Y[0,21] = 1.0$

$Y[0,22] = 1.0$

$Y[0,28] = 1.0$

$Y[0,29] = 1.0$

$Y[0,31] = 1.0$

$Y[0,34] = 1.0$

$Y[0,35] = 1.0$

$Y[0,38] = 1.0$

$Y[0,43] = 1.0$

$Y[0,45] = 1.0$

$Y[0,48] = 1.0$

$========$

APPENDIX C

SOURCE CODES

C.1 Python implementation for Algorithm-5

```
def max_degree(self, grph):
    nn = max(grph.degree, key=lambda x: x[1])
    grph.remove_node(nn[0])
    return nn[0]

def sort_nodes_to_inc_span(self, grph):
    return [self.max_degree(grph) for i in range(len(grph.nodes()))]
```

C.2 Python implementation for Algorithm-6

```
def max_degree_b(self, grph, seen):
    max_val = max(grph.degree, key=lambda x: x[1])[1]
    nns = [n for n in grph.degree if n[1] == max_val]
    nns_ = [n for n in nns if n[0] not in seen]
    if len(nns_)>0:
        nn = nns_[0]
    else:
        nn = nns[0]
    neighbors = grph.neighbors(nn[0])
    grph.remove_node(nn[0])
    seen |= set(list(neighbors))
    return (nn[0], seen)

def sort_nodes_to_inc_span(self, grph):
    result=list()
    seen = set()
    for _ in range(len(grph.degree)):
        elem, seen = self.max_degree_b(grph, seen)
        result.append(elem)
    return result
```

C.2.1 Python implementation for Algorithm-7

```
def max_degree_b(self, grph, seen, X_u):
    dd = {gd[0]:gd[1]*X_u[gd[0]] for gd in grph.degree}
    max_val = max(dd.items(), key=lambda x: x[1])[1]
    nns = [n for n in dd.items() if n[1] == max_val]
    nns_ = [n for n in nns if n[0] not in seen]
    if len(nns_)>0:
        nn = nns_[0]
    else:
        nn = nns[0]
    neighbors = grph.neighbors(nn[0])
    grph.remove_node(nn[0])
    seen |= set(list(neighbors))
    return (nn[0], seen)

def sort_nodes_to_inc_span(self, grph, X_u):
```

```
result=list()
seen = set()
for _ in range(len(grph.degree)):
    elem, seen = self.max_degree_b(grph, seen, X_u)
    result.append(elem)
return result
```

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